# SUPPLEMENT: DOPE: DOWNSTREAM OBJECTIVE-PRESERVING ENCODING OF REGRESSION TABLES INTO KILOBYTE GENERATORS

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## ABSTRACT

This supplement holds the method details, the release-gate specification and its conformance test, the data and selection protocol, the statistics, every five-seed contrast and level behind the main paper, the controls, the fidelity and empirical-privacy panels, the ablations, the baseline coverage, the per-lineage tables, and the extended related work. A blank retention cell is an unavailable fit or an auditor that is uninformative on one of the three sample seeds; it is never a zero. BeyondArena is a separate fit check and is never pooled with the PMLB panel.

## A METHOD DETAILS

### A.1 ARTIFACT CONTENTS AND SAMPLING

Equation (1) of the main paper is the conditional mean; the joint draw is separate. The profile reported in the main paper samples with the symbolic kernel stored in the artifact; the separate neural-joint sampler is unused. The stored dependence is one choice from the compile search: independence, a Chow–Liu tree, a triangular autoregression, a sparse Gaussian copula, a factor model, a truncated vine, or a two-component mixture, and the reported artifacts span several of these families.

Training screens the inputs by absolute rank correlation with the target and keeps at most twelve. The artifact stores a marginal and a transform for each feature, the selected dependence, the residual, a noise model, and a per-feature missing rate. A draw samples a latent vector from the stored dependence, maps each coordinate through the standard normal distribution function, inverts the marginal, and inverts the feature transform; the residual then supplies $\hat{y}$. When the training target is not binary, the stored noise is heteroscedastic: a linear fit of the log absolute residual on up to four screened features, with the scale clamped between the lower and upper deciles of those absolute residuals. The sampled target adds Gaussian noise of that scale and is clamped to the unit interval of the training transform. Each feature is then replaced, independently, by a missing mark with its stored probability. Before encoding, the compiler quantizes the marginals, the dependence, the residual, and the noise model. The charged byte count of a DOPE artifact is the encoded model file plus `projection.json`; a baseline is charged the files its sampler reads, as listed in Appendix C.3.

### A.2 TRAINING FREEZE, COMPILE SEARCH, AND THE REPORTED PROFILE

Training maps the screened inputs through the width-16 $\tanh$ layer and runs AdamW on the regression mean-squared error, with gradient clipping at $5$. The hidden weights and biases are then frozen, and the intercept, the linear skip, and the hidden readout are refit on the frozen features. Coefficients whose absolute value is below $10^{-5}$ are dropped from the skip.

The compile search records the proxy $0.45u + 0.25d + 0.30j$ and keeps the candidate with the smallest normalized gate shortfall, then the smallest artifact, then the lowest candidate id. It reads

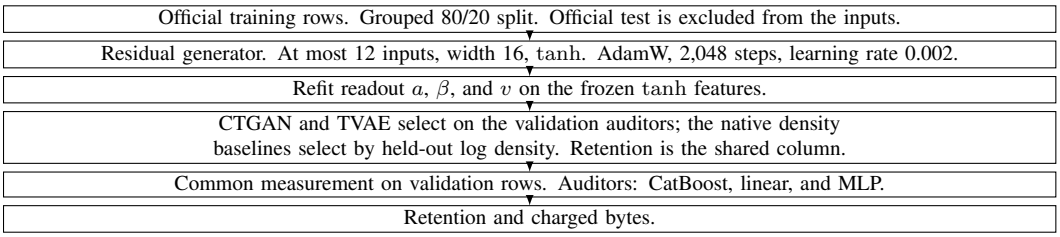



Figure 1: **Pipeline of the reported profile.** The official test is outside the inputs. No lineage denominator and no Holm family.

the fit rows only. A candidate record carries no release certificate, and no global DOPE family has been selected.

The reported research profile, `features12_steps2048`, was chosen among four bounded research profiles (twelve or twenty-four screened inputs, 512 or 2,048 steps) on a six-lineage discovery panel and checked on six disjoint confirmation lineages, by validation retention at fit seed 11. Those twelve lineages belong to the hundred-lineage panel, so the reported profile carries a mild selection on validation rows. The untuned trainer default, used for any other profile, sets the hidden width between 4 and 12 and trains for 96 steps; the main paper reports it as a separate arm. Other neural shapes (micro-TVAEs, TinyMAT, joint TVAE, a masked transformer, and a TabSyn candidate) are outside this measurement, and the direct-rank TabDDPM candidate is a research comparator with no claim on the 10,240-byte tier.

### A.3 Dataset sketch and router vector

An index can store a fixed sketch of a training table, which a separate dataset-action router consumes; neither enters the generator reported in the main paper. `embed-dataset` min-max normalizes each training column onto the unit interval, leaves a missing entry missing, and never reads the official test. Each column becomes twelve content summaries: a missing rate, moments, quantiles, a unique-value share, a tail ratio, and the absolute association with the target. Columns are sorted by those summaries, variance first. A ten-number header records the task, the log row count, the log column count, and the shape of the target. Each of the twelve summaries is then reduced across columns to a six-number distribution (minimum, lower quartile, median, upper quartile, maximum, and mean), which is the $12 \times 6$ block. The sketch then keeps the first $64$ columns and stores each column's twelve summaries, padding a shorter table with zeros, and absolute correlations among the kept columns collapse to one further six-number distribution. The width is therefore $10 + 12 \times 6 + 64 \times 12 + 6 = 856$. The router vector appends $24$ candidate states, each a hidden state of width $128$ plus $10$ action predictions, to that sketch, for $856 + 24 \times (128 + 10) = 4{,}168$ dimensions. A neighbour found with the sketch is still fit with the reported profile and still charged in bytes.

## B Release gate

### B.1 MFS-v2, the historical formula

MFS-v2 is the release score that historical receipts pin. With component $c_i \in [0, 1]$, weight $w_i > 0$, $\epsilon = 10^{-6}$, and $W = \sum_i w_i$,

$$\text{MFS-v2} = 100 \exp\left( \frac{1}{W} \sum_i w_i \ln(\epsilon + c_i) \right)$$

when the hard-gate map is non-empty, every hard gate passed, and every positive-weight component is finite and inside $[0, 1]$; otherwise the scalar is null. Its weights are utility transfer $0.30$, driver fidelity $0.20$, distribution fidelity $0.20$, structure fidelity $0.15$, coverage realism $0.10$, and compactness $0.05$. Privacy has soft weight zero, and a membership failure, an attribute failure, or a copied row is a gate that utility cannot offset. Compactness is $\text{clamp}(1 - \text{bytes}/10240, 0, 1)$, and an artifact past $10{,}240$ bytes fails the byte gate.

### B.2 What the components measure

Utility transfer is a one-sided lower bound on retention, clamped into $[0, 1]$; it is distinct from the median retention in the main paper. Driver fidelity is an equal mixture of feature-importance rank agreement and top-feature-set overlap between the real and the synthetic table. Distribution fidelity is half the minimum marginal fidelity, plus a quarter of the sliced-Wasserstein fidelity and a quarter of the maximum-mean-discrepancy fidelity. Structure fidelity is the minimum dependence fidelity. The driver term stored beside the panel is the same equal mixture, applied to absolute correlations with the target; it differs from the permutation-importance auditor in the v2 definition, and this paper reports no component value from it.

### B.3 MFS-v3 and representation closeness

MFS-v3 adds representation closeness to the geometric mean, as in equation (2) of the main paper. v3 utility transfer is the same one-sided bound on the Kumo large retention: the model is conditioned on the synthetic rows and scored on real holdout rows. Mitra and TabICL must be measured on the same mapped tables for the gate. Their retentions stay out of that bound, and a CatBoost retention cannot fill this component. Real fit rows, real holdout rows, and synthetic rows share one map, fit on the real fit rows only: numeric columns go to the open unit interval by an empirical mid-rank quantile, categorical levels are pre-encoded to fixed codes in that interval, and column names become HMAC digests. The map withholds the minimum and the maximum, and a second fit on the synthetic rows is a protocol failure.

Representation closeness asks whether Kumo large places the mapped synthetic table near the mapped real table. The distance is half the Euclidean length between the two unit-length vectors, so it lies in $[0, 1]$, and closeness is one minus that distance divided by the distance to a matched null, clamped to $[0, 1]$. The synthetic table must also sit closer than an independent-marginal table of the same size under the same map. The matched null is chosen among three independent-marginal tables drawn on the shared map, seeded by the first eight bytes, little-endian, of the SHA digest of the dataset, the method, the configuration, the sample seed, and the candidate index $0$, $1$, or $2$; candidates are ordered by ordinary linear mean squared error on the mapped holdout, then by index, and the middle candidate is the matched null. Those losses are fixed before any encoder runs. Mitra and TabICL must show the same sign as Kumo large on the gap against the independent-marginal table; a sign flip, a zero match distance, or a missing vector makes the component null. Neither is averaged into representation closeness. The licensed encoder set is Kumo large (OpenMDW and Apache), Mitra (Apache), and TabICL (BSD); Foundation, TabPFN, TabDPT, the project's Hyperion, and target-aware blocks are refused. TabDPT is refused because its pretraining tables overlap this panel; whether the pretraining corpora of the three admitted encoders overlap PMLB is unknown, so their retentions are reported as transfer measurements rather than as independent evidence.

A recipient keeps the generator and a hash manifest: a salt identifier, the normalizer name, the encoder pin, one digest per column, a dtype code, and the byte length. The manifest carries no row, header, category string, extremum, or encoder weight. A public sidecar that echoes a source string or an extremum fails the cleartext gate, and an exact copy of a real row fails its own gate. The near-copy floor is the $0.01$ quantile, with linear interpolation, of squared Euclidean distances from each distinct real fit row to the nearest other distinct real fit row on the shared map; a synthetic row below that floor, an undefined floor, or a charge above $10{,}240$ bytes makes the scalar null.

### B.4 Executable conformance

The release score uses the contract's weighted geometric mean with a component floor, clipped to $[0, 100]$; the components lie in $[0, 1]$ and a failed gate yields a null scalar. The matched null is the median linear-MSE candidate among three deterministic independent-marginal tables. This is a median selector; a learnability band is outside this implementation. The near-copy floor is the declared quantile of nearest-other squared distances among distinct real fit rows on the shared map, with linear interpolation, and undefined floors fail the gate. The scorer tests in `research/benchmark/tests/test_mfs_v3.py` verify exactly $100$ for a perfect profile, the byte boundary, and each prerequisite; the panel tests verify median null selection and the

Table 1: **MFS-v3 contract fields as implemented.** Each row is read from `production/kpi-contract-v3.json` by the scorer. No Holm family.

| | |
|---|---|
| Scalar | $100 \exp(\sum_i w_i \log \max(s_i, \epsilon) / \sum_i w_i)$, clipped to $[0, 100]$ |
| Weights | compactness = 0.05, coverage_realism = 0.05, distribution_fidelity = 0.15, driver_fidelity = 0.15, representation_closeness = 0.15, structure_fidelity = 0.1, utility_transfer = 0.35 |
| Epsilon | $1 \times 10^{-6}$ |
| Bytes | $b \in \mathbb{Z},\ 0 < b \leq 10240$ |
| Prerequisites | Real-vs-real control; required sizes; artifact sampling; locked metric implementations. |
| Matched null | Median linear MSE of three independent-marginal candidates; candidate-index tie break. |
| Near-copy floor | $q = 0.01$ of nearest-other distinct real-row squared distances; linear interpolation. |

Table 2: **Invented applications of the bounded MFS-v3 formula.** Arithmetic illustrations only. No Holm family.

| Invented input | Scalar |
|---|---|
| All seven components equal to one | 100 |
| Compactness component set to zero | 50.12 |
| Byte gate failed | null |

quantile floor. Table 1 lists the implemented fields, and Table 2 applies the bounded formula to invented components.

### B.5 Clean-license transfer on the density block

The density block was also scored by Kumo large, Mitra, and TabICL on the shared map, with the three sample seeds of the main panel. These auditors are separate from the CatBoost, linear, and MLP rows, and their Holm adjustment is one family per auditor of size three (that auditor against the three density comparators). Kumo large averages 16 inverse-transformed quantile forecasts. The receipt scores 0 of 388 method–lineage cells (97 lineages $\times$ 4 methods), so the scalar is null. Cleartext is unscanned on 388 method–lineage cells.

## C Data, selection, cost, and availability

### C.1 Data lock and partitions

The data lock admits 104 MIT-licensed entries and prepares 100. 4 lineages have no split object because the same source-row digest appears in two partitions (`check_nonoverlap` raises on that overlap), and another 3,755 catalog entries lack a redistribution grant and are excluded. The prepared cut is a grouped $80/20$ split of the official training rows, seed 1729. Worker directories hold `train.csv` and `validation.csv` only; the official test is held elsewhere and its contents are unread. The fit view has 72 to 800 rows; sources above the quota use a hybrid coreset that keeps the observed feature and target range. Of these prepared lineages, 20 names begin with feynman, which the PMLB catalog describes as synthetic physics, 14 begin with strogatz, which that catalog describes as simulated dynamics, 25 contain the Friedman-style fri marker, 6 contain bng and are Bayesian-network generators, and 35 are the remainder. This paper does not re-verify which of the remainder are observational measurements. Distinct lineage identifiers are not a claim of independent source families. The legacy lineage-bootstrap intervals resample lineages independently. The printed family-cluster intervals use a two-stage bootstrap. Lineages in the four simulated name families (feynman, strogatz, fri, and bng) form one cluster per family, and every other lineage is a singleton cluster. Each draw resamples clusters with replacement, then resamples lineages with replacement inside each drawn cluster. A verified parent map is absent from the

Table 3: **Kumo large, Mitra, and TabICL on the density block at $4n$.** Each cell is a retention; the denominator is the paired informative lineages in that row. Holm family size is three per auditor. W/T/L counts lineages; a tie is a difference of zero.

| Auditor | Comparator | $n$ | DOPE | Other | Difference | W/T/L | Holm $p$ |
|---|---|---|---|---|---|---|---|
| Kumo large | Gaussian copula | 97 | **0.863** [0.809, 0.880] | 0.747 [0.668, 0.778] | 0.059 [0.0098, 0.116] | 62/0/35 | $5.70 \times 10^{-4}$ |
| Kumo large | Chow–Liu | 97 | **0.863** [0.809, 0.880] | 0.562 [0.501, 0.629] | 0.247 [0.195, 0.306] | 85/0/12 | $1.34 \times 10^{-11}$ |
| Kumo large | Indep. marginals | 97 | **0.863** [0.809, 0.880] | -0.000819 [-0.00189, 7.58e-05] | 0.874 [0.834, 0.888] | 94/0/3 | $7.46 \times 10^{-17}$ |
| Mitra | Gaussian copula | 97 | **0.858** [0.803, 0.887] | 0.713 [0.585, 0.771] | 0.066 [0.023, 0.135] | 65/0/32 | $1.34 \times 10^{-4}$ |
| Mitra | Chow–Liu | 97 | **0.858** [0.803, 0.887] | 0.583 [0.527, 0.643] | 0.242 [0.182, 0.293] | 85/0/12 | $2.71 \times 10^{-11}$ |
| Mitra | Indep. marginals | 97 | **0.858** [0.803, 0.887] | -0.00907 [-0.014, -0.00377] | 0.890 [0.840, 0.909] | 94/0/3 | $6.01 \times 10^{-17}$ |
| TabICL | Gaussian copula | 97 | **0.871** [0.835, 0.893] | 0.754 [0.672, 0.787] | 0.079 [0.00947, 0.116] | 62/0/35 | $1.61 \times 10^{-4}$ |
| TabICL | Chow–Liu | 97 | **0.871** [0.835, 0.893] | 0.587 [0.484, 0.650] | 0.267 [0.200, 0.302] | 87/0/10 | $7.12 \times 10^{-12}$ |
| TabICL | Indep. marginals | 97 | **0.871** [0.835, 0.893] | -0.00446 [-0.012, 0.0012] | 0.891 [0.846, 0.919] | 92/0/5 | $5.83 \times 10^{-17}$ |

Holm family: 3 tests per auditor; DOPE versus Gaussian copula, Chow–Liu, and independent marginals.

Table 4: **Pinned representation vectors.** Each row names the activation that the density-panel receipt reads as that auditor's table vector. No Holm family.

| Auditor | Pinned vector |
|---|---|
| Kumo large | `row_project_output_mean_labeled_rows` |
| Mitra | `final_layer_norm_input_mean_self_query` |
| TabICL | `tf_icl_output_mean_self_query` |

ledger, so those intervals put remainder lineages that share a parent in separate clusters, including the cpu_small lineages. Wilcoxon and TOST retain independent-lineage assumptions.

The fit view is not the full PMLB table. Frozen catalog metadata draws each official training and test file by quota stratification on the target distribution and high-variance feature quantile bins, aiming at 1000 training rows and 1000 test rows, with at most 2000 rows in total and at least 101 rows. A source smaller than that quota keeps every row and still reserves a test share of at least 0.2. Of the 100 prepared lineages, 60 were already inside the quota. The other lineages use a hybrid coreset that keeps the observed feature and target range; the bin counts are 1 lineage at 11 bins, and 39 lineages at 50 bins. The generator then groups identical official-training rows and assigns those groups with seed 1729 so that about 80% of the rows become the fit view and the rest become the validation holdout. Duplicate rows are not split across that cut. On these lineages the fit view has 72 to 800 rows. The bound keeps every lineage on one worker view and leaves the official test sealed. It is not a second draw at scoring time.

### C.2 Selection rule for each method

Table 6 states what chose each configuration and which rows it read. Only DOPE's compile search and the density baselines' fixed or density-selected configurations avoid the scoring auditors; CTGAN and TVAE select on the same validation rows and the same linear and MLP auditors that score retention, and DOPE's research profile was chosen on a discovery panel inside the evaluated lineages. The sealed test run removes all of these overlaps for its estimate.

### C.3 Compute ledger and budget protocol

Table 7 lists fit time, peak memory, GPU models, GPU hours, and charged bytes for receipts that store a fit timer. Peak RAM is resident bytes in mebibytes, peak VRAM is `peak_gpu_used_mib`, and GPU hours sum the walls on which the receipt recorded a GPU process. No receipt stores a separate CPU timer. Sample-timer medians (maxima) are 186.7 (197.0) for CTGAN, 186.1 (196.6) for TVAE, and 342.4 (511.7) for Forest-Flow. Gaussian copula, Chow–Liu, and independent-marginal attempts store one unsplit `wall_seconds`, with medians 1.71, 0.168, and 0.162 and maxima 39.9, 1.29, and 0.429. The logged replay medians 4.56 and 14.4 are replay walls, distinct from the scored runs. Forest-Flow medians use the 12 native-grid operation walls; 953 sums `fit_seconds` and 8,801 sums receipt operations. Byte charges count the files a sampler reads: the DOPE model plus `projection.json`; every TabSyn file except intermediate diffusion-epoch checkpoints, which the sampler never reads; and the stored per-lineage charge for every other baseline.

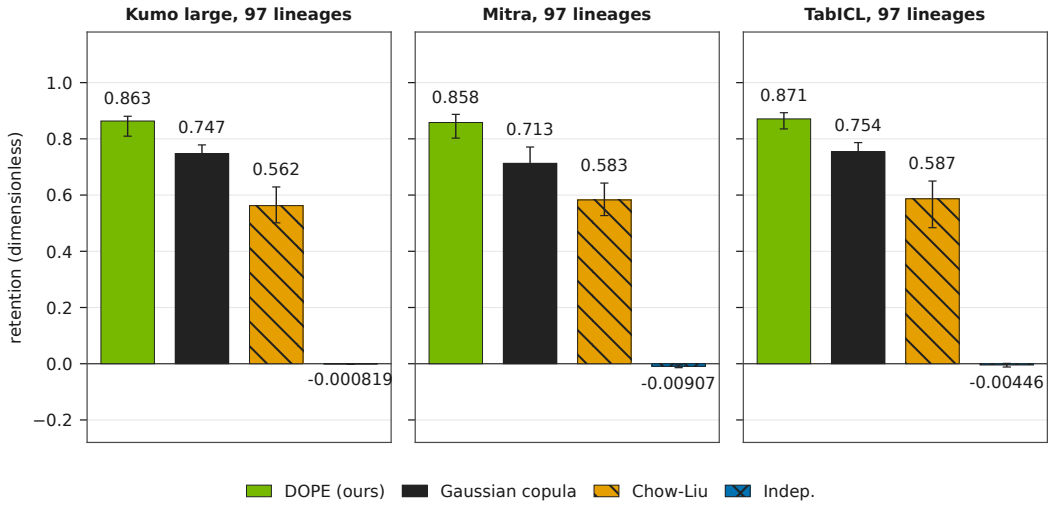



Figure 2: **Kumo large, Mitra, and TabICL retention at $4n$.** Each bar is the lineage median and its lineage-bootstrap interval; the denominator is the paired informative lineages in that panel. Holm family size is three per auditor.

Table 5: **Test-split provenance.** File-name counts only. No retention denominator and no Holm family.

| Item | Record |
|---|---|
| Catalog SHA-256 | `ab9fda8d2dea46067b70a42812e9d3c1d9dc6ba025780100df34377e81aa1120` |
| Eligible / prepared / overlap exclusions | 104 / 100 / 4 |
| Unknown-rights catalog entries held out | 3755 |
| Split | `official_test_grouped_training_80_20`, seed 1729 |
| Overlap lineage ids | `590210721973399a`, `9fe58a656fdb458a`, `d06a584f07d0acd5`, `d0ba629ed07b5d63` |
| Final evaluation authorized | false |
| S3 worker directories / `train.csv` / `validation.csv` | 100 / 100 / 100 |
| `test.csv` under S3 workers | 0 |
| BeyondArena workers / worker `test.csv` | 5 / 0 |
| Evaluator `test.csv` files, contents not read | 5 |

The frozen Forest-Flow budget keeps the 600-second fit cap and every timed-out attempt receipt; a timeout cell has missing quality metrics and counts as no DOPE win. The October 8 disposition addendum was written after 75 observed timeouts and is therefore no retroactive pre-registration. predeclare_v2 instead fixes a common budget (`B1`: 1,800 seconds per fit on one 16-core slot and one GPU) for every method and every lineage, applied non-selectively; `docs/whitepaper/BASELINE_PROTOCOL.md` links the frozen checkpoint and every disposition.

### C.4 Public data and artifact availability

A credential-free path to the source data is

```
python3 -m research.benchmark.fetch_pmlb --list
python3 -m research.benchmark.fetch_pmlb DATASET_ID OUTPUT_DIRECTORY
```

It authenticates PMLB data against an upstream Git LFS SHA pin at a fixed public revision. The public manifest lists every prepared lineage's PMLB identifier and the historical train, validation, and test split hashes from the committed data lock. The historical staging used a hybrid coreset and target-stratified selection whose exact row indices were never captured, so the public fetch reproduces the source datasets and the split identities but cannot rebuild the exact historical partitions; publishing the fit and validation partitions themselves (PMLB is MIT-licensed) closes that gap and is scheduled with the artifact release. The originally scored model files and synthetic rows have no published bundle yet; the committed validation ledgers, the original metadata, and the compressed numeric logs, authenticated by the input pins and rebuilt by `just paper`, reproduce every reduction and figure.

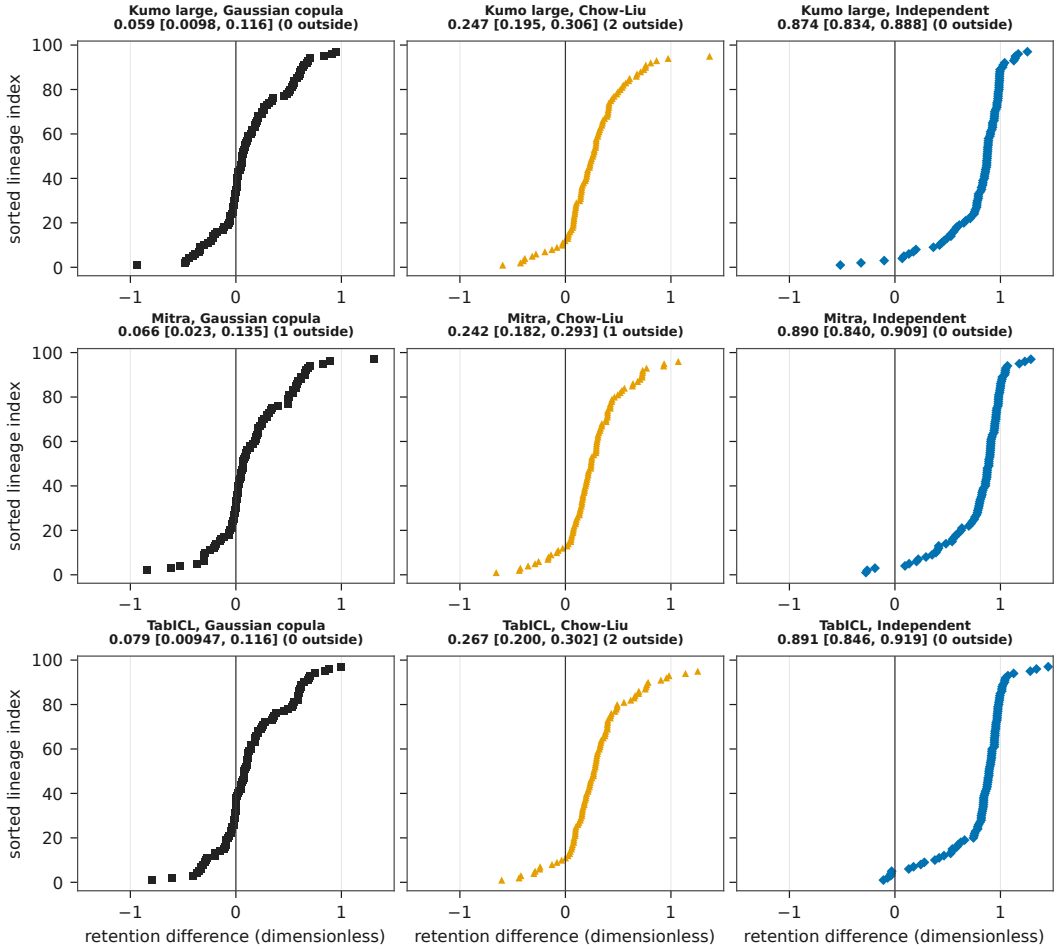



Figure 3: **Paired retention differences for Kumo large, Mitra, and TabICL, sorted by lineage.** The denominator is the paired informative lineages in that panel. Holm family size is three per auditor.

### C.5 Reproducibility and hashes

Regenerate every `generated/` file and the figure PDFs with `just paper` from the commit that contains these files. The same recipe writes the main PDF, the anonymous main PDF, the supplement PDF, and the anonymous supplement PDF. That recipe reads the committed validation ledgers, original fit metadata, and compressed numeric replay logs. The replay cost, loss curves, and BeyondArena extracts have file and field mappings in `generated/original-field-map.json`. Those original metadata snapshots were captured after the historical runs; they leave historical custody and the scored artifacts in place. Pinned LICENSE extracts are reread when their local store is available. Official test files are sealed until the single pre-registered run. The catalog hash is ab9fda8d2dea46067b70a42812e9d3c1 d9dc6ba025780100df34377e81aa1120. The grouped training split uses seed 1729. The historical DOPE and legacy comparison fits use fit seed 11 and sample seeds 101, 211, and 307; the five-seed DOPE refit and its pins are in `research/benchmark/review_fixes/predeclare_v2.json`. Retained panels document the additional ARF and Forest-Flow fit seeds and the measured TabSyn and TabDDPM subsets separately. `python3 research/benchmark/verify_paper_numbers.py` exits nonzero when a displayed macro disagrees with those ledgers. The remaining baselines, the shadow-model and artifact attacks, and the full ablation grid enter the tables only after their runs close. The fidelity table is a separate empirical ledger on the grouped validation split. Empirical resemblance is no differential-privacy guarantee and no HIPAA de-identification.

Table 6: **Selection rule for each method.** The scored column is retention on the validation cut. No Holm family.

| Method | Criterion | Data used | Budget |
|---|---|---|---|
| DOPE | Profile chosen on a six-lineage discovery panel and checked on six confirmation lineages; compile search keeps the smallest gate shortfall, then the smallest artifact. | Discovery and confirmation lineages' validation cut; compile search reads fit rows only. | At most twelve inputs, width 16, 2,048 steps, five fit seeds. |
| DOPE untuned default | No selection. | Fit rows only. | Trainer default, five fit seeds. |
| Gaussian copula | Held-out mean log density. Copulas 0.14.1 `GaussianMultivariate`. | Validation cut, distinct from the retention auditors. | One library configuration. |
| CTGAN | Mean of linear and MLP training-mean skill. | The validation rows and the two auditors used at scoring. | Author implementation; stopped runs omitted. |
| TVAE | Same skill as CTGAN. | Same rows and auditors as CTGAN. | Author implementation; stopped runs omitted. |
| Chow–Liu | Native: held-out mean log density over a fixed trial set. Default: no search. | Native: validation cut. Default: fit view, eight bins, Laplace smoothing 1.0. Written for this comparison. | Fixed trial set. |
| Independent marginals | Native: held-out mean log density over a fixed trial set. Default: no search. | Native: validation cut. Default: fit-view marginals. Written for this comparison. | Fixed trial set. |
| ARF | Author default; separately, the native objective. | Author default reads no validation rows; the native arm reads the validation cut. | Five fit seeds. |
| TabSyn | No selection. | Author architecture with a scaled schedule. | 200 VAE and at most 1,000 diffusion epochs (the authors' early stop can end a fit sooner), five fit seeds from 11 to 71. |
| Forest-Flow | Author regression objective on a frozen grid. | 6 confirmation lineages. | Fit cap 600 seconds. |

Table 7: **Fit time, memory, GPU, and charged bytes where a fit timer was stored.** The denominator is the receipts that stored a timer. No Holm family.

| Method | Fit s | RAM / VRAM | Hardware | GPU h | Bytes |
|---|---|---|---|---|---|
| DOPE | 7.25(13.5) | 920.7/ 1067 | one NVIDIA GeForce RTX 4090 and two NVIDIA GeForce RTX 3090 GPUs | 0.193 | 1,895.5(758–22,233) |
| CTGAN | 213.5(268.1) | 1557.8/ 1101 | NVIDIA GeForce RTX 4090 | 2.52 | 741,834(528,075–1,428,695) |
| TVAE | 214.2(253.2) | 1252.9/ 1069 | NVIDIA GeForce RTX 4090 | 2.45 | 317,141.5(157,578–741,659) |
| Forest-Flow | 185.9(380.6) | 6112.0/ 3181 | NVIDIA GeForce RTX 4090 | 2.44 | 325,848,913.5(115,202,253–1,085,155,062) |

PMLB is MIT. Every dataset entry in the data lock records SPDX MIT and the note that the source license is MIT (PMLB). CTGAN and TVAE share one pinned LICENSE file: Business Source License 1.1, licensor DataCebo, Inc., licensed work CTGAN. The Gaussian copula uses the pinned Copulas LICENSE: Business Source License 1.1, licensor DataCebo, Inc., licensed work Copulas. TabSyn is Apache License 2.0. TabDDPM is MIT License, Copyright (c) 2022 Akim Kotelnikov. Forest-Diffusion is MIT License, Copyright (c) 2023 Samsung Electronics Co., Ltd. All Rights Reserved. ARF is MIT License, Copyright (c) 2023 Kristin Blesch, Marvin Wright. Chow–Liu and the independent marginals are implementations written for this comparison, under this repository's license: MIT License, Copyright (c) 2026 Dope contributors. The SDV repository LICENSE is absent from the pinned snapshot, so this paragraph leaves that license unrecorded.

The rows below are the SHA-256 digests of each ledger and lock the paper reads, plus the commit and the file hash of the harness that measured the grouped-validation fidelity ledger. Regenerating

the macros never rewrites a ledger hash.

Table 8: Ledger and lock digests. No Holm family.

| Ledger or lock | SHA-256 |
|---|---|
| Code commit of the measured harness | 17f3294d139e19f0ec831605f8750368eebc30bb |
| Harness SHA-256 | a952062c0f83805f6442a440a5ae15293a843349 d423eaa106a9c56a20d14c0f |
| research/benchmark/results/paper-original-metadata-v1/index.json | 6d73d5f7b37849526a98a451b41a49e045ef21e9 11058d920940c5ea989f7fd9 |
| docs/whitepaper/generated/original-field-map.json | 5b164d434414d3389f639d3645f40b11a999b32c 56daa2cfe36d25b2205082b6 |
| docs/whitepaper/scripts/paper_receipts.py | a2d6d45e3ff06a38ed61ab029fadfd14529978bc fc89618aa9ceb3fbe57238ee |
| research/benchmark/expanded_validation_metrics.py | a952062c0f83805f6442a440a5ae15293a843349 d423eaa106a9c56a20d14c0f |
| research/benchmark/methods.lock.json | 899c355cb55769710ed369fea6b082589b318588 60fb74929bf4190de28aca33 |
| research/benchmark/results/expanded-validation-diagnostics.json | cd20ac0d07a1dcab7c5e46db195adb463172b517 4387bb71a054b2e5f340beeb |
| research/benchmark/results/s3-coreset-procedure.json | 73251e43cf783687427569931bf6127f42e9323 74af33137c57fd8fe8544704 |
| research/benchmark/results/s3-lineage-record.json | bbd0852c49f595315104261efa4eaa6f50db257e 323b8c002aac6737a2dbe7d7 |
| research/benchmark/results/density-matched-population-validation.json | 5264b88a40ad5efb21d9b119789caeb13e71e911 a17f1d8b57f1d8e8ac31732a |
| research/benchmark/results/sdv-matched-population-validation.json | 4dc367eb78159a0386882d23ae1f99b0aa2cc9f4 25d1aae499b6f3453222a44e |
| research/benchmark/results/s3-matched-forest-confirmation-validation.json | 97a25be902954cad16c5b5802d0e0ba468dea104 221efa2465434c94e6774a1f |
| research/benchmark/results/s3-data.lock.json | 857b61324d0a5b7dbbb98e0c147fcc79c5bb8616 e7ca61afd6785469c17f038f |
| research/benchmark/results/arf-native-closure-watch-v1.receipt.json | 7e8eeb1ba94ca747367e98d5a016d2ed3ed4c40c 5a98ca76d7a5fe27ce9c173c |
| research/benchmark/results/beyondarena-s3-inventory.json | 504166764631dbdf5e02d116334af5cc587731fc f3efe130d62240e6dd512d64 |
| docs/whitepaper/generated/fit-trace.json | d8c31b9e4eb59cb1a087cf5dcdde4a91a73eca00 f3704fbff647a7da6ce65522 |
| docs/whitepaper/generated/loss-curves.json | 0124915497c98e560ef99c1e5ecf7ad876f91744 39ff9c763cf1cf906b4aebe1 |
| docs/whitepaper/generated/replay-cost.json | faf77e7ce1ed1d22127701bf68bbfe2f8d183f89 cbd3475dd80094888c36f086 |
| docs/whitepaper/generated/provenance.json | dea64ea61eaacac0e287298bba3ba76aa7900819 aa00feb3a5d66cdcb4fb8a17 |
| docs/whitepaper/generated/beyond-fit.json | a1447f908bc9c51876d345597039607c734b507b f85aa8e08db0639e80a1f036 |

# D STATISTICS

## D.1 RETENTION ALGEBRA AND VARIANCE

With $r$ the retention, the ratio rearranges to

$$L_{\text{synthetic}} = L_{\text{real}} + (1 - r)\big(L_{\text{null}} - L_{\text{real}}\big).$$

Invented losses $1$, $0.01$, and retention $0.94$ give a synthetic-trained loss of $0.0694$, which is $6.94$ times the real-trained mean squared error. The CatBoost median in this paper is not this example. Let $D = L_{\text{null}} - L_{\text{real}}$, so $R = (L_{\text{null}} - L_{\text{synthetic}})/D$ when $D \neq 0$. The partial derivatives are

$$\frac{\partial R}{\partial L_{\text{null}}} = \frac{L_{\text{synthetic}} - L_{\text{real}}}{D^2}, \qquad \frac{\partial R}{\partial L_{\text{synthetic}}} = -\frac{1}{D}, \qquad \frac{\partial R}{\partial L_{\text{real}}} = \frac{R}{D}.$$

If the three losses have covariance $\Sigma$, the delta-method variance of one lineage's retention is $\nabla R^{\top} \Sigma \nabla R$, which diverges as $D$ approaches zero. The informativeness rule drops a lineage whose real improvement is below $0.01$ times the absolute null loss as a stability cut; the tables report bootstrap intervals rather than this standard error.

## D.2 INFORMATIVENESS THRESHOLD

The primary threshold is $0.01$ times the absolute null loss. Table 9 lists the CatBoost median by threshold and the informative counts; each column uses its own informative count, so the Chow–Liu column differs from the density-block paired median of $0.586$ in Table 22. Thresholds 0, 0.001, and 0.05 keep that median and that count. At 0.1 the count is 96 and the median is 0.942.

Table 9: **Informativeness threshold and auditor counts.** The denominator for each median is the informative count beside it. No Holm family.

| Threshold | DOPE | Gaussian | Chow–Liu | Indep. |
|---|---|---|---|---|
| 0 | 0.940 (97) | 0.656 (99) | 0.582 (99) | -0.012 (99) |
| 0.001 | 0.940 (97) | 0.656 (99) | 0.582 (99) | -0.012 (99) |
| 0.01 | 0.940 (97) | 0.656 (99) | 0.582 (99) | -0.012 (99) |
| 0.05 | 0.940 (97) | 0.656 (99) | 0.582 (99) | -0.012 (99) |
| 0.1 | 0.942 (96) | 0.668 (98) | 0.584 (98) | -0.012 (98) |

| Threshold | CatBoost | Linear | MLP |
|---|---|---|---|
| 0.001 | 97 | 93 | 75 |
| 0.01 | 97 | 92 | 75 |
| 0.05 | 97 | 90 | 73 |

Reading the threshold counts from top to bottom, CatBoost is 97, 97, 97; linear is 93, 92, 90; and the MLP is 75, 75, 73.

### D.3 Bootstrap, families, and equivalence

The v2 analyses follow `research/benchmark/review_fixes/predeclare_v2.json`, committed before any five-seed result existed. A fit value is the median of its three sample-seed retentions; a lineage value is the median of its complete fits (at least three for a five-seed arm); the panel level is the median of lineage values. Paired contrasts use the Hodges–Lehmann pseudo-median of lineage differences, which is the location the two-sided Wilcoxon signed-rank test addresses, and report the median difference beside it. Intervals come from a nested percentile bootstrap with $10{,}000$ draws: the four simulated name prefixes are clusters and every other lineage is its own cluster; a draw resamples clusters, then lineages inside each drawn cluster, then fit seeds inside each drawn lineage, separately for each arm. Holm families are declared per size: the primary family of 12 tests, the density family of 9, a deep-generator family of 12, an other-generator family of 9, and a control family of 6; a member without data keeps its slot. The fit-seed-11 tables of supplement E keep the first version's per-size families (nine density, six neural, six ARF, and three Forest-Flow tests), which each caption names. Equivalence requires the $90\%$ nested interval of the Hodges–Lehmann difference to lie inside $\pm 0.02$; a $20\%$ trimmed-mean (Yuen) test is a sensitivity. The v1 contract tested the mean paired difference with two one-sided $t$ tests at $\pm 0.02$ and unadjusted $\alpha = 0.05$; paired retention differences have heavy tails when the denominator is small, so that estimand is kept only as a continuity column below. The Wilcoxon, Yuen, and v1 mean tests treat lineages as independent; only the bootstrap intervals resample the family clusters, and every verdict reads those intervals.

A Friedman test and a Nemenyi critical difference compare the four density methods on their common complete-informative set, with rank 1 for the higher retention; on CatBoost the Holm sign-test values are $2.15 \times 10^{-15}$, $6.72 \times 10^{-24}$, and $4.95 \times 10^{-27}$. A Bayesian signed-rank procedure would be an alternative to these frequentist tests and is not computed here.

## E Five-seed panel

Table 10 gives the retention level of every arm, auditor, and size, and Table 11 every DOPE-minus-comparator contrast with its family-adjusted $p$-value and both equivalence rules. Table 12 gives the seed-exchangeability diagnostics: how often seed 11 is the best of five, the estimate from seeds 23 to 71 alone, and the binary-by-seed diagnostic on the lineages common to every cell. The historical-binary refit reproduces the samples of the historical seed-11 population byte for byte on $86$ of $98$ lineages. The later binary is the one an earlier four-seed run used; it writes the same seed-11 artifacts as the historical binary on $100$ of $100$ lineages, and its samples match the refit at seeds 23 to 71 on $400$ of $400$ fits. Every retention cell in these tables, for DOPE and for every baseline including Forest-Flow, was scored again from its stored synthetic sample file by one pinned evaluator with auditor seed 1729; a cell whose sample file was missing stays out. Each statistic draws its own bootstrap stream, so two rows built from identical samples, such as the seeds-23-to-71 sensitivity and the later-binary four-seed arm, can differ in the third decimal of an interval. The fit-seed-11 privacy and detector panels in Appendix G come from separate earlier

runs, which record the Forest-Flow cells as unavailable. The historical density, SDV, Forest-Flow, and fit-seed-11 DOPE receipts had used the sample seed as the auditor seed, while the ARF, TabSyn, and control receipts had used 1729; the uniform re-score removes that difference, so the five-seed tables and the historical block tables below differ slightly for this reason as well as for the extra fit seeds.

Table 10: **Retention level of every arm.** Median of lineage values; a five-seed arm needs three complete fits per lineage. No Holm family.

| Arm | Size | Auditor | Lineages | Fits | Median | 95% nested interval |
|---|---|---|---|---|---|---|
| DOPE (ours) | $n$ | CatBoost | 99 | 495 | 0.824 | [0.725, 0.942] |
| DOPE (ours) | $n$ | Linear | 94 | 470 | 0.986 | [0.964, 0.993] |
| DOPE (ours) | $n$ | MLP | 88 | 440 | 0.969 | [0.895, 0.992] |
| DOPE (ours) | $4n$ | CatBoost | 99 | 495 | 0.898 | [0.804, 0.972] |
| DOPE (ours) | $4n$ | Linear | 94 | 470 | 0.991 | [0.972, 0.997] |
| DOPE (ours) | $4n$ | MLP | 88 | 440 | 1.133 | [1.029, 1.246] |
| DOPE, untuned default | $n$ | CatBoost | 99 | 495 | 0.636 | [0.387, 0.825] |
| DOPE, untuned default | $n$ | Linear | 94 | 470 | 0.986 | [0.974, 0.992] |
| DOPE, untuned default | $n$ | MLP | 88 | 440 | 0.887 | [0.801, 0.957] |
| DOPE, untuned default | $4n$ | CatBoost | 99 | 495 | 0.704 | [0.424, 0.877] |
| DOPE, untuned default | $4n$ | Linear | 94 | 470 | 0.993 | [0.983, 0.998] |
| DOPE, untuned default | $4n$ | MLP | 88 | 440 | 0.991 | [0.960, 1.058] |
| TabSyn (scaled schedule) | $n$ | CatBoost | 99 | 493 | 0.863 | [0.751, 0.928] |
| TabSyn (scaled schedule) | $n$ | Linear | 94 | 469 | 0.932 | [0.791, 0.983] |
| TabSyn (scaled schedule) | $n$ | MLP | 88 | 439 | 0.884 | [0.734, 0.976] |
| TabSyn (scaled schedule) | $4n$ | CatBoost | 99 | 493 | 0.924 | [0.814, 0.965] |
| TabSyn (scaled schedule) | $4n$ | Linear | 94 | 469 | 0.952 | [0.827, 0.987] |
| TabSyn (scaled schedule) | $4n$ | MLP | 88 | 439 | 1.013 | [0.954, 1.142] |
| ARF (author default) | $n$ | CatBoost | 99 | 495 | 0.718 | [0.570, 0.844] |
| ARF (author default) | $n$ | Linear | 94 | 470 | 0.903 | [0.794, 0.965] |
| ARF (author default) | $n$ | MLP | 88 | 440 | 0.828 | [0.631, 0.886] |
| ARF (author default) | $4n$ | CatBoost | 99 | 495 | 0.822 | [0.689, 0.894] |
| ARF (author default) | $4n$ | Linear | 94 | 470 | 0.938 | [0.846, 0.982] |
| ARF (author default) | $4n$ | MLP | 88 | 440 | 0.991 | [0.886, 1.090] |
| ARF (native selected) | $n$ | CatBoost | 99 | 495 | 0.671 | [0.551, 0.827] |
| ARF (native selected) | $n$ | Linear | 94 | 470 | 0.894 | [0.742, 0.966] |
| ARF (native selected) | $n$ | MLP | 88 | 440 | 0.805 | [0.583, 0.897] |
| ARF (native selected) | $4n$ | CatBoost | 99 | 495 | 0.804 | [0.647, 0.885] |
| ARF (native selected) | $4n$ | Linear | 94 | 470 | 0.920 | [0.828, 0.977] |
| ARF (native selected) | $4n$ | MLP | 88 | 440 | 0.966 | [0.843, 1.027] |
| Gaussian copula | $n$ | CatBoost | 99 | 99 | 0.610 | [0.310, 0.778] |
| Gaussian copula | $n$ | Linear | 94 | 94 | 0.961 | [0.915, 0.990] |
| Gaussian copula | $n$ | MLP | 88 | 88 | 0.800 | [0.618, 0.890] |
| Gaussian copula | $4n$ | CatBoost | 99 | 99 | 0.669 | [0.369, 0.867] |
| Gaussian copula | $4n$ | Linear | 94 | 94 | 0.987 | [0.963, 0.996] |
| Gaussian copula | $4n$ | MLP | 88 | 88 | 0.978 | [0.892, 1.072] |
| Chow-Liu tree | $n$ | CatBoost | 99 | 99 | 0.466 | [0.246, 0.578] |
| Chow-Liu tree | $n$ | Linear | 94 | 94 | 0.636 | [0.267, 0.756] |
| Chow-Liu tree | $n$ | MLP | 88 | 88 | 0.454 | [0.180, 0.631] |
| Chow-Liu tree | $4n$ | CatBoost | 99 | 99 | 0.576 | [0.294, 0.638] |
| Chow-Liu tree | $4n$ | Linear | 94 | 94 | 0.659 | [0.282, 0.763] |
| Chow-Liu tree | $4n$ | MLP | 88 | 88 | 0.660 | [0.347, 0.804] |
| Independent marginals | $n$ | CatBoost | 99 | 99 | −0.046 | [−0.083, −0.027] |
| Independent marginals | $n$ | Linear | 94 | 94 | −0.050 | [−0.082, −0.015] |
| Independent marginals | $n$ | MLP | 88 | 88 | −0.215 | [−0.484, −0.069] |
| Independent marginals | $4n$ | CatBoost | 99 | 99 | −0.012 | [−0.040, −0.002] |
| Independent marginals | $4n$ | Linear | 94 | 94 | 0.002 | [−0.029, 0.019] |
| Independent marginals | $4n$ | MLP | 88 | 88 | −0.049 | [−0.161, −0.018] |
| TVAE (native selected) | $n$ | CatBoost | 21 | 21 | 0.606 | [0.412, 0.739] |
| TVAE (native selected) | $n$ | Linear | 21 | 21 | 0.764 | [0.552, 0.862] |
| TVAE (native selected) | $n$ | MLP | 20 | 20 | 0.721 | [0.254, 0.796] |
| TVAE (native selected) | $4n$ | CatBoost | 21 | 21 | 0.674 | [0.452, 0.776] |
| TVAE (native selected) | $4n$ | Linear | 21 | 21 | 0.796 | [0.581, 0.875] |
| TVAE (native selected) | $4n$ | MLP | 20 | 20 | 0.812 | [0.644, 0.877] |
| CTGAN (native selected) | $n$ | CatBoost | 21 | 21 | −0.074 | [−0.477, 0.118] |
| CTGAN (native selected) | $n$ | Linear | 21 | 21 | −0.056 | [−0.384, 0.118] |
| CTGAN (native selected) | $n$ | MLP | 20 | 20 | −0.247 | [−1.072, 0.104] |
| CTGAN (native selected) | $4n$ | CatBoost | 21 | 21 | −0.080 | [−0.282, 0.223] |
| CTGAN (native selected) | $4n$ | Linear | 21 | 21 | −0.033 | [−0.454, 0.124] |
| CTGAN (native selected) | $4n$ | MLP | 20 | 20 | 0.033 | [−0.338, 0.328] |
| Forest-Flow (historical) | $n$ | CatBoost | 6 | 6 | 0.977 | [0.807, 0.999] |
| Forest-Flow (historical) | $n$ | Linear | 5 | 5 | 0.981 | [0.754, 1.083] |
| Forest-Flow (historical) | $n$ | MLP | 6 | 6 | 0.885 | [0.578, 1.000] |
| Forest-Flow (historical) | $4n$ | CatBoost | 6 | 6 | 1.011 | [0.905, 1.041] |
| Forest-Flow (historical) | $4n$ | Linear | 5 | 5 | 0.985 | [0.977, 1.001] |
| Forest-Flow (historical) | $4n$ | MLP | 6 | 6 | 1.054 | [0.929, 2.732] |
| Predictor-only control | $n$ | CatBoost | 99 | 495 | 0.596 | [0.340, 0.726] |
| Predictor-only control | $n$ | Linear | 94 | 470 | 0.970 | [0.916, 0.994] |

| Arm | Size | Auditor | Lineages | Fits | Median | 95% nested interval |
|---|---|---|---|---|---|---|
| Predictor-only control | $n$ | MLP | 88 | 440 | 0.849 | [0.729, 0.913] |
| Predictor-only control | $4n$ | CatBoost | 99 | 495 | 0.631 | [0.375, 0.784] |
| Predictor-only control | $4n$ | Linear | 94 | 470 | 0.981 | [0.955, 0.997] |
| Predictor-only control | $4n$ | MLP | 88 | 440 | 0.947 | [0.883, 1.018] |
| Real rows resampled | $4n$ | CatBoost | 99 | 99 | 1.018 | [1.005, 1.028] |
| Real rows resampled | $4n$ | Linear | 94 | 94 | 0.998 | [0.991, 1.000] |
| Real rows resampled | $4n$ | MLP | 88 | 88 | 1.174 | [1.051, 1.234] |
| Chow-Liu tree (default) | $n$ | CatBoost | 99 | 99 | 0.564 | [0.455, 0.677] |
| Chow-Liu tree (default) | $n$ | Linear | 94 | 94 | 0.661 | [0.608, 0.787] |
| Chow-Liu tree (default) | $n$ | MLP | 88 | 88 | 0.523 | [0.393, 0.684] |
| Chow-Liu tree (default) | $4n$ | CatBoost | 99 | 99 | 0.636 | [0.545, 0.754] |
| Chow-Liu tree (default) | $4n$ | Linear | 94 | 94 | 0.727 | [0.640, 0.810] |
| Chow-Liu tree (default) | $4n$ | MLP | 88 | 88 | 0.755 | [0.649, 0.949] |
| CTGAN (author default) | $n$ | CatBoost | 17 | 17 | −0.284 | [−0.562, −0.054] |
| CTGAN (author default) | $n$ | Linear | 17 | 17 | −0.322 | [−1.564, −0.070] |
| CTGAN (author default) | $n$ | MLP | 16 | 16 | −0.399 | [−0.817, −0.147] |
| CTGAN (author default) | $4n$ | CatBoost | 17 | 17 | −0.237 | [−0.509, −0.022] |
| CTGAN (author default) | $4n$ | Linear | 17 | 17 | −0.294 | [−1.606, −0.031] |
| CTGAN (author default) | $4n$ | MLP | 16 | 16 | −0.170 | [−0.658, −0.003] |
| DOPE, later binary, seed 11 | $n$ | CatBoost | 99 | 99 | 0.883 | [0.820, 0.970] |
| DOPE, later binary, seed 11 | $n$ | Linear | 94 | 94 | 0.979 | [0.964, 0.994] |
| DOPE, later binary, seed 11 | $n$ | MLP | 88 | 88 | 0.988 | [0.924, 0.995] |
| DOPE, later binary, seed 11 | $4n$ | CatBoost | 99 | 99 | 0.944 | [0.893, 0.985] |
| DOPE, later binary, seed 11 | $4n$ | Linear | 94 | 94 | 0.994 | [0.972, 0.998] |
| DOPE, later binary, seed 11 | $4n$ | MLP | 88 | 88 | 1.175 | [1.044, 1.349] |
| DOPE, later binary, seeds 23-71 | $n$ | CatBoost | 99 | 396 | 0.791 | [0.686, 0.930] |
| DOPE, later binary, seeds 23-71 | $n$ | Linear | 94 | 376 | 0.984 | [0.960, 0.992] |
| DOPE, later binary, seeds 23-71 | $n$ | MLP | 88 | 352 | 0.958 | [0.883, 0.990] |
| DOPE, later binary, seeds 23-71 | $4n$ | CatBoost | 99 | 396 | 0.862 | [0.772, 0.960] |
| DOPE, later binary, seeds 23-71 | $4n$ | Linear | 94 | 376 | 0.989 | [0.970, 0.996] |
| DOPE, later binary, seeds 23-71 | $4n$ | MLP | 88 | 352 | 1.125 | [1.026, 1.229] |
| DOPE, historical seed 11 | $n$ | CatBoost | 97 | 97 | 0.880 | [0.822, 0.971] |
| DOPE, historical seed 11 | $n$ | Linear | 92 | 92 | 0.984 | [0.968, 0.994] |
| DOPE, historical seed 11 | $n$ | MLP | 87 | 87 | 0.988 | [0.923, 0.996] |
| DOPE, historical seed 11 | $4n$ | CatBoost | 97 | 97 | 0.949 | [0.893, 0.987] |
| DOPE, historical seed 11 | $4n$ | Linear | 92 | 92 | 0.994 | [0.972, 0.997] |
| DOPE, historical seed 11 | $4n$ | MLP | 87 | 87 | 1.174 | [1.045, 1.349] |
| DOPE historical, 12 inputs, 512 steps | $n$ | CatBoost | 97 | 97 | 0.750 | [0.415, 0.923] |
| DOPE historical, 12 inputs, 512 steps | $n$ | Linear | 92 | 92 | 0.982 | [0.957, 0.990] |
| DOPE historical, 12 inputs, 512 steps | $n$ | MLP | 87 | 87 | 0.960 | [0.831, 0.988] |
| DOPE historical, 12 inputs, 512 steps | $4n$ | CatBoost | 97 | 97 | 0.801 | [0.445, 0.958] |
| DOPE historical, 12 inputs, 512 steps | $4n$ | Linear | 92 | 92 | 0.984 | [0.966, 0.994] |
| DOPE historical, 12 inputs, 512 steps | $4n$ | MLP | 87 | 87 | 1.046 | [0.998, 1.201] |
| DOPE historical, 24 inputs, 512 steps | $n$ | CatBoost | 93 | 93 | 0.762 | [0.416, 0.924] |
| DOPE historical, 24 inputs, 512 steps | $n$ | Linear | 88 | 88 | 0.982 | [0.964, 0.991] |
| DOPE historical, 24 inputs, 512 steps | $n$ | MLP | 84 | 84 | 0.964 | [0.868, 0.993] |
| DOPE historical, 24 inputs, 512 steps | $4n$ | CatBoost | 93 | 93 | 0.802 | [0.457, 0.958] |
| DOPE historical, 24 inputs, 512 steps | $4n$ | Linear | 88 | 88 | 0.986 | [0.968, 0.996] |
| DOPE historical, 24 inputs, 512 steps | $4n$ | MLP | 84 | 84 | 1.035 | [0.999, 1.204] |
| DOPE historical, 24 inputs, 2048 steps | $n$ | CatBoost | 93 | 93 | 0.884 | [0.747, 0.971] |
| DOPE historical, 24 inputs, 2048 steps | $n$ | Linear | 88 | 88 | 0.977 | [0.953, 0.995] |
| DOPE historical, 24 inputs, 2048 steps | $n$ | MLP | 84 | 84 | 0.990 | [0.949, 0.999] |
| DOPE historical, 24 inputs, 2048 steps | $4n$ | CatBoost | 93 | 93 | 0.931 | [0.814, 0.985] |
| DOPE historical, 24 inputs, 2048 steps | $4n$ | Linear | 88 | 88 | 0.990 | [0.967, 0.997] |
| DOPE historical, 24 inputs, 2048 steps | $4n$ | MLP | 84 | 84 | 1.124 | [1.045, 1.266] |
| Forest-Flow default (historical) | $n$ | CatBoost | 6 | 6 | 0.977 | [0.807, 0.999] |
| Forest-Flow default (historical) | $n$ | Linear | 5 | 5 | 0.981 | [0.754, 1.083] |
| Forest-Flow default (historical) | $n$ | MLP | 6 | 6 | 0.885 | [0.578, 1.000] |
| Forest-Flow default (historical) | $4n$ | CatBoost | 6 | 6 | 1.011 | [0.905, 1.041] |
| Forest-Flow default (historical) | $4n$ | Linear | 5 | 5 | 0.985 | [0.977, 1.001] |
| Forest-Flow default (historical) | $4n$ | MLP | 6 | 6 | 1.054 | [0.929, 2.732] |
| Gaussian copula (default) | $n$ | CatBoost | 99 | 99 | 0.565 | [0.260, 0.809] |
| Gaussian copula (default) | $n$ | Linear | 94 | 94 | 0.921 | [0.742, 0.962] |
| Gaussian copula (default) | $n$ | MLP | 88 | 88 | 0.786 | [0.573, 0.931] |
| Gaussian copula (default) | $4n$ | CatBoost | 99 | 99 | 0.605 | [0.305, 0.879] |
| Gaussian copula (default) | $4n$ | Linear | 94 | 94 | 0.956 | [0.878, 0.983] |
| Gaussian copula (default) | $4n$ | MLP | 88 | 88 | 0.935 | [0.851, 1.033] |
| Independent marginals (default) | $n$ | CatBoost | 99 | 99 | −0.044 | [−0.077, −0.028] |
| Independent marginals (default) | $n$ | Linear | 94 | 94 | −0.049 | [−0.092, −0.005] |
| Independent marginals (default) | $n$ | MLP | 88 | 88 | −0.184 | [−0.449, −0.053] |
| Independent marginals (default) | $4n$ | CatBoost | 99 | 99 | −0.016 | [−0.040, −0.006] |
| Independent marginals (default) | $4n$ | Linear | 94 | 94 | 0.001 | [−0.027, 0.021] |
| Independent marginals (default) | $4n$ | MLP | 88 | 88 | −0.047 | [−0.132, −0.018] |
| Predictor-only, split 2027 | $n$ | CatBoost | 99 | 99 | 0.530 | [0.322, 0.729] |
| Predictor-only, split 2027 | $n$ | Linear | 97 | 97 | 0.990 | [0.951, 0.998] |
| Predictor-only, split 2027 | $n$ | MLP | 84 | 84 | 0.828 | [0.731, 0.910] |
| Predictor-only, split 2027 | $4n$ | CatBoost | 99 | 99 | 0.600 | [0.365, 0.778] |
| Predictor-only, split 2027 | $4n$ | Linear | 97 | 97 | 0.998 | [0.987, 1.001] |
| Predictor-only, split 2027 | $4n$ | MLP | 84 | 84 | 0.950 | [0.883, 1.031] |

| Arm | Size | Auditor | Lineages | Fits | Median | 95% nested interval |
|---|---|---|---|---|---|---|
| Predictor-only, split 2999 | $n$ | CatBoost | 100 | 100 | 0.570 | [0.339, 0.722] |
| Predictor-only, split 2999 | $n$ | Linear | 92 | 92 | 0.993 | [0.972, 0.999] |
| Predictor-only, split 2999 | $n$ | MLP | 86 | 86 | 0.844 | [0.764, 0.937] |
| Predictor-only, split 2999 | $4n$ | CatBoost | 100 | 100 | 0.647 | [0.368, 0.777] |
| Predictor-only, split 2999 | $4n$ | Linear | 92 | 92 | 0.998 | [0.979, 1.002] |
| Predictor-only, split 2999 | $4n$ | MLP | 86 | 86 | 0.963 | [0.882, 1.065] |
| Predictor-only, split 4099 | $n$ | CatBoost | 100 | 100 | 0.562 | [0.308, 0.716] |
| Predictor-only, split 4099 | $n$ | Linear | 95 | 95 | 0.987 | [0.961, 0.995] |
| Predictor-only, split 4099 | $n$ | MLP | 86 | 86 | 0.837 | [0.732, 0.942] |
| Predictor-only, split 4099 | $4n$ | CatBoost | 100 | 100 | 0.640 | [0.362, 0.777] |
| Predictor-only, split 4099 | $4n$ | Linear | 95 | 95 | 0.993 | [0.974, 0.999] |
| Predictor-only, split 4099 | $4n$ | MLP | 86 | 86 | 0.965 | [0.917, 1.021] |
| Predictor-only, split 8191 | $n$ | CatBoost | 100 | 100 | 0.611 | [0.337, 0.752] |
| Predictor-only, split 8191 | $n$ | Linear | 96 | 96 | 0.978 | [0.960, 0.996] |
| Predictor-only, split 8191 | $n$ | MLP | 85 | 85 | 0.810 | [0.727, 0.903] |
| Predictor-only, split 8191 | $4n$ | CatBoost | 100 | 100 | 0.677 | [0.359, 0.787] |
| Predictor-only, split 8191 | $4n$ | Linear | 96 | 96 | 0.996 | [0.969, 0.999] |
| Predictor-only, split 8191 | $4n$ | MLP | 85 | 85 | 0.951 | [0.884, 1.018] |
| TVAE (author default) | $n$ | CatBoost | 18 | 18 | 0.300 | [0.072, 0.547] |
| TVAE (author default) | $n$ | Linear | 18 | 18 | 0.354 | [−0.480, 0.657] |
| TVAE (author default) | $n$ | MLP | 17 | 17 | 0.179 | [−1.050, 0.569] |
| TVAE (author default) | $4n$ | CatBoost | 18 | 18 | 0.283 | [0.175, 0.596] |
| TVAE (author default) | $4n$ | Linear | 18 | 18 | 0.429 | [0.156, 0.682] |
| TVAE (author default) | $4n$ | MLP | 17 | 17 | 0.333 | [−0.417, 0.612] |

Table 11: **All DOPE-minus-comparator contrasts.** Nested bootstrap 95% intervals. The Holm family size is in parentheses; families are declared per size in predeclare_v2. HL eq. is the pre-declared equivalence rule (90% interval of the Hodges–Lehmann difference inside $\pm 0.02$); Yuen eq. is the trimmed-mean sensitivity.

| Size | Auditor | Comparator | $n$ | Median diff. | Hodges–Lehmann | W/T/L | Holm $p$ (family) | HL eq. | Yuen eq. |
|---|---|---|---|---|---|---|---|---|---|
| $n$ | CatBoost | TabSyn (scaled schedule) | 99 | 0.004 [−0.054, 0.058] | 0.002 [−0.057, 0.056] | 53/0/46 | 1.000 (12) | no | no |
| $n$ | Linear | TabSyn (scaled schedule) | 94 | 0.053 [0.006, 0.168] | 0.098 [0.014, 0.244] | 74/0/20 | $3.03 \times 10^{-8}$ (12) | no | no |
| $n$ | MLP | TabSyn (scaled schedule) | 88 | 0.061 [0.003, 0.160] | 0.100 [0.019, 0.221] | 64/0/24 | $1.60 \times 10^{-5}$ (12) | no | no |
| $n$ | CatBoost | ARF (author default) | 99 | 0.084 [0.004, 0.165] | 0.090 [0.006, 0.190] | 71/0/28 | 0.001 (12) | no | no |
| $n$ | Linear | ARF (author default) | 94 | 0.065 [0.003, 0.190] | 0.090 [0.008, 0.251] | 74/0/20 | $3.53 \times 10^{-6}$ (12) | no | no |
| $n$ | MLP | ARF (author default) | 88 | 0.111 [0.050, 0.226] | 0.144 [0.061, 0.332] | 74/0/14 | $5.54 \times 10^{-8}$ (12) | no | no |
| $n$ | CatBoost | ARF (native selected) | 99 | 0.083 [0.009, 0.201] | 0.109 [0.025, 0.220] | 71/0/28 | $1.71 \times 10^{-4}$ (9) | no | no |
| $n$ | Linear | ARF (native selected) | 94 | 0.077 [0.010, 0.198] | 0.104 [0.019, 0.256] | 76/0/18 | $1.10 \times 10^{-7}$ (9) | no | no |
| $n$ | MLP | ARF (native selected) | 88 | 0.142 [0.063, 0.274] | 0.175 [0.081, 0.363] | 76/0/12 | $3.06 \times 10^{-10}$ (9) | no | no |
| $n$ | CatBoost | Gaussian copula | 99 | 0.196 [0.090, 0.243] | 0.202 [0.110, 0.266] | 88/0/11 | $1.41 \times 10^{-14}$ (9) | no | no |
| $n$ | Linear | Gaussian copula | 94 | 0.008 [−0.005, 0.038] | 0.020 [−0.005, 0.049] | 59/0/35 | 0.010 (9) | no | no |
| $n$ | MLP | Gaussian copula | 88 | 0.132 [0.090, 0.185] | 0.157 [0.097, 0.242] | 73/0/15 | $2.68 \times 10^{-9}$ (9) | no | no |
| $n$ | CatBoost | Chow-Liu tree | 99 | 0.313 [0.240, 0.513] | 0.369 [0.268, 0.660] | 87/0/12 | $1.24 \times 10^{-15}$ (9) | no | no |
| $n$ | Linear | Chow-Liu tree | 94 | 0.342 [0.226, 0.624] | 0.481 [0.282, 0.819] | 91/0/3 | $1.97 \times 10^{-15}$ (9) | no | no |
| $n$ | MLP | Chow-Liu tree | 88 | 0.489 [0.340, 0.755] | 0.620 [0.406, 1.079] | 83/0/5 | $4.89 \times 10^{-14}$ (9) | no | no |
| $n$ | CatBoost | Independent marginals | 99 | 0.866 [0.767, 0.982] | 0.849 [0.745, 0.964] | 99/0/0 | $5.13 \times 10^{-17}$ (9) | no | no |
| $n$ | Linear | Independent marginals | 94 | 1.002 [0.974, 1.065] | 1.023 [0.975, 1.097] | 94/0/0 | $3.05 \times 10^{-16}$ (9) | no | no |
| $n$ | MLP | Independent marginals | 88 | 1.164 [1.039, 1.473] | 1.265 [1.078, 1.549] | 87/0/1 | $1.17 \times 10^{-14}$ (9) | no | no |
| $n$ | CatBoost | TVAE (native selected) | 21 | 0.195 [0.089, 0.395] | 0.221 [0.115, 0.347] | 19/0/2 | — | no | no |
| $n$ | Linear | TVAE (native selected) | 21 | 0.250 [0.128, 0.427] | 0.284 [0.158, 0.549] | 20/0/1 | — | no | no |
| $n$ | MLP | TVAE (native selected) | 20 | 0.221 [0.109, 0.507] | 0.254 [0.118, 0.596] | 17/0/3 | — | no | no |
| $n$ | CatBoost | CTGAN (native selected) | 21 | 0.785 [0.706, 1.342] | 0.880 [0.731, 1.204] | 21/0/0 | — | no | no |

| Size | Auditor | Comparator | $n$ | Median diff. | Hodges–Lehmann | W/T/L | Holm $p$ (family) | HL eq. | Yuen eq. |
|---|---|---|---|---|---|---|---|---|---|
| $n$ | Linear | CTGAN (native selected) | 21 | $1.052$ $[0.849, 1.342]$ | $1.104$ $[0.833, 1.732]$ | 21/0/0 | — | no | no |
| $n$ | MLP | CTGAN (native selected) | 20 | $1.135$ $[0.772, 1.776]$ | $1.295$ $[0.789, 2.228]$ | 19/0/1 | — | no | no |
| $n$ | CatBoost | Forest-Flow (historical) | 6 | $-0.045$ $[-0.463, 0.020]$ | $-0.206$ $[-0.385, 0.003]$ | 1/0/5 | — | no | no |
| $n$ | Linear | Forest-Flow (historical) | 5 | $0.024$ $[-0.005, 0.074]$ | $0.037$ $[-0.034, 0.074]$ | 5/0/0 | — | no | no |
| $n$ | MLP | Forest-Flow (historical) | 6 | $0.005$ $[-0.041, 0.123]$ | $0.010$ $[-0.086, 0.123]$ | 3/0/3 | — | no | no |
| $n$ | CatBoost | DOPE, untuned default | 99 | $0.101$ $[0.035, 0.155]$ | $0.127$ $[0.052, 0.186]$ | 89/0/10 | — | no | no |
| $n$ | Linear | DOPE, untuned default | 94 | $-0.002$ $[-0.016, 0.003]$ | $-0.007$ $[-0.027, 0.002]$ | 42/0/52 | — | no | yes |
| $n$ | MLP | DOPE, untuned default | 88 | $0.034$ $[0.003, 0.069]$ | $0.050$ $[0.008, 0.082]$ | 63/0/25 | — | no | no |
| $n$ | CatBoost | Predictor-only control | 99 | $0.215$ $[0.128, 0.271]$ | $0.214$ $[0.148, 0.269]$ | 93/0/6 | $4.36 \times 10^{-15}$ (6) | no | no |
| $n$ | Linear | Predictor-only control | 94 | $0.003$ $[-0.014, 0.041]$ | $0.016$ $[-0.022, 0.054]$ | 55/0/39 | $0.142$ (6) | no | no |
| $n$ | MLP | Predictor-only control | 88 | $0.106$ $[0.044, 0.168]$ | $0.134$ $[0.067, 0.195]$ | 73/0/15 | $1.96 \times 10^{-8}$ (6) | no | no |
| $n$ | CatBoost | Chow-Liu tree (default) | 99 | $0.239$ $[0.175, 0.313]$ | $0.261$ $[0.185, 0.352]$ | 86/0/13 | — | no | no |
| $n$ | Linear | Chow-Liu tree (default) | 94 | $0.282$ $[0.177, 0.346]$ | $0.327$ $[0.202, 0.481]$ | 90/0/4 | — | no | no |
| $n$ | MLP | Chow-Liu tree (default) | 88 | $0.389$ $[0.242, 0.502]$ | $0.490$ $[0.311, 0.769]$ | 82/0/6 | — | no | no |
| $n$ | CatBoost | CTGAN (author default) | 17 | $1.093$ $[0.787, 1.446]$ | $1.108$ $[0.861, 1.707]$ | 17/0/0 | $1.53 \times 10^{-4}$ (12) | no | no |
| $n$ | Linear | CTGAN (author default) | 17 | $1.314$ $[0.981, 2.569]$ | $1.610$ $[1.151, 2.510]$ | 17/0/0 | $1.53 \times 10^{-4}$ (12) | no | no |
| $n$ | MLP | CTGAN (author default) | 16 | $1.314$ $[0.891, 1.790]$ | $1.378$ $[1.030, 2.528]$ | 15/0/1 | $0.001$ (12) | no | no |
| $n$ | CatBoost | DOPE, later binary, seed 11 | 99 | $-0.020$ $[-0.058, -0.003]$ | $-0.033$ $[-0.074, -0.009]$ | 14/10/75 | — | no | no |
| $n$ | Linear | DOPE, later binary, seed 11 | 94 | $0.000$ $[-0.001, 0.003]$ | $0.002$ $[-0.006, 0.013]$ | 40/19/35 | — | yes | yes |
| $n$ | MLP | DOPE, later binary, seed 11 | 88 | $-0.003$ $[-0.010, 0.000]$ | $-0.007$ $[-0.021, 0.000]$ | 25/16/47 | — | yes | yes |
| $n$ | CatBoost | DOPE, later binary, seeds 23-71 | 99 | $0.008$ $[0.000, 0.011]$ | $0.010$ $[0.000, 0.020]$ | 79/0/20 | — | yes | yes |
| $n$ | Linear | DOPE, later binary, seeds 23-71 | 94 | $0.000$ $[-0.002, 0.002]$ | $0.000$ $[-0.005, 0.006]$ | 44/0/50 | — | yes | yes |
| $n$ | MLP | DOPE, later binary, seeds 23-71 | 88 | $0.003$ $[-0.001, 0.006]$ | $0.003$ $[-0.005, 0.012]$ | 54/0/34 | — | yes | yes |
| $n$ | CatBoost | DOPE, historical seed 11 | 97 | $-0.022$ $[-0.060, -0.003]$ | $-0.034$ $[-0.076, -0.009]$ | 15/8/74 | — | no | no |
| $n$ | Linear | DOPE, historical seed 11 | 92 | $0.000$ $[-0.002, 0.002]$ | $0.001$ $[-0.006, 0.011]$ | 38/16/38 | — | yes | yes |
| $n$ | MLP | DOPE, historical seed 11 | 87 | $-0.003$ $[-0.011, 0.000]$ | $-0.007$ $[-0.023, 0.000]$ | 25/15/47 | — | yes | yes |
| $n$ | CatBoost | DOPE historical, 12 inputs, 512 steps | 97 | $0.032$ $[0.006, 0.092]$ | $0.060$ $[0.012, 0.143]$ | 72/0/25 | — | no | no |
| $n$ | Linear | DOPE historical, 12 inputs, 512 steps | 92 | $0.002$ $[-0.006, 0.007]$ | $0.001$ $[-0.010, 0.012]$ | 49/0/43 | — | yes | yes |
| $n$ | MLP | DOPE historical, 12 inputs, 512 steps | 87 | $0.007$ $[-0.004, 0.016]$ | $0.015$ $[-0.005, 0.038]$ | 50/0/37 | — | no | no |
| $n$ | CatBoost | DOPE historical, 24 inputs, 512 steps | 93 | $0.032$ $[0.006, 0.087]$ | $0.053$ $[0.012, 0.126]$ | 69/0/24 | — | no | no |
| $n$ | Linear | DOPE historical, 24 inputs, 512 steps | 88 | $0.003$ $[-0.005, 0.008]$ | $0.003$ $[-0.009, 0.012]$ | 48/0/40 | — | yes | yes |
| $n$ | MLP | DOPE historical, 24 inputs, 512 steps | 84 | $0.002$ $[-0.015, 0.014]$ | $0.010$ $[-0.015, 0.029]$ | 45/0/39 | — | no | no |
| $n$ | CatBoost | DOPE historical, 24 inputs, 2048 steps | 93 | $-0.013$ $[-0.030, -0.001]$ | $-0.019$ $[-0.041, -0.006]$ | 24/6/63 | — | no | no |
| $n$ | Linear | DOPE historical, 24 inputs, 2048 steps | 88 | $0.000$ $[-0.005, 0.001]$ | $0.000$ $[-0.012, 0.012]$ | 35/14/39 | — | yes | yes |
| $n$ | MLP | DOPE historical, 24 inputs, 2048 steps | 84 | $-0.005$ $[-0.020, 0.000]$ | $-0.014$ $[-0.040, -0.003]$ | 21/13/50 | — | no | no |
| $n$ | CatBoost | Forest-Flow default (historical) | 6 | $-0.045$ $[-0.463, 0.020]$ | $-0.206$ $[-0.366, 0.003]$ | 1/0/5 | — | no | no |
| $n$ | Linear | Forest-Flow default (historical) | 5 | $0.024$ $[-0.005, 0.074]$ | $0.037$ $[-0.043, 0.074]$ | 5/0/0 | — | no | no |
| $n$ | MLP | Forest-Flow default (historical) | 6 | $0.005$ $[-0.041, 0.123]$ | $0.010$ $[-0.074, 0.123]$ | 3/0/3 | — | no | no |
| $n$ | CatBoost | Gaussian copula (default) | 99 | $0.222$ $[0.092, 0.283]$ | $0.229$ $[0.121, 0.303]$ | 89/0/10 | — | no | no |

| Size | Auditor | Comparator | $n$ | Median diff. | Hodges–Lehmann | W/T/L | Holm $p$ (family) | HL eq. | Yuen eq. |
|---|---|---|---|---|---|---|---|---|---|
| $n$ | Linear | Gaussian copula (default) | 94 | 0.048 [0.013, 0.150] | 0.104 [0.025, 0.224] | 71/0/23 | — | no | no |
| $n$ | MLP | Gaussian copula (default) | 88 | 0.129 [0.069, 0.231] | 0.178 [0.085, 0.309] | 74/0/14 | — | no | no |
| $n$ | CatBoost | Independent marginals (default) | 99 | 0.875 [0.757, 0.981] | 0.842 [0.739, 0.962] | 99/0/0 | — | no | no |
| $n$ | Linear | Independent marginals (default) | 94 | 1.011 [0.969, 1.061] | 1.017 [0.965, 1.089] | 93/0/1 | — | no | no |
| $n$ | MLP | Independent marginals (default) | 88 | 1.121 [1.039, 1.441] | 1.229 [1.061, 1.449] | 86/0/2 | — | no | no |
| $n$ | CatBoost | Predictor-only, split 2027 | 98 | 0.214 [0.147, 0.281] | 0.230 [0.160, 0.301] | 92/0/6 | — | no | no |
| $n$ | Linear | Predictor-only, split 2027 | 93 | −0.006 [−0.032, 0.009] | −0.010 [−0.043, 0.012] | 40/0/53 | — | no | no |
| $n$ | MLP | Predictor-only, split 2027 | 78 | 0.104 [0.020, 0.162] | 0.128 [0.040, 0.198] | 61/0/17 | — | no | no |
| $n$ | CatBoost | Predictor-only, split 2999 | 99 | 0.200 [0.111, 0.263] | 0.216 [0.137, 0.287] | 86/0/13 | — | no | no |
| $n$ | Linear | Predictor-only, split 2999 | 89 | −0.007 [−0.023, 0.008] | −0.006 [−0.034, 0.015] | 35/0/54 | — | no | yes |
| $n$ | MLP | Predictor-only, split 2999 | 82 | 0.074 [0.011, 0.159] | 0.121 [0.030, 0.191] | 61/0/21 | — | no | no |
| $n$ | CatBoost | Predictor-only, split 4099 | 99 | 0.223 [0.126, 0.275] | 0.226 [0.152, 0.299] | 86/0/13 | — | no | no |
| $n$ | Linear | Predictor-only, split 4099 | 91 | −0.005 [−0.016, 0.012] | −0.003 [−0.033, 0.025] | 40/0/51 | — | no | yes |
| $n$ | MLP | Predictor-only, split 4099 | 84 | 0.098 [0.005, 0.159] | 0.119 [0.041, 0.197] | 58/0/26 | — | no | no |
| $n$ | CatBoost | Predictor-only, split 8191 | 99 | 0.179 [0.091, 0.252] | 0.199 [0.121, 0.263] | 85/0/14 | — | no | no |
| $n$ | Linear | Predictor-only, split 8191 | 91 | 0.004 [−0.010, 0.027] | 0.007 [−0.016, 0.037] | 50/0/41 | — | no | yes |
| $n$ | MLP | Predictor-only, split 8191 | 81 | 0.101 [0.024, 0.170] | 0.133 [0.044, 0.201] | 67/0/14 | — | no | no |
| $n$ | CatBoost | TVAE (author default) | 18 | 0.452 [0.329, 0.778] | 0.504 [0.365, 0.747] | 18/0/0 | $9.16 \times 10^{-5}$ (12) | no | no |
| $n$ | Linear | TVAE (author default) | 18 | 0.593 [0.326, 1.897] | 0.628 [0.432, 12.069] | 18/0/0 | $9.16 \times 10^{-5}$ (12) | no | no |
| $n$ | MLP | TVAE (author default) | 17 | 0.632 [0.270, 1.242] | 0.748 [0.378, 1.925] | 16/0/1 | $2.44 \times 10^{-4}$ (12) | no | no |
| $4n$ | CatBoost | TabSyn (scaled schedule) | 99 | 0.006 [−0.054, 0.045] | −0.003 [−0.053, 0.053] | 53/0/46 | 1.000 (12) | no | no |
| $4n$ | Linear | TabSyn (scaled schedule) | 94 | 0.037 [0.002, 0.136] | 0.069 [0.005, 0.210] | 74/0/20 | $8.23 \times 10^{-7}$ (12) | no | no |
| $4n$ | MLP | TabSyn (scaled schedule) | 88 | 0.005 [−0.006, 0.154] | 0.054 [−0.006, 0.277] | 52/0/36 | 0.109 (12) | no | no |
| $4n$ | CatBoost | ARF (author default) | 99 | 0.035 [−0.021, 0.130] | 0.048 [−0.034, 0.149] | 60/0/39 | 0.229 (12) | no | no |
| $4n$ | Linear | ARF (author default) | 94 | 0.045 [−0.002, 0.155] | 0.062 [−0.011, 0.193] | 70/0/24 | $2.87 \times 10^{-4}$ (12) | no | no |
| $4n$ | MLP | ARF (author default) | 88 | 0.117 [0.020, 0.237] | 0.137 [0.041, 0.397] | 70/0/18 | $3.26 \times 10^{-7}$ (12) | no | no |
| $4n$ | CatBoost | ARF (native selected) | 99 | 0.051 [−0.005, 0.162] | 0.083 [0.006, 0.191] | 66/0/33 | 0.004 (9) | no | no |
| $4n$ | Linear | ARF (native selected) | 94 | 0.073 [0.003, 0.164] | 0.083 [0.006, 0.217] | 73/0/21 | $1.89 \times 10^{-6}$ (9) | no | no |
| $4n$ | MLP | ARF (native selected) | 88 | 0.106 [0.034, 0.306] | 0.191 [0.060, 0.431] | 71/0/17 | $2.48 \times 10^{-8}$ (9) | no | no |
| $4n$ | CatBoost | Gaussian copula | 99 | 0.164 [0.062, 0.242] | 0.191 [0.089, 0.273] | 83/0/16 | $1.46 \times 10^{-13}$ (9) | no | no |
| $4n$ | Linear | Gaussian copula | 94 | 0.002 [−0.014, 0.015] | 0.007 [−0.023, 0.030] | 53/0/41 | 0.183 (9) | no | yes |
| $4n$ | MLP | Gaussian copula | 88 | 0.170 [0.047, 0.264] | 0.234 [0.126, 0.390] | 77/0/11 | $9.26 \times 10^{-13}$ (9) | no | no |
| $4n$ | CatBoost | Chow-Liu tree | 99 | 0.270 [0.201, 0.517] | 0.356 [0.242, 0.728] | 91/0/8 | $5.59 \times 10^{-15}$ (9) | no | no |
| $4n$ | Linear | Chow-Liu tree | 94 | 0.333 [0.209, 0.659] | 0.495 [0.266, 0.816] | 91/0/3 | $1.91 \times 10^{-15}$ (9) | no | no |
| $4n$ | MLP | Chow-Liu tree | 88 | 0.566 [0.353, 0.795] | 0.658 [0.435, 1.166] | 88/0/0 | $2.24 \times 10^{-15}$ (9) | no | no |
| $4n$ | CatBoost | Independent marginals | 99 | 0.913 [0.826, 0.978] | 0.881 [0.790, 0.964] | 99/0/0 | $5.13 \times 10^{-17}$ (9) | no | no |
| $4n$ | Linear | Independent marginals | 94 | 0.982 [0.937, 1.014] | 0.982 [0.926, 1.008] | 94/0/0 | $3.05 \times 10^{-16}$ (9) | no | no |
| $4n$ | MLP | Independent marginals | 88 | 1.220 [1.094, 1.467] | 1.327 [1.143, 1.629] | 88/0/0 | $2.24 \times 10^{-15}$ (9) | no | no |
| $4n$ | CatBoost | TVAE (native selected) | 21 | 0.201 [0.080, 0.351] | 0.201 [0.097, 0.339] | 19/0/2 | — | no | no |

| Size | Auditor | Comparator | $n$ | Median diff. | Hodges–Lehmann | W/T/L | Holm $p$ (family) | HL eq. | Yuen eq. |
|---|---|---|---|---|---|---|---|---|---|
| $4n$ | Linear | TVAE (native selected) | 21 | 0.213 [0.130, 0.414] | 0.262 [0.145, 0.473] | 20/0/1 | — | no | no |
| $4n$ | MLP | TVAE (native selected) | 20 | 0.286 [0.141, 0.694] | 0.426 [0.189, 1.231] | 19/0/1 | — | no | no |
| $4n$ | CatBoost | CTGAN (native selected) | 21 | 0.792 [0.705, 1.302] | 0.912 [0.731, 1.298] | 21/0/0 | — | no | no |
| $4n$ | Linear | CTGAN (native selected) | 21 | 0.983 [0.843, 1.426] | 1.066 [0.812, 1.695] | 21/0/0 | — | no | no |
| $4n$ | MLP | CTGAN (native selected) | 20 | 1.086 [0.707, 2.268] | 1.439 [0.788, 2.272] | 20/0/0 | — | no | no |
| $4n$ | CatBoost | Forest-Flow (historical) | 6 | −0.035 [−0.477, −0.004] | −0.231 [−0.382, −0.005] | 0/0/6 | — | no | no |
| $4n$ | Linear | Forest-Flow (historical) | 5 | 0.000 [−0.060, 0.037] | 0.002 [−0.051, 0.117] | 3/0/2 | — | no | no |
| $4n$ | MLP | Forest-Flow (historical) | 6 | −0.027 [−0.411, −0.001] | −0.080 [−0.562, 0.004] | 1/0/5 | — | no | no |
| $4n$ | CatBoost | DOPE, untuned default | 99 | 0.110 [0.033, 0.166] | 0.134 [0.053, 0.208] | 87/0/12 | — | no | no |
| $4n$ | Linear | DOPE, untuned default | 94 | −0.001 [−0.017, 0.006] | −0.003 [−0.026, 0.005] | 43/0/51 | — | no | yes |
| $4n$ | MLP | DOPE, untuned default | 88 | 0.080 [0.028, 0.132] | 0.138 [0.052, 0.251] | 69/0/19 | — | no | no |
| $4n$ | CatBoost | Predictor-only control | 99 | 0.194 [0.069, 0.252] | 0.197 [0.112, 0.267] | 92/0/7 | $2.80 \times 10^{-14}$ (6) | no | no |
| $4n$ | Linear | Predictor-only control | 94 | 0.001 [−0.015, 0.031] | 0.008 [−0.027, 0.046] | 51/0/43 | 0.230 (6) | no | yes |
| $4n$ | MLP | Predictor-only control | 88 | 0.165 [0.077, 0.248] | 0.227 [0.118, 0.378] | 79/0/9 | $4.73 \times 10^{-13}$ (6) | no | no |
| $4n$ | CatBoost | Real rows resampled | 99 | −0.132 [−0.222, −0.050] | −0.179 [−0.276, −0.074] | 10/0/89 | $2.02 \times 10^{-13}$ (6) | no | no |
| $4n$ | Linear | Real rows resampled | 94 | −0.008 [−0.023, 0.003] | −0.008 [−0.046, 0.008] | 33/0/61 | 0.050 (6) | no | yes |
| $4n$ | MLP | Real rows resampled | 88 | −0.032 [−0.067, 0.000] | −0.037 [−0.101, 0.070] | 25/0/63 | 0.046 (6) | no | no |
| $4n$ | CatBoost | Chow-Liu tree (default) | 99 | 0.211 [0.096, 0.283] | 0.230 [0.125, 0.348] | 82/0/17 | — | no | no |
| $4n$ | Linear | Chow-Liu tree (default) | 94 | 0.259 [0.136, 0.355] | 0.317 [0.183, 0.466] | 87/0/7 | — | no | no |
| $4n$ | MLP | Chow-Liu tree (default) | 88 | 0.394 [0.209, 0.570] | 0.489 [0.284, 0.799] | 84/0/4 | — | no | no |
| $4n$ | CatBoost | CTGAN (author default) | 17 | 0.978 [0.810, 1.489] | 1.097 [0.863, 1.760] | 17/0/0 | $1.53 \times 10^{-4}$ (12) | no | no |
| $4n$ | Linear | CTGAN (author default) | 17 | 1.277 [0.987, 2.628] | 1.463 [1.093, 2.148] | 17/0/0 | $1.53 \times 10^{-4}$ (12) | no | no |
| $4n$ | MLP | CTGAN (author default) | 16 | 1.454 [1.030, 2.301] | 1.541 [1.148, 2.552] | 16/0/0 | $2.14 \times 10^{-4}$ (12) | no | no |
| $4n$ | CatBoost | DOPE, later binary, seed 11 | 99 | −0.021 [−0.061, 0.000] | −0.037 [−0.079, −0.008] | 19/15/65 | — | no | no |
| $4n$ | Linear | DOPE, later binary, seed 11 | 94 | 0.000 [−0.002, 0.000] | 0.000 [−0.010, 0.007] | 39/16/39 | — | yes | yes |
| $4n$ | MLP | DOPE, later binary, seed 11 | 88 | −0.012 [−0.018, 0.000] | −0.020 [−0.042, −0.002] | 24/12/52 | — | no | no |
| $4n$ | CatBoost | DOPE, later binary, seeds 23-71 | 99 | 0.007 [0.000, 0.012] | 0.011 [0.000, 0.021] | 76/0/23 | — | yes | yes |
| $4n$ | Linear | DOPE, later binary, seeds 23-71 | 94 | 0.000 [−0.001, 0.003] | 0.000 [−0.003, 0.007] | 45/0/49 | — | yes | yes |
| $4n$ | MLP | DOPE, later binary, seeds 23-71 | 88 | 0.005 [0.000, 0.008] | 0.007 [−0.004, 0.019] | 56/0/32 | — | yes | yes |
| $4n$ | CatBoost | DOPE, historical seed 11 | 97 | −0.028 [−0.061, 0.000] | −0.037 [−0.081, −0.008] | 20/12/65 | — | no | no |
| $4n$ | Linear | DOPE, historical seed 11 | 92 | 0.000 [−0.003, 0.001] | 0.000 [−0.011, 0.006] | 38/14/40 | — | yes | yes |
| $4n$ | MLP | DOPE, historical seed 11 | 87 | −0.012 [−0.021, 0.000] | −0.020 [−0.042, −0.001] | 24/11/52 | — | no | no |
| $4n$ | CatBoost | DOPE historical, 12 inputs, 512 steps | 97 | 0.036 [0.008, 0.107] | 0.061 [0.013, 0.162] | 73/0/24 | — | no | no |
| $4n$ | Linear | DOPE historical, 12 inputs, 512 steps | 92 | 0.001 [−0.005, 0.007] | 0.003 [−0.008, 0.013] | 49/0/43 | — | yes | yes |
| $4n$ | MLP | DOPE historical, 12 inputs, 512 steps | 87 | 0.025 [0.004, 0.071] | 0.058 [0.015, 0.142] | 62/0/25 | — | no | no |
| $4n$ | CatBoost | DOPE historical, 24 inputs, 512 steps | 93 | 0.033 [0.006, 0.075] | 0.051 [0.013, 0.137] | 70/0/23 | — | no | no |
| $4n$ | Linear | DOPE historical, 24 inputs, 512 steps | 88 | 0.001 [−0.005, 0.009] | 0.003 [−0.010, 0.016] | 48/0/40 | — | yes | yes |
| $4n$ | MLP | DOPE historical, 24 inputs, 512 steps | 84 | 0.024 [0.002, 0.064] | 0.047 [0.005, 0.124] | 57/0/27 | — | no | no |
| $4n$ | CatBoost | DOPE historical, 24 inputs, 2048 steps | 93 | −0.012 [−0.033, 0.000] | −0.020 [−0.045, −0.004] | 27/11/55 | — | no | no |

| Size | Auditor | Comparator | $n$ | Median diff. | Hodges–Lehmann | W/T/L | Holm $p$ (family) | HL eq. | Yuen eq. |
|---|---|---|---|---|---|---|---|---|---|
| $4n$ | Linear | DOPE historical, 24 inputs, 2048 steps | 88 | $0.000$ $[-0.006, 0.004]$ | $0.000$ $[-0.015, 0.012]$ | 38/11/39 | — | yes | yes |
| $4n$ | MLP | DOPE historical, 24 inputs, 2048 steps | 84 | $-0.011$ $[-0.033, 0.000]$ | $-0.023$ $[-0.050, 0.000]$ | 24/9/51 | — | no | no |
| $4n$ | CatBoost | Forest-Flow default (historical) | 6 | $-0.035$ $[-0.477, -0.004]$ | $-0.231$ $[-0.408, -0.005]$ | 0/0/6 | — | no | no |
| $4n$ | Linear | Forest-Flow default (historical) | 5 | $0.000$ $[-0.033, 0.037]$ | $0.002$ $[-0.051, 0.117]$ | 3/0/2 | — | no | no |
| $4n$ | MLP | Forest-Flow default (historical) | 6 | $-0.027$ $[-0.411, -0.001]$ | $-0.080$ $[-0.562, 0.004]$ | 1/0/5 | — | no | no |
| $4n$ | CatBoost | Gaussian copula (default) | 99 | $0.201$ $[0.062, 0.264]$ | $0.215$ $[0.091, 0.302]$ | 84/0/15 | — | no | no |
| $4n$ | Linear | Gaussian copula (default) | 94 | $0.034$ $[0.003, 0.079]$ | $0.061$ $[0.008, 0.169]$ | 68/0/26 | — | no | no |
| $4n$ | MLP | Gaussian copula (default) | 88 | $0.185$ $[0.100, 0.351]$ | $0.299$ $[0.144, 0.618]$ | 78/0/10 | — | no | no |
| $4n$ | CatBoost | Independent marginals (default) | 99 | $0.917$ $[0.819, 0.975]$ | $0.884$ $[0.790, 0.966]$ | 99/0/0 | — | no | no |
| $4n$ | Linear | Independent marginals (default) | 94 | $0.978$ $[0.947, 1.012]$ | $0.982$ $[0.924, 1.008]$ | 92/0/2 | — | no | no |
| $4n$ | MLP | Independent marginals (default) | 88 | $1.188$ $[1.056, 1.411]$ | $1.305$ $[1.121, 1.580]$ | 88/0/0 | — | no | no |
| $4n$ | CatBoost | Predictor-only, split 2027 | 98 | $0.215$ $[0.097, 0.296]$ | $0.229$ $[0.141, 0.300]$ | 92/0/6 | — | no | no |
| $4n$ | Linear | Predictor-only, split 2027 | 93 | $-0.009$ $[-0.034, 0.002]$ | $-0.015$ $[-0.060, 0.001]$ | 35/0/58 | — | no | no |
| $4n$ | MLP | Predictor-only, split 2027 | 78 | $0.103$ $[0.034, 0.205]$ | $0.194$ $[0.067, 0.362]$ | 60/0/18 | — | no | no |
| $4n$ | CatBoost | Predictor-only, split 2999 | 99 | $0.177$ $[0.062, 0.266]$ | $0.198$ $[0.107, 0.280]$ | 83/0/16 | — | no | no |
| $4n$ | Linear | Predictor-only, split 2999 | 89 | $-0.007$ $[-0.031, 0.009]$ | $-0.010$ $[-0.060, 0.013]$ | 34/0/55 | — | no | no |
| $4n$ | MLP | Predictor-only, split 2999 | 82 | $0.144$ $[0.045, 0.263]$ | $0.202$ $[0.075, 0.413]$ | 64/0/18 | — | no | no |
| $4n$ | CatBoost | Predictor-only, split 4099 | 99 | $0.188$ $[0.088, 0.262]$ | $0.209$ $[0.130, 0.291]$ | 89/0/10 | — | no | no |
| $4n$ | Linear | Predictor-only, split 4099 | 91 | $-0.003$ $[-0.018, 0.014]$ | $-0.002$ $[-0.028, 0.025]$ | 40/0/51 | — | no | yes |
| $4n$ | MLP | Predictor-only, split 4099 | 84 | $0.113$ $[0.011, 0.199]$ | $0.154$ $[0.041, 0.290]$ | 61/0/23 | — | no | no |
| $4n$ | CatBoost | Predictor-only, split 8191 | 99 | $0.172$ $[0.086, 0.245]$ | $0.190$ $[0.099, 0.269]$ | 85/0/14 | — | no | no |
| $4n$ | Linear | Predictor-only, split 8191 | 91 | $-0.004$ $[-0.020, 0.011]$ | $-0.004$ $[-0.031, 0.022]$ | 39/0/52 | — | no | yes |
| $4n$ | MLP | Predictor-only, split 8191 | 81 | $0.122$ $[0.040, 0.219]$ | $0.178$ $[0.064, 0.319]$ | 67/0/14 | — | no | no |
| $4n$ | CatBoost | TVAE (author default) | 18 | $0.470$ $[0.299, 0.785]$ | $0.505$ $[0.357, 0.739]$ | 18/0/0 | $9.16 \times 10^{-5}$ (12) | no | no |
| $4n$ | Linear | TVAE (author default) | 18 | $0.562$ $[0.321, 0.832]$ | $0.575$ $[0.393, 7.242]$ | 18/0/0 | $9.16 \times 10^{-5}$ (12) | no | no |
| $4n$ | MLP | TVAE (author default) | 17 | $0.761$ $[0.375, 1.880]$ | $0.932$ $[0.557, 2.021]$ | 17/0/0 | $1.53 \times 10^{-4}$ (12) | no | no |

Table 13 repeats the first version's source-family sensitivity on the five-seed panel under the inherited family rule: the excluding-simulated rows drop the four simulated families, and inside one family the family is fixed while its lineages, and each lineage's fit seeds, are resampled.

Table 13: **Source-family sensitivity, five-seed panel.** A DOPE row is DOPE's median retention within the scope; a comparator row is the Hodges–Lehmann DOPE-minus-comparator difference on the lineages both arms complete there. The excluding-simulated rows drop the four simulated families; inside one family the family is fixed and its lineages are resampled, and fit seeds are resampled inside each lineage separately for each arm. A one-seed comparator contributes its single fit. Linear and MLP contrasts are in the family file. These rows are a sensitivity outside every Holm family.

| Scope | Size | Auditor | Row | $n$ | Estimate | 95% nested CI | W/T/L |
|---|---|---|---|---|---|---|---|
| excl. simulated | $n$ | CatBoost | DOPE (ours) | 34 | $0.839$ | $[0.728, 0.913]$ | — |
| excl. simulated | $n$ | CatBoost | − ARF (author default) | 34 | $0.055$ | $[-0.036, 0.117]$ | 20/0/14 |
| excl. simulated | $n$ | CatBoost | − TabSyn (scaled schedule) | 34 | $-0.012$ | $[-0.079, 0.057]$ | 16/0/18 |
| excl. simulated | $n$ | CatBoost | − Gaussian copula | 34 | $0.140$ | $[0.076, 0.201]$ | 30/0/4 |
| excl. simulated | $n$ | CatBoost | − Chow-Liu tree | 34 | $0.471$ | $[0.346, 0.656]$ | 33/0/1 |
| excl. simulated | $n$ | CatBoost | − Independent marginals | 34 | $0.866$ | $[0.788, 0.923]$ | 34/0/0 |
| excl. simulated | $n$ | Linear | DOPE (ours) | 32 | $0.992$ | $[0.961, 0.996]$ | — |
| excl. simulated | $n$ | MLP | DOPE (ours) | 31 | $0.946$ | $[0.892, 0.994]$ | — |
| excl. simulated | $4n$ | CatBoost | DOPE (ours) | 34 | $0.900$ | $[0.785, 0.968]$ | — |
| excl. simulated | $4n$ | CatBoost | − ARF (author default) | 34 | $0.015$ | $[-0.069, 0.081]$ | 17/0/17 |

| Scope | Size | Auditor | Row | $n$ | Estimate | 95% nested CI | W/T/L |
|---|---|---|---|---|---|---|---|
| excl. simulated | $4n$ | CatBoost | − TabSyn (scaled schedule) | 34 | $-0.013$ | $[-0.078, 0.051]$ | 17/0/17 |
| excl. simulated | $4n$ | CatBoost | − Gaussian copula | 34 | $0.142$ | $[0.073, 0.203]$ | 25/0/9 |
| excl. simulated | $4n$ | CatBoost | − Chow-Liu tree | 34 | $0.488$ | $[0.330, 0.696]$ | 34/0/0 |
| excl. simulated | $4n$ | CatBoost | − Independent marginals | 34 | $0.876$ | $[0.782, 0.950]$ | 34/0/0 |
| excl. simulated | $4n$ | Linear | DOPE (ours) | 32 | $0.992$ | $[0.970, 1.000]$ | — |
| excl. simulated | $4n$ | MLP | DOPE (ours) | 31 | $1.160$ | $[1.019, 1.257]$ | — |
| feynman | $n$ | CatBoost | DOPE (ours) | 20 | $0.951$ | $[0.927, 0.983]$ | — |
| feynman | $n$ | CatBoost | − ARF (author default) | 20 | $0.064$ | $[0.026, 0.101]$ | 18/0/2 |
| feynman | $n$ | CatBoost | − TabSyn (scaled schedule) | 20 | $0.007$ | $[-0.019, 0.024]$ | 14/0/6 |
| feynman | $n$ | CatBoost | − Gaussian copula | 20 | $0.170$ | $[0.112, 0.236]$ | 20/0/0 |
| feynman | $n$ | CatBoost | − Chow-Liu tree | 20 | $0.325$ | $[0.236, 0.540]$ | 20/0/0 |
| feynman | $n$ | CatBoost | − Independent marginals | 20 | $0.994$ | $[0.958, 1.021]$ | 20/0/0 |
| feynman | $n$ | Linear | DOPE (ours) | 20 | $0.990$ | $[0.984, 0.999]$ | — |
| feynman | $n$ | MLP | DOPE (ours) | 20 | $0.971$ | $[0.944, 0.994]$ | — |
| feynman | $4n$ | CatBoost | DOPE (ours) | 20 | $0.980$ | $[0.954, 0.993]$ | — |
| feynman | $4n$ | CatBoost | − ARF (author default) | 20 | $0.048$ | $[0.013, 0.089]$ | 18/0/2 |
| feynman | $4n$ | CatBoost | − TabSyn (scaled schedule) | 20 | $-0.004$ | $[-0.038, 0.014]$ | 12/0/8 |
| feynman | $4n$ | CatBoost | − Gaussian copula | 20 | $0.159$ | $[0.106, 0.210]$ | 20/0/0 |
| feynman | $4n$ | CatBoost | − Chow-Liu tree | 20 | $0.297$ | $[0.214, 0.490]$ | 20/0/0 |
| feynman | $4n$ | CatBoost | − Independent marginals | 20 | $0.976$ | $[0.945, 0.988]$ | 20/0/0 |
| feynman | $4n$ | Linear | DOPE (ours) | 20 | $0.994$ | $[0.984, 0.998]$ | — |
| feynman | $4n$ | MLP | DOPE (ours) | 20 | $1.018$ | $[0.997, 1.136]$ | — |
| strogatz | $n$ | CatBoost | DOPE (ours) | 14 | $0.775$ | $[0.493, 0.975]$ | — |
| strogatz | $n$ | CatBoost | − ARF (author default) | 14 | $0.006$ | $[-0.126, 0.191]$ | 8/0/6 |
| strogatz | $n$ | CatBoost | − TabSyn (scaled schedule) | 14 | $-0.048$ | $[-0.147, 0.118]$ | 5/0/9 |
| strogatz | $n$ | CatBoost | − Gaussian copula | 14 | $0.239$ | $[0.063, 0.546]$ | 10/0/4 |
| strogatz | $n$ | CatBoost | − Chow-Liu tree | 14 | $1.127$ | $[0.386, 2.814]$ | 11/0/3 |
| strogatz | $n$ | CatBoost | − Independent marginals | 14 | $0.844$ | $[0.654, 1.033]$ | 14/0/0 |
| strogatz | $n$ | Linear | DOPE (ours) | 13 | $0.968$ | $[0.818, 0.996]$ | — |
| strogatz | $n$ | MLP | DOPE (ours) | 11 | $0.974$ | $[0.895, 1.008]$ | — |
| strogatz | $4n$ | CatBoost | DOPE (ours) | 14 | $0.836$ | $[0.541, 0.975]$ | — |
| strogatz | $4n$ | CatBoost | − ARF (author default) | 14 | $-0.034$ | $[-0.132, 0.097]$ | 5/0/9 |
| strogatz | $4n$ | CatBoost | − TabSyn (scaled schedule) | 14 | $-0.023$ | $[-0.133, 0.131]$ | 6/0/8 |
| strogatz | $4n$ | CatBoost | − Gaussian copula | 14 | $0.287$ | $[0.044, 0.624]$ | 10/0/4 |
| strogatz | $4n$ | CatBoost | − Chow-Liu tree | 14 | $1.534$ | $[0.364, 2.959]$ | 12/0/2 |
| strogatz | $4n$ | CatBoost | − Independent marginals | 14 | $0.877$ | $[0.672, 1.084]$ | 14/0/0 |
| strogatz | $4n$ | Linear | DOPE (ours) | 13 | $0.942$ | $[0.778, 0.997]$ | — |
| strogatz | $4n$ | MLP | DOPE (ours) | 11 | $1.384$ | $[1.019, 1.859]$ | — |
| fri | $n$ | CatBoost | DOPE (ours) | 25 | $0.632$ | $[0.553, 0.786]$ | — |
| fri | $n$ | CatBoost | − ARF (author default) | 25 | $0.282$ | $[0.134, 0.440]$ | 20/0/5 |
| fri | $n$ | CatBoost | − TabSyn (scaled schedule) | 25 | $-0.041$ | $[-0.169, 0.120]$ | 12/0/13 |
| fri | $n$ | CatBoost | − Gaussian copula | 25 | $0.324$ | $[0.236, 0.400]$ | 24/0/1 |
| fri | $n$ | CatBoost | − Chow-Liu tree | 25 | $0.223$ | $[0.124, 0.318]$ | 19/0/6 |
| fri | $n$ | CatBoost | − Independent marginals | 25 | $0.682$ | $[0.599, 0.771]$ | 25/0/0 |
| fri | $n$ | Linear | DOPE (ours) | 24 | $0.973$ | $[0.904, 0.999]$ | — |
| fri | $n$ | MLP | DOPE (ours) | 21 | $0.817$ | $[0.636, 0.983]$ | — |
| fri | $4n$ | CatBoost | DOPE (ours) | 25 | $0.774$ | $[0.613, 0.847]$ | — |
| fri | $4n$ | CatBoost | − ARF (author default) | 25 | $0.245$ | $[0.079, 0.411]$ | 17/0/8 |
| fri | $4n$ | CatBoost | − TabSyn (scaled schedule) | 25 | $-0.032$ | $[-0.164, 0.115]$ | 12/0/13 |
| fri | $4n$ | CatBoost | − Gaussian copula | 25 | $0.338$ | $[0.246, 0.421]$ | 25/0/0 |
| fri | $4n$ | CatBoost | − Chow-Liu tree | 25 | $0.203$ | $[0.099, 0.299]$ | 21/0/4 |
| fri | $4n$ | CatBoost | − Independent marginals | 25 | $0.757$ | $[0.664, 0.852]$ | 25/0/0 |
| fri | $4n$ | Linear | DOPE (ours) | 24 | $0.996$ | $[0.981, 1.022]$ | — |
| fri | $4n$ | MLP | DOPE (ours) | 21 | $1.245$ | $[0.973, 1.584]$ | — |
| bng | $n$ | CatBoost | DOPE (ours) | 6 | $0.923$ | $[0.095, 0.990]$ | — |
| bng | $n$ | CatBoost | − ARF (author default) | 6 | $0.110$ | $[-0.322, 0.197]$ | 5/0/1 |
| bng | $n$ | CatBoost | − TabSyn (scaled schedule) | 6 | $0.230$ | $[0.043, 0.940]$ | 6/0/0 |
| bng | $n$ | CatBoost | − Gaussian copula | 6 | $0.082$ | $[-0.038, 0.316]$ | 4/0/2 |
| bng | $n$ | CatBoost | − Chow-Liu tree | 6 | $0.159$ | $[-0.007, 0.445]$ | 4/0/2 |
| bng | $n$ | CatBoost | − Independent marginals | 6 | $0.944$ | $[0.664, 1.184]$ | 6/0/0 |
| bng | $n$ | Linear | DOPE (ours) | 5 | $0.984$ | $[0.855, 0.999]$ | — |
| bng | $n$ | MLP | DOPE (ours) | 5 | $1.149$ | $[0.975, 1.624]$ | — |
| bng | $4n$ | CatBoost | DOPE (ours) | 6 | $0.962$ | $[0.246, 1.018]$ | — |
| bng | $4n$ | CatBoost | − ARF (author default) | 6 | $-0.042$ | $[-0.461, 0.103]$ | 3/0/3 |
| bng | $4n$ | CatBoost | − TabSyn (scaled schedule) | 6 | $0.242$ | $[0.053, 1.313]$ | 6/0/0 |
| bng | $4n$ | CatBoost | − Gaussian copula | 6 | $-0.024$ | $[-0.112, 0.068]$ | 3/0/3 |
| bng | $4n$ | CatBoost | − Chow-Liu tree | 6 | $0.077$ | $[-0.035, 0.493]$ | 4/0/2 |
| bng | $4n$ | CatBoost | − Independent marginals | 6 | $0.980$ | $[0.568, 1.118]$ | 6/0/0 |
| bng | $4n$ | Linear | DOPE (ours) | 5 | $0.983$ | $[0.883, 1.003]$ | — |
| bng | $4n$ | MLP | DOPE (ours) | 5 | $1.134$ | $[1.026, 3.107]$ | — |

### E.1 FIT-SEED-11 REVIEW PANELS

The tables below are the fit-seed-11 review panels that preceded the five-seed refit. A method with two configurations keeps one row per configuration. Their intervals are the family-cluster bootstrap of the median, their equivalence column is the v1 mean-difference test, and their Holm families are

Table 12: **Seed exchangeability and the binary-by-seed diagnostic, CatBoost.** The count of lineages where seed 11 is the best of five is descriptive: a seed acts on every lineage at once, so the lineages are not independent trials and no $p$-value is given. The $2\times2$ cells are medians on the lineages common to all cells, with the median on each cell's own lineages beside it. No Holm family.

| Diagnostic | Count | Value |
|---|---|---|
| Seed 11 highest of five (CatBoost, $4n$) | 47 of 99 | 19.8 if seeds were exchangeable; descriptive, no $p$-value |
| Seeds 23–71 only, CatBoost, $n$ | 99 lineages | $0.791\ [0.692, 0.930]$ |
| Seeds 23–71 only, Linear, $n$ | 94 lineages | $0.984\ [0.961, 0.992]$ |
| Seeds 23–71 only, MLP, $n$ | 88 lineages | $0.958\ [0.875, 0.990]$ |
| Seeds 23–71 only, CatBoost, $4n$ | 99 lineages | $0.862\ [0.771, 0.958]$ |
| Seeds 23–71 only, Linear, $4n$ | 94 lineages | $0.989\ [0.970, 0.996]$ |
| Seeds 23–71 only, MLP, $4n$ | 88 lineages | $1.125\ [1.025, 1.235]$ |
| $2\times2$ cell: historical binary, seed 11 | 97 common (99 own) | 0.949 (own lineages 0.944) |
| $2\times2$ cell: historical binary, seeds 23–71 | 97 common (99 own) | 0.862 (own lineages 0.862) |
| $2\times2$ cell: later binary, seed 11 | 97 common (99 own) | 0.949 (own lineages 0.944) |
| $2\times2$ cell: later binary, seeds 23–71 | 97 common (99 own) | 0.862 (own lineages 0.862) |
| $2\times2$ cell: historical seed-11 population | 97 common (97 own) | 0.949 (own lineages 0.949) |

split by size: nine density tests, six neural tests, six ARF tests, and three Forest-Flow tests per size. Forest-Flow cells store no null loss, so its loss-path columns in Table 19 are empty while its informative count matches the size table. The sensitivity column of that table is the capped cohort, a different denominator from the headline.

Table 14: **DOPE minus comparator at both sizes, fit seed 11.** Median paired retention difference with its family-cluster bootstrap 95% interval, the mean difference, and the v1 mean-difference equivalence test at $\pm0.02$. W/T/L counts lineages. The Wilcoxon and equivalence tests treat lineages as independent. Holm families are separate per size: density 9, neural 6, ARF 6, and Forest-Flow 3 tests.

| Size | Auditor | Comparator | Configuration | $n$ | Median diff. | Hierarchical CI | Mean | TOST | W/T/L | Holm $p$ |
|---|---|---|---|---|---|---|---|---|---|---|
| $1n$ | catboost | GaussianCopula | native_selected | 97 | 0.2178 | [0.1028, 0.3340] | 0.2483 | no | 90/0/7 | 0.000 |
| $1n$ | catboost | Chow-Liu | native_selected | 97 | 0.3635 | [0.2986, 0.5314] | 7.7460 | no | 93/0/4 | 0.000 |
| $1n$ | catboost | independent_marginals | native_selected | 97 | 0.9238 | [0.8575, 1.0178] | 1.1580 | no | 97/0/0 | 0.000 |
| $1n$ | linear | GaussianCopula | native_selected | 92 | 0.0088 | [-0.0118, 0.0329] | 0.2382 | no | 53/0/39 | 0.071 |
| $1n$ | linear | Chow-Liu | native_selected | 92 | 0.3309 | [0.2275, 0.6482] | 1.5301 | no | 86/0/6 | 0.000 |
| $1n$ | linear | independent_marginals | native_selected | 92 | 0.9955 | [0.9691, 1.0693] | 1.1054 | no | 92/0/0 | 0.000 |
| $1n$ | mlp | GaussianCopula | native_selected | 75 | 0.1453 | [0.0479, 0.2445] | 0.1971 | no | 64/0/11 | 0.000 |
| $1n$ | mlp | Chow-Liu | native_selected | 75 | 0.5339 | [0.3343, 0.8171] | 1.1319 | no | 75/0/0 | 0.000 |
| $1n$ | mlp | independent_marginals | native_selected | 75 | 1.0462 | [0.9834, 1.2772] | 1.3698 | no | 75/0/0 | 0.000 |
| $4n$ | catboost | GaussianCopula | native_selected | 97 | 0.2237 | [0.0723, 0.3530] | 0.2613 | no | 86/0/11 | 0.000 |
| $4n$ | catboost | Chow-Liu | native_selected | 97 | 0.3370 | [0.2823, 0.5048] | 8.0937 | no | 94/0/3 | 0.000 |
| $4n$ | catboost | independent_marginals | native_selected | 97 | 0.9556 | [0.8911, 0.9875] | 1.2876 | no | 96/0/1 | 0.000 |
| $4n$ | linear | GaussianCopula | native_selected | 92 | 0.0037 | [-0.0144, 0.0138] | 0.1832 | no | 49/0/43 | 0.373 |
| $4n$ | linear | Chow-Liu | native_selected | 92 | 0.3399 | [0.1947, 0.6595] | 1.4415 | no | 86/0/6 | 0.000 |
| $4n$ | linear | independent_marginals | native_selected | 92 | 0.9879 | [0.9427, 1.0142] | 1.0440 | no | 92/0/0 | 0.000 |
| $4n$ | mlp | GaussianCopula | native_selected | 75 | 0.1850 | [0.0811, 0.2624] | 0.3295 | no | 65/0/10 | 0.000 |
| $4n$ | mlp | Chow-Liu | native_selected | 75 | 0.5301 | [0.3183, 0.8832] | 0.9903 | no | 74/0/1 | 0.000 |
| $4n$ | mlp | independent_marginals | native_selected | 75 | 1.1043 | [1.0367, 1.3283] | 1.3100 | no | 75/0/0 | 0.000 |
| $1n$ | catboost | CTGAN | native_selected | 21 | 0.8816 | [0.7166, 1.4415] | 1.0872 | no | 21/0/0 | 0.000 |
| $1n$ | catboost | TVAE | native_selected | 21 | 0.2381 | [0.1133, 0.4819] | 0.2867 | no | 19/0/2 | 0.000 |
| $1n$ | linear | CTGAN | native_selected | 21 | 1.0516 | [0.8722, 1.3740] | 1.4244 | no | 21/0/0 | 0.000 |
| $1n$ | linear | TVAE | native_selected | 21 | 0.2510 | [0.1501, 0.4434] | 1.3010 | no | 20/0/1 | 0.000 |
| $1n$ | mlp | CTGAN | native_selected | 19 | 0.9336 | [0.6818, 2.0568] | 2.1780 | no | 19/0/0 | 0.000 |
| $1n$ | mlp | TVAE | native_selected | 19 | 0.2725 | [0.1706, 0.7092] | 0.5985 | no | 18/0/1 | 0.000 |
| $4n$ | catboost | CTGAN | native_selected | 21 | 0.8807 | [0.7020, 1.3326] | 1.0962 | no | 21/0/0 | 0.000 |
| $4n$ | catboost | TVAE | native_selected | 21 | 0.2361 | [0.0868, 0.4471] | 0.2673 | no | 19/0/2 | 0.000 |
| $4n$ | linear | CTGAN | native_selected | 21 | 1.0485 | [0.8718, 1.4642] | 1.3489 | no | 21/0/0 | 0.000 |
| $4n$ | linear | TVAE | native_selected | 21 | 0.2629 | [0.1298, 0.4183] | 1.0124 | no | 20/0/1 | 0.000 |
| $4n$ | mlp | CTGAN | native_selected | 19 | 1.0788 | [0.8605, 2.0296] | 1.8412 | no | 19/0/0 | 0.000 |
| $4n$ | mlp | TVAE | native_selected | 19 | 0.2842 | [0.1227, 0.8120] | 0.7771 | no | 17/0/2 | 0.000 |
| $1n$ | catboost | ARF | author_default | 97 | 0.1086 | [0.0026, 0.2394] | 0.1144 | no | 73/0/24 | 0.000 |
| $1n$ | catboost | ARF | native_selected | 97 | 0.1010 | [0.0236, 0.2404] | 0.1354 | no | 76/0/21 | 0.000 |
| $1n$ | linear | ARF | author_default | 92 | 0.0409 | [0.0049, 0.1752] | 0.1786 | no | 64/0/28 | 0.000 |

| Size | Auditor | Comparator | Configuration | $n$ | Median diff. | Hierarchical CI | Mean | TOST | W/T/L | Holm $p$ |
|---|---|---|---|---|---|---|---|---|---|---|
| $1n$ | linear | ARF | native_selected | 92 | 0.0649 | [0.0059, 0.2012] | 0.2123 | no | 68/0/24 | 0.000 |
| $1n$ | mlp | ARF | author_default | 75 | 0.1371 | [0.0311, 0.2860] | 0.2632 | no | 61/0/14 | 0.000 |
| $1n$ | mlp | ARF | native_selected | 75 | 0.1423 | [0.0464, 0.2643] | 0.3097 | no | 64/0/11 | 0.000 |
| $4n$ | catboost | ARF | author_default | 97 | 0.0767 | [0.0103, 0.1957] | 0.2059 | no | 69/0/28 | 0.000 |
| $4n$ | catboost | ARF | native_selected | 97 | 0.0954 | [0.0166, 0.2137] | 0.1737 | no | 70/0/27 | 0.000 |
| $4n$ | linear | ARF | author_default | 92 | 0.0274 | [-0.0020, 0.1195] | 0.0924 | no | 61/0/31 | 0.000 |
| $4n$ | linear | ARF | native_selected | 92 | 0.0369 | [0.0008, 0.1639] | 0.1249 | no | 65/0/27 | 0.000 |
| $4n$ | mlp | ARF | author_default | 75 | 0.1153 | [0.0340, 0.3709] | 0.2894 | no | 57/0/18 | 0.000 |
| $4n$ | mlp | ARF | native_selected | 75 | 0.1429 | [0.0283, 0.3673] | 0.3089 | no | 58/0/17 | 0.000 |
| $1n$ | catboost | ForestDiffusion/Forest-Flow | native_selected | 6 | -0.0198 | [-0.4355, 0.0060] | -0.0917 | no | 1/0/5 | 0.469 |
| $1n$ | linear | ForestDiffusion/Forest-Flow | native_selected | 5 | 0.0275 | [-0.1396, 0.0723] | -0.0040 | no | 3/0/2 | 1.000 |
| $1n$ | mlp | ForestDiffusion/Forest-Flow | native_selected | 4 | 0.0196 | [-0.0405, 0.2906] | 0.0723 | no | 2/0/2 | 1.000 |
| $4n$ | catboost | ForestDiffusion/Forest-Flow | native_selected | 6 | -0.0269 | [-0.4550, 0.0080] | -0.1089 | no | 1/0/5 | 0.375 |
| $4n$ | linear | ForestDiffusion/Forest-Flow | native_selected | 5 | 0.0115 | [-0.2599, 0.3016] | 0.0118 | no | 3/0/2 | 0.812 |
| $4n$ | mlp | ForestDiffusion/Forest-Flow | native_selected | 4 | -0.0317 | [-0.2275, -0.0102] | -0.0753 | no | 0/0/4 | 0.375 |

Table 15: **TabSyn and DOPE retention levels, fit seed 11.** TabSyn runs the scaled schedule (200 VAE and 1,000 diffusion epochs). A lineage median needs three informative sample seeds at size $n$ or $4n$; own-lineage rows use every informative TabSyn lineage, and matched rows use the lineages shared with informative DOPE features12_steps2048. Intervals are the family-cluster bootstrap. Lineages absent from the admitted cohort: 02a45777900441ce; absence is no win. No Holm family.

| Method | Configuration | Cohort | Size | Auditor | $n$ lineages | Median | Hierarchical CI |
|---|---|---|---|---|---|---|---|
| TabSyn | scaled_200_vae_1000_diffusion | own lineages | $1n$ | catboost | 98 | 0.8557 | [0.6955, 0.9459] |
| TabSyn | scaled_200_vae_1000_diffusion | own lineages | $1n$ | linear | 93 | 0.9107 | [0.7705, 0.9890] |
| TabSyn | scaled_200_vae_1000_diffusion | own lineages | $1n$ | mlp | 87 | 0.9170 | [0.7414, 0.9844] |
| TabSyn | scaled_200_vae_1000_diffusion | own lineages | $4n$ | catboost | 98 | 0.8966 | [0.8188, 0.9779] |
| TabSyn | scaled_200_vae_1000_diffusion | own lineages | $4n$ | linear | 93 | 0.9585 | [0.8297, 0.9897] |
| TabSyn | scaled_200_vae_1000_diffusion | own lineages | $4n$ | mlp | 87 | 1.0314 | [0.9689, 1.1865] |
| TabSyn | scaled_200_vae_1000_diffusion | matched lineages | $1n$ | catboost | 96 | 0.8599 | [0.7076, 0.9459] |
| TabSyn | scaled_200_vae_1000_diffusion | matched lineages | $1n$ | linear | 91 | 0.9280 | [0.7705, 0.9896] |
| TabSyn | scaled_200_vae_1000_diffusion | matched lineages | $1n$ | mlp | 72 | 0.9438 | [0.8201, 0.9857] |
| TabSyn | scaled_200_vae_1000_diffusion | matched lineages | $4n$ | catboost | 96 | 0.9009 | [0.8161, 0.9788] |
| TabSyn | scaled_200_vae_1000_diffusion | matched lineages | $4n$ | linear | 91 | 0.9595 | [0.8310, 0.9897] |
| TabSyn | scaled_200_vae_1000_diffusion | matched lineages | $4n$ | mlp | 72 | 1.0355 | [0.9992, 1.1968] |
| DOPE | features12_steps2048 | matched lineages | $1n$ | catboost | 96 | 0.8852 | [0.8224, 0.9690] |
| DOPE | features12_steps2048 | matched lineages | $1n$ | linear | 91 | 0.9812 | [0.9659, 0.9942] |
| DOPE | features12_steps2048 | matched lineages | $1n$ | mlp | 72 | 0.9732 | [0.9490, 0.9898] |
| DOPE | features12_steps2048 | matched lineages | $4n$ | catboost | 96 | 0.9386 | [0.8922, 0.9831] |
| DOPE | features12_steps2048 | matched lineages | $4n$ | linear | 91 | 0.9940 | [0.9718, 0.9967] |
| DOPE | features12_steps2048 | matched lineages | $4n$ | mlp | 72 | 1.0684 | [1.0048, 1.2714] |

Table 16: **DOPE minus TabSyn, fit seed 11.** Median paired retention difference on the matched lineages with its family-cluster bootstrap interval, the mean difference, the v1 mean-difference equivalence test at $\pm 0.02$, and W/T/L. The Holm family has three tests per size (one comparator, three auditors), separate from every other comparator block; the Wilcoxon and equivalence tests treat lineages as independent.

| Size | Auditor | Comparator | Configuration | $n$ | Median diff. | Hierarchical CI | Mean | TOST | W/T/L | Holm $p$ |
|---|---|---|---|---|---|---|---|---|---|---|
| $1n$ | catboost | TabSyn | scaled_200_vae_1000_diffusion | 96 | 0.0122 | [-0.0033, 0.0851] | 0.0338 | no | 59/0/37 | 0.042 |
| $1n$ | linear | TabSyn | scaled_200_vae_1000_diffusion | 91 | 0.0348 | [-0.0008, 0.1265] | 58.8632 | no | 61/0/30 | 0.000 |
| $1n$ | mlp | TabSyn | scaled_200_vae_1000_diffusion | 72 | 0.0333 | [-0.0078, 0.1272] | 0.2127 | no | 43/0/29 | 0.011 |
| $4n$ | catboost | TabSyn | scaled_200_vae_1000_diffusion | 96 | 0.0157 | [-0.0033, 0.0890] | 0.0737 | no | 56/0/40 | 0.056 |
| $4n$ | linear | TabSyn | scaled_200_vae_1000_diffusion | 91 | 0.0199 | [-0.0018, 0.1364] | 68.9447 | no | 60/0/31 | 0.000 |
| $4n$ | mlp | TabSyn | scaled_200_vae_1000_diffusion | 72 | -0.0041 | [-0.0384, 0.0738] | 0.1739 | no | 33/0/39 | 0.682 |

Table 17: **v1 mean-difference equivalence, fit seed 11.** Mean DOPE-minus-comparator retention difference, its 90% $t$ interval, and the larger of the two one-sided $p$-values against $\pm 0.02$. The test treats lineages as independent, and outliers move the mean; it is kept for continuity with the first version. Holm families as in the paired table; the equivalence $p$ is unadjusted.

| Size | Auditor | Comparator | Configuration | Mean diff. | 90% CI | TOST $p$ | Equivalent at $\pm 0.02$ |
|---|---|---|---|---|---|---|---|
| $1n$ | catboost | GaussianCopula | native_selected | 0.2483 | [0.1779, 0.3188] | 1.000 | no |
| $1n$ | catboost | Chow-Liu | native_selected | 7.7460 | [-3.7576, 19.2497] | 0.866 | no |
| $1n$ | catboost | independent_marginals | native_selected | 1.1580 | [0.7097, 1.6062] | 1.000 | no |
| $1n$ | linear | GaussianCopula | native_selected | 0.2382 | [0.0337, 0.4427] | 0.960 | no |
| $1n$ | linear | Chow-Liu | native_selected | 1.5301 | [0.7702, 2.2899] | 0.999 | no |
| $1n$ | linear | independent_marginals | native_selected | 1.1054 | [0.9906, 1.2202] | 1.000 | no |
| $1n$ | mlp | GaussianCopula | native_selected | 0.1971 | [0.1217, 0.2725] | 1.000 | no |

| Size | Auditor | Comparator | Configuration | Mean diff. | 90% CI | TOST $p$ | Equivalent at $\pm 0.02$ |
|---|---|---|---|---|---|---|---|
| $1n$ | mlp | Chow-Liu | native_selected | 1.1319 | [0.7996, 1.4642] | 1.000 | no |
| $1n$ | mlp | independent_marginals | native_selected | 1.3698 | [1.1741, 1.5655] | 1.000 | no |
| $4n$ | catboost | GaussianCopula | native_selected | 0.2613 | [0.2007, 0.3220] | 1.000 | no |
| $4n$ | catboost | Chow-Liu | native_selected | 8.0937 | [-3.9560, 20.1434] | 0.866 | no |
| $4n$ | catboost | independent_marginals | native_selected | 1.2876 | [0.6528, 1.9224] | 0.999 | no |
| $4n$ | linear | GaussianCopula | native_selected | 0.1832 | [0.0184, 0.3480] | 0.948 | no |
| $4n$ | linear | Chow-Liu | native_selected | 1.4415 | [0.7544, 2.1285] | 1.000 | no |
| $4n$ | linear | independent_marginals | native_selected | 1.0440 | [0.9496, 1.1384] | 1.000 | no |
| $4n$ | mlp | GaussianCopula | native_selected | 0.3295 | [0.2309, 0.4282] | 1.000 | no |
| $4n$ | mlp | Chow-Liu | native_selected | 0.9903 | [0.6981, 1.2825] | 1.000 | no |
| $4n$ | mlp | independent_marginals | native_selected | 1.3100 | [1.2007, 1.4194] | 1.000 | no |
| $1n$ | catboost | CTGAN | native_selected | 1.0872 | [0.8392, 1.3352] | 1.000 | no |
| $1n$ | catboost | TVAE | native_selected | 0.2867 | [0.1907, 0.3827] | 1.000 | no |
| $1n$ | linear | CTGAN | native_selected | 1.4244 | [0.9705, 1.8783] | 1.000 | no |
| $1n$ | linear | TVAE | native_selected | 1.3010 | [-0.2842, 2.8862] | 0.911 | no |
| $1n$ | mlp | CTGAN | native_selected | 2.1780 | [1.0282, 3.3279] | 0.998 | no |
| $1n$ | mlp | TVAE | native_selected | 0.5985 | [0.2704, 0.9265] | 0.997 | no |
| $4n$ | catboost | CTGAN | native_selected | 1.0962 | [0.8335, 1.3588] | 1.000 | no |
| $4n$ | catboost | TVAE | native_selected | 0.2673 | [0.1704, 0.3642] | 1.000 | no |
| $4n$ | linear | CTGAN | native_selected | 1.3489 | [0.9775, 1.7204] | 1.000 | no |
| $4n$ | linear | TVAE | native_selected | 1.0124 | [-0.1398, 2.1646] | 0.923 | no |
| $4n$ | mlp | CTGAN | native_selected | 1.8412 | [1.1388, 2.5436] | 1.000 | no |
| $4n$ | mlp | TVAE | native_selected | 0.7771 | [0.2190, 1.3353] | 0.985 | no |
| $1n$ | catboost | ARF | author_default | 0.1144 | [0.0426, 0.1862] | 0.984 | no |
| $1n$ | catboost | ARF | native_selected | 0.1354 | [0.0651, 0.2057] | 0.996 | no |
| $1n$ | linear | ARF | author_default | 0.1786 | [0.0606, 0.2965] | 0.986 | no |
| $1n$ | linear | ARF | native_selected | 0.2123 | [0.0869, 0.3378] | 0.994 | no |
| $1n$ | mlp | ARF | author_default | 0.2632 | [0.1518, 0.3746] | 1.000 | no |
| $1n$ | mlp | ARF | native_selected | 0.3097 | [0.1406, 0.4789] | 0.997 | no |
| $4n$ | catboost | ARF | author_default | 0.2059 | [0.0480, 0.3638] | 0.973 | no |
| $4n$ | catboost | ARF | native_selected | 0.1737 | [0.1098, 0.2376] | 1.000 | no |
| $4n$ | linear | ARF | author_default | 0.0924 | [0.0269, 0.1579] | 0.965 | no |
| $4n$ | linear | ARF | native_selected | 0.1249 | [0.0554, 0.1943] | 0.993 | no |
| $4n$ | mlp | ARF | author_default | 0.2894 | [0.1767, 0.4022] | 1.000 | no |
| $4n$ | mlp | ARF | native_selected | 0.3089 | [0.1650, 0.4527] | 0.999 | no |
| $1n$ | catboost | ForestDiffusion/Forest-Flow | native_selected | -0.0917 | [-0.2333, 0.0500] | 0.823 | no |
| $1n$ | linear | ForestDiffusion/Forest-Flow | native_selected | -0.0040 | [-0.0820, 0.0740] | 0.342 | no |
| $1n$ | mlp | ForestDiffusion/Forest-Flow | native_selected | 0.0723 | [-0.1050, 0.2497] | 0.731 | no |
| $4n$ | catboost | ForestDiffusion/Forest-Flow | native_selected | -0.1089 | [-0.2579, 0.0401] | 0.859 | no |
| $4n$ | linear | ForestDiffusion/Forest-Flow | native_selected | 0.0118 | [-0.1779, 0.2015] | 0.466 | no |
| $4n$ | mlp | ForestDiffusion/Forest-Flow | native_selected | -0.0753 | [-0.1953, 0.0447] | 0.821 | no |

Table 18: **Source-family sensitivity, fit seed 11.** A DOPE row is the median DOPE retention within one source family; a comparator row is the median DOPE-minus-comparator difference on that family, under CatBoost. The excluding-simulated rows drop the four simulated families. Intervals are the family-cluster bootstrap. These rows are a sensitivity, printed without Holm $p$-values.

| Family | Size | Auditor | Contrast | $n$ | Median | Hierarchical CI |
|---|---|---|---|---|---|---|
| feynman | $1n$ | catboost | DOPE | 20 | 0.9703 | [0.9454, 0.9906] |
| feynman | $1n$ | catboost | GaussianCopula | 20 | 0.2002 | [0.1097, 0.2549] |
| feynman | $1n$ | catboost | Chow-Liu | 20 | 0.3253 | [0.2200, 0.4252] |
| feynman | $1n$ | catboost | independent_marginals | 20 | 1.0201 | [0.9776, 1.0254] |
| feynman | $1n$ | linear | DOPE | 20 | 0.9938 | [0.9806, 0.9975] |
| feynman | $1n$ | mlp | DOPE | 19 | 0.9713 | [0.9609, 0.9885] |
| feynman | $4n$ | catboost | DOPE | 20 | 0.9894 | [0.9713, 0.9944] |
| feynman | $4n$ | catboost | GaussianCopula | 20 | 0.1894 | [0.0993, 0.2840] |
| feynman | $4n$ | catboost | Chow-Liu | 20 | 0.2829 | [0.2346, 0.3576] |
| feynman | $4n$ | catboost | independent_marginals | 20 | 0.9813 | [0.9777, 0.9947] |
| feynman | $4n$ | linear | DOPE | 20 | 0.9941 | [0.9854, 0.9982] |
| feynman | $4n$ | mlp | DOPE | 19 | 0.9965 | [0.9860, 1.0067] |
| strogatz | $1n$ | catboost | DOPE | 14 | 0.7819 | [0.4900, 0.9793] |
| strogatz | $1n$ | catboost | GaussianCopula | 14 | 0.1925 | [-0.0086, 0.3450] |
| strogatz | $1n$ | catboost | Chow-Liu | 14 | 0.8451 | [0.2765, 2.3318] |
| strogatz | $1n$ | catboost | independent_marginals | 14 | 0.8779 | [0.6389, 1.0237] |
| strogatz | $1n$ | linear | DOPE | 13 | 0.9687 | [0.8345, 0.9960] |
| strogatz | $1n$ | mlp | DOPE | 9 | 0.9967 | [0.8086, 1.0923] |
| strogatz | $4n$ | catboost | DOPE | 14 | 0.8854 | [0.5626, 0.9845] |
| strogatz | $4n$ | catboost | GaussianCopula | 14 | 0.1303 | [-0.0086, 0.3896] |
| strogatz | $4n$ | catboost | Chow-Liu | 14 | 0.8739 | [0.2276, 2.9286] |
| strogatz | $4n$ | catboost | independent_marginals | 14 | 0.9465 | [0.6883, 1.0851] |
| strogatz | $4n$ | linear | DOPE | 13 | 0.9718 | [0.7502, 0.9969] |
| strogatz | $4n$ | mlp | DOPE | 9 | 1.0815 | [0.9862, 1.4876] |
| fri | $1n$ | catboost | DOPE | 24 | 0.8292 | [0.7550, 0.8791] |
| fri | $1n$ | catboost | GaussianCopula | 24 | 0.4997 | [0.3340, 0.5624] |
| fri | $1n$ | catboost | Chow-Liu | 24 | 0.3401 | [0.2105, 0.4416] |
| fri | $1n$ | catboost | independent_marginals | 24 | 0.8660 | [0.7905, 0.8976] |

| Family | Size | Auditor | Contrast | $n$ | Median | Hierarchical CI |
|---|---|---|---|---|---|---|
| fri | $1n$ | linear | DOPE | 23 | 0.9675 | [0.8787, 1.0009] |
| fri | $1n$ | mlp | DOPE | 15 | 0.9889 | [0.8894, 1.0310] |
| fri | $4n$ | catboost | DOPE | 24 | 0.9054 | [0.8485, 0.9276] |
| fri | $4n$ | catboost | GaussianCopula | 24 | 0.5583 | [0.3530, 0.5913] |
| fri | $4n$ | catboost | Chow-Liu | 24 | 0.2964 | [0.1981, 0.4561] |
| fri | $4n$ | catboost | independent_marginals | 24 | 0.9152 | [0.8394, 0.9537] |
| fri | $4n$ | linear | DOPE | 23 | 0.9960 | [0.9668, 1.0359] |
| fri | $4n$ | mlp | DOPE | 15 | 1.5792 | [1.0250, 1.6797] |
| bng | $1n$ | catboost | DOPE | 6 | 0.9118 | [0.0978, 1.1020] |
| bng | $1n$ | catboost | GaussianCopula | 6 | 0.1041 | [0.0472, 0.1614] |
| bng | $1n$ | catboost | Chow-Liu | 6 | 0.1781 | [0.0312, 0.6353] |
| bng | $1n$ | catboost | independent_marginals | 6 | 1.0364 | [0.5939, 1.3518] |
| bng | $1n$ | linear | DOPE | 5 | 0.9718 | [0.9305, 1.0382] |
| bng | $1n$ | mlp | DOPE | 4 | 1.0475 | [0.9751, 1.6088] |
| bng | $4n$ | catboost | DOPE | 6 | 0.9334 | [0.2662, 1.0918] |
| bng | $4n$ | catboost | GaussianCopula | 6 | 0.0387 | [-0.0122, 0.1742] |
| bng | $4n$ | catboost | Chow-Liu | 6 | 0.0761 | [0.0430, 0.5724] |
| bng | $4n$ | catboost | independent_marginals | 6 | 0.9060 | [0.4007, 1.2632] |
| bng | $4n$ | linear | DOPE | 5 | 0.9941 | [0.9173, 1.0458] |
| bng | $4n$ | mlp | DOPE | 4 | 1.2962 | [0.9871, 1.7049] |
| other | $1n$ | catboost | DOPE | 33 | 0.9020 | [0.8417, 0.9725] |
| other | $1n$ | catboost | GaussianCopula | 33 | 0.1651 | [0.0633, 0.2597] |
| other | $1n$ | catboost | Chow-Liu | 33 | 0.4339 | [0.2997, 0.6052] |
| other | $1n$ | catboost | independent_marginals | 33 | 0.9217 | [0.8303, 0.9973] |
| other | $1n$ | linear | DOPE | 31 | 0.9812 | [0.9713, 0.9952] |
| other | $1n$ | mlp | DOPE | 28 | 0.9322 | [0.8545, 0.9864] |
| other | $4n$ | catboost | DOPE | 33 | 0.9373 | [0.8908, 0.9878] |
| other | $4n$ | catboost | GaussianCopula | 33 | 0.1912 | [0.0762, 0.2495] |
| other | $4n$ | catboost | Chow-Liu | 33 | 0.4514 | [0.3214, 0.6687] |
| other | $4n$ | catboost | independent_marginals | 33 | 0.9293 | [0.8699, 1.0026] |
| other | $4n$ | linear | DOPE | 31 | 0.9951 | [0.9662, 1.0017] |
| other | $4n$ | mlp | DOPE | 28 | 1.0955 | [1.0357, 1.1953] |
| excluding simulated | $1n$ | catboost | GaussianCopula | 33 | 0.1651 | [0.0698, 0.2637] |
| excluding simulated | $1n$ | catboost | Chow-Liu | 33 | 0.4339 | [0.2997, 0.6052] |
| excluding simulated | $1n$ | catboost | independent_marginals | 33 | 0.9217 | [0.8575, 0.9973] |
| excluding simulated | $4n$ | catboost | GaussianCopula | 33 | 0.1912 | [0.0762, 0.2495] |
| excluding simulated | $4n$ | catboost | Chow-Liu | 33 | 0.4514 | [0.2957, 0.6687] |
| excluding simulated | $4n$ | catboost | independent_marginals | 33 | 0.9293 | [0.8699, 1.0026] |

Table 19: **Informative-lineage denominators, fit seed 11, CatBoost.** Informative is the lineage count behind each retention level. Complete-loss rows have three finite stored null, TRTR, and TSTR losses, and the noninformative, undefined, capped, and sensitivity counts are taken among them. No Holm family.

| Method | Configuration | Size | Informative | Complete-loss rows | Non-informative | Undefined | Capped | Sensitivity $n$ | Capped median |
|---|---|---|---|---|---|---|---|---|---|
| DOPE | features12_steps2048 | $1n$ | 97 | 98 | 1 | 0 | 1 | 98 | 0.8862 |
| DOPE | features12_steps2048 | $4n$ | 97 | 98 | 1 | 0 | 1 | 98 | 0.9359 |
| GaussianCopula | native_selected | $1n$ | 99 | 100 | 1 | 0 | 1 | 100 | 0.5996 |
| GaussianCopula | native_selected | $4n$ | 99 | 100 | 1 | 0 | 1 | 100 | 0.6658 |
| Chow-Liu | native_selected | $1n$ | 99 | 100 | 1 | 0 | 12 | 100 | 0.4809 |
| Chow-Liu | native_selected | $4n$ | 99 | 100 | 1 | 0 | 11 | 100 | 0.5782 |
| independent_marginals | native_selected | $1n$ | 99 | 100 | 1 | 0 | 3 | 100 | -0.0440 |
| independent_marginals | native_selected | $4n$ | 99 | 100 | 1 | 0 | 1 | 100 | -0.0120 |
| CTGAN | native_selected | $1n$ | 21 | 22 | 1 | 0 | 2 | 22 | -0.0889 |
| CTGAN | native_selected | $4n$ | 21 | 22 | 1 | 0 | 2 | 22 | -0.0801 |
| TVAE | native_selected | $1n$ | 21 | 22 | 1 | 0 | 0 | 22 | 0.6094 |
| TVAE | native_selected | $4n$ | 21 | 22 | 1 | 0 | 0 | 22 | 0.6582 |
| ARF | author_default | $1n$ | 99 | 100 | 1 | 0 | 1 | 100 | 0.7505 |
| ARF | author_default | $4n$ | 99 | 100 | 1 | 0 | 1 | 100 | 0.8314 |
| ARF | native_selected | $1n$ | 99 | 100 | 1 | 0 | 1 | 100 | 0.7410 |
| ARF | native_selected | $4n$ | 99 | 100 | 1 | 0 | 1 | 100 | 0.8237 |
| ForestDiffusion/Forest-Flow | native_selected | $1n$ | 6 | 0 | 0 | 0 | 0 | 0 | — |
| ForestDiffusion/Forest-Flow | native_selected | $4n$ | 6 | 0 | 0 | 0 | 0 | 0 | — |

Table 21: **First-version headline contrasts at $4n$, fit seed 11.** Paired median DOPE-minus-baseline retention difference with W/T/L over lineages; a tie is a difference of zero. The ARF rows use the family-cluster interval of the review panel and its size-$4n$ Holm family of six. The density, neural, and Forest-Flow rows show each block's highest-median baseline with a lineage-bootstrap interval and the block's Holm family (nine, six, and three tests). The five-seed panel supersedes these rows; they are the traceability record of the first version.

| Block | Auditor | Baseline | $n$ | Difference | W/T/L | Holm $p$ |
|---|---|---|---|---|---|---|
| Full panel | CatBoost | Author-default ARF | 97 | 0.0767 [0.0103, 0.1957] | 69/0/28 | $1.31 \times 10^{-5}$ |
| Full panel | Linear | Author-default ARF | 92 | 0.0274 [-0.0020, 0.1195] | 61/0/31 | $8.53 \times 10^{-5}$ |
| Full panel | MLP | Author-default ARF | 75 | 0.1153 [0.0340, 0.3709] | 57/0/18 | $4.56 \times 10^{-8}$ |
| Density | CatBoost | Gaussian copula | 97 | 0.224 [0.149, 0.290] | 86/0/11 | $1.35 \times 10^{-13}$ |
| Density | Linear | Gaussian copula | 92 | 0.00366 [-0.00438, 0.012] | 49/0/43 | 0.373 |
| Density | MLP | Gaussian copula | 75 | 0.185 [0.128, 0.228] | 65/0/10 | $4.96 \times 10^{-11}$ |
| Neural | CatBoost | TVAE | 21 | 0.236 [0.087, 0.431] | 19/0/2 | $1.57 \times 10^{-4}$ |
| Neural | Linear | TVAE | 21 | 0.263 [0.166, 0.410] | 20/0/1 | $1.57 \times 10^{-4}$ |
| Neural | MLP | TVAE | 19 | 0.284 [0.153, 0.734] | 17/0/2 | $1.57 \times 10^{-4}$ |
| Forest | CatBoost | Forest-Flow | 6 | -0.027 [-0.308, 0.00804] | 1/0/5 | 0.375 |
| Forest | Linear | Forest-Flow | 5 | 0.012 [-0.260, 0.302] | 3/0/2 | 0.812 |
| Forest | MLP | Forest-Flow | 4 | -0.032 [-0.227, -0.010] | 0/0/4 | 0.375 |

Table 20: **CatBoost auditor RMSE, fit seed 11.** Median over regression lineages of the validation RMSE of the auditor trained on synthetic rows and of the auditor trained on the real fit rows. RMSE is in each lineage's original target units: the square root of the median sample-seed mean squared error, which the evaluator computes on the target min–max projected and clipped to $[0, 1]$ by the training split, times that split's target range. A median therefore mixes lineage scales and is no cross-method ranking. No Holm family.

| Method | Configuration | Size | Regression lineages | Synthetic RMSE | Real TRTR RMSE |
|---|---|---|---|---|---|
| DOPE | features12_steps2048 | $1n$ | 98 | 0.0887 | 0.0522 |
| DOPE | features12_steps2048 | $4n$ | 98 | 0.0786 | 0.0522 |
| GaussianCopula | native_selected | $1n$ | 100 | 0.1166 | 0.0529 |
| GaussianCopula | native_selected | $4n$ | 100 | 0.1124 | 0.0529 |
| Chow-Liu | native_selected | $1n$ | 100 | 0.1359 | 0.0529 |
| Chow-Liu | native_selected | $4n$ | 100 | 0.1280 | 0.0529 |
| independent_marginals | native_selected | $1n$ | 100 | 0.1796 | 0.0529 |
| independent_marginals | native_selected | $4n$ | 100 | 0.1760 | 0.0529 |
| CTGAN | native_selected | $1n$ | 22 | 0.1862 | 0.0678 |
| CTGAN | native_selected | $4n$ | 22 | 0.1780 | 0.0678 |
| TVAE | native_selected | $1n$ | 22 | 0.1314 | 0.0678 |
| TVAE | native_selected | $4n$ | 22 | 0.1257 | 0.0678 |
| ARF | author_default | $1n$ | 100 | 0.0971 | 0.0529 |
| ARF | author_default | $4n$ | 100 | 0.0866 | 0.0529 |
| ARF | native_selected | $1n$ | 100 | 0.1019 | 0.0529 |
| ARF | native_selected | $4n$ | 100 | 0.0958 | 0.0529 |
| ForestDiffusion/Forest-Flow | native_selected | $1n$ | 0 | — | — |
| ForestDiffusion/Forest-Flow | native_selected | $4n$ | 0 | — | — |

## E.2 HISTORICAL BLOCK TABLES

The density, neural, and Forest-Flow block tables below are the fit-seed-11 receipts behind the first version of this paper; their intervals are lineage-bootstrap intervals of the median, and they are kept for continuity. The paired median difference is distinct from the difference of two medians. The two DOPE artifacts above the cap in that population are 4544_GeographicalOriginalofMusic at 22,233 bytes and 588_fri_c4_1000_100 at 19,019 bytes, and 678_visualizing_environmental has a fit whose CatBoost group is uninformative, so the CatBoost median uses 97 lineages while the byte count uses 100. The displayed byte range runs from 758 to 22,233. CatBoost reads RMSE $0.053$ for the real-trained auditor and $0.081$ for the synthetic-trained one, with training-mean skill $0.881$ and $0.792$; RMSE is the square root of the stored mean squared error, and conventional held-out $R^2$ is unstored for these rows. Table 21 collects the first version's headline contrasts. On its $6$ Forest-Flow lineages the CatBoost medians are $0.940$ for DOPE, whose artifacts have a median of 2,067 bytes, and $1.003$ [$0.926$, $1.026$] for Forest-Flow. The Nemenyi critical differences for the four density methods are $0.476$ under CatBoost and $0.489$ under the linear auditor.

Table 22: **Density block at $4n$, fit seed 11.** Each cell is a retention; the denominator is the paired informative lineages in that row. Holm family size is nine. W/T/L counts lineages; a tie is a difference of zero.

| Auditor | Comparator | $n$ | DOPE | Other | Difference | W/T/L | Holm $p$ |
|---|---|---|---|---|---|---|---|
| CatBoost | Gaussian copula | 97 | **0.940** [0.914, 0.976] | 0.656 [0.431, 0.803] | 0.224 [0.149, 0.290] | 86/0/11 | $1.35 \times 10^{-13}$ |
| CatBoost | Chow–Liu | 97 | **0.940** [0.914, 0.976] | 0.586 [0.465, 0.637] | 0.337 [0.282, 0.451] | 94/0/3 | $4.85 \times 10^{-16}$ |
| CatBoost | Indep. marginals | 97 | **0.940** [0.914, 0.976] | -0.012 [-0.020, -0.00649] | 0.956 [0.918, 0.980] | 96/0/1 | $1.17 \times 10^{-16}$ |
| Linear | Gaussian copula | 92 | 0.994 [0.981, 0.996] | 0.989 [0.968, 0.994] | 0.00366 [-0.00438, 0.012] | 49/0/43 | 0.373 |
| Linear | Chow–Liu | 92 | **0.994** [0.981, 0.996] | 0.661 [0.356, 0.738] | 0.340 [0.244, 0.425] | 86/0/6 | $7.12 \times 10^{-15}$ |
| Linear | Indep. marginals | 92 | **0.994** [0.981, 0.996] | 0.00206 [-0.023, 0.00583] | 0.988 [0.975, 1.001] | 92/0/0 | $5.70 \times 10^{-16}$ |
| MLP | Gaussian copula | 75 | **1.061** [1.024, 1.125] | 0.928 [0.865, 0.953] | 0.185 [0.128, 0.228] | 65/0/10 | $4.96 \times 10^{-11}$ |
| MLP | Chow–Liu | 75 | **1.061** [1.024, 1.125] | 0.692 [0.582, 0.754] | 0.530 [0.350, 0.651] | 74/0/1 | $2.11 \times 10^{-13}$ |
| MLP | Indep. marginals | 75 | **1.061** [1.024, 1.125] | -0.028 [-0.047, -0.013] | 1.104 [1.047, 1.179] | 75/0/0 | $2.11 \times 10^{-13}$ |

Table 23: **Neural block at $4n$, fit seed 11.** The denominator is the paired informative lineages in the cell; CatBoost uses 21. Holm family size is six. W/T/L counts lineages; a tie is a difference of zero.

| Auditor | Comparator | $n$ | DOPE | Other | Difference | W/T/L | Holm $p$ |
|---|---|---|---|---|---|---|---|
| CatBoost | CTGAN | 21 | **0.953** [0.896, 0.990] | -0.081 [-0.202, 0.103] | 0.881 [0.702, 1.150] | 21/0/0 | $5.72 \times 10^{-6}$ |
| CatBoost | TVAE | 21 | **0.953** [0.896, 0.990] | 0.669 [0.539, 0.787] | 0.236 [0.087, 0.431] | 19/0/2 | $1.57 \times 10^{-4}$ |
| Linear | CTGAN | 21 | **0.999** [0.996, 1.015] | -0.033 [-0.225, 0.080] | 1.049 [0.916, 1.236] | 21/0/0 | $5.72 \times 10^{-6}$ |
| Linear | TVAE | 21 | **0.999** [0.996, 1.015] | 0.796 [0.605, 0.867] | 0.263 [0.166, 0.410] | 20/0/1 | $1.57 \times 10^{-4}$ |
| MLP | CTGAN | 19 | **1.026** [0.998, 1.263] | -0.149 [-0.471, 0.083] | 1.079 [0.877, 2.030] | 19/0/0 | $1.53 \times 10^{-5}$ |
| MLP | TVAE | 19 | **1.026** [0.998, 1.263] | 0.753 [0.478, 0.852] | 0.284 [0.153, 0.734] | 17/0/2 | $1.57 \times 10^{-4}$ |

# F CONTROLS

The real-row control draws the published training rows with replacement until the table has four times the training row count. The predictor-only control fits a ridge regression on the training rows, samples each feature independently from its training marginal, and adds Gaussian residual noise; neither control uses the DOPE encoding, and the predictor-only gap bundles DOPE's dependence model with DOPE's nonlinear residual. The paired column is DOPE minus the control. In the fit-seed-11 control table below, the size-$n$ Holm family is the three predictor-only auditor tests, because the real-row control is defined at $4n$ only, and the size-$4n$ family has six tests; the five-seed panel keeps all six declared control slots at each size. Extra predictor fit seeds are reported and are outside the Holm family, and split-seed rows are a separate sensitivity.

Table 28: **DOPE minus each control, fit seed 11.** A paired row is the median DOPE-minus-control retention difference with its family-cluster bootstrap 95% interval, the mean difference, the v1 mean-difference equivalence test at $\pm 0.02$, and W/T/L over lineages; the Wilcoxon and equivalence tests treat lineages as independent. Predictor fit-seed and split-seed rows give the control's own CatBoost retention level. The size-$n$ Holm family is the three predictor-only tests, because the real-row control exists at $4n$ only; the size-$4n$ family has six tests. Fit-seed and split-seed rows are outside both Holm families.

| Size | Auditor | Control | $n$ | Median diff. or level | Hierarchical CI | Mean | TOST | W/T/L | Holm $p$ |
|---|---|---|---|---|---|---|---|---|---|
| $1n$ | catboost | predictor_only | 97 | 0.2528 | [0.1449, 0.3903] | 0.2286 | no | 87/0/10 | 0.000 |
| $1n$ | linear | predictor_only | 92 | -0.0026 | [-0.0176, 0.0137] | 0.0156 | no | 42/0/50 | 0.523 |
| $1n$ | mlp | predictor_only | 73 | 0.0739 | [-0.0009, 0.1498] | 0.1331 | no | 52/0/21 | 0.000 |
| $4n$ | catboost | real_bootstrap_4n | 97 | -0.0806 | [-0.1276, -0.0290] | -0.1922 | no | 14/0/83 | 0.000 |
| $4n$ | catboost | predictor_only | 97 | 0.2680 | [0.0784, 0.3687] | 0.2265 | no | 84/0/13 | 0.000 |
| $4n$ | linear | real_bootstrap_4n | 92 | -0.0058 | [-0.0203, 0.0028] | 0.0028 | no | 34/0/58 | 0.147 |
| $4n$ | linear | predictor_only | 92 | -0.0051 | [-0.0181, 0.0077] | -0.0023 | no | 39/0/53 | 0.229 |
| $4n$ | mlp | real_bootstrap_4n | 73 | -0.0522 | [-0.1495, 0.0057] | -0.0256 | no | 23/0/50 | 0.087 |
| $4n$ | mlp | predictor_only | 73 | 0.1296 | [0.0356, 0.1867] | 0.2230 | no | 55/0/18 | 0.000 |
| $1n$ | catboost | predictor fit 11 | 99 | 0.5962 | [0.3397, 0.7479] | — | — | — | — |
| $4n$ | catboost | predictor fit 11 | 99 | 0.6374 | [0.3756, 0.7905] | — | — | — | — |
| $1n$ | catboost | predictor fit 23 | 99 | 0.5664 | [0.3364, 0.7239] | — | — | — | — |
| $4n$ | catboost | predictor fit 23 | 99 | 0.6363 | [0.3854, 0.7705] | — | — | — | — |
| $1n$ | catboost | predictor fit 37 | 99 | 0.5497 | [0.3611, 0.7305] | — | — | — | — |
| $4n$ | catboost | predictor fit 37 | 99 | 0.6263 | [0.3547, 0.7705] | — | — | — | — |
| $1n$ | catboost | predictor fit 53 | 99 | 0.6074 | [0.3341, 0.7333] | — | — | — | — |
| $4n$ | catboost | predictor fit 53 | 99 | 0.6164 | [0.3525, 0.7823] | — | — | — | — |
| $1n$ | catboost | predictor fit 71 | 99 | 0.5933 | [0.3386, 0.7163] | — | — | — | — |
| $4n$ | catboost | predictor fit 71 | 99 | 0.6722 | [0.3677, 0.8007] | — | — | — | — |
| $1n$ | catboost | split 2027 | 99 | 0.5300 | [0.3215, 0.7287] | — | — | — | — |
| $4n$ | catboost | split 2027 | 99 | 0.5997 | [0.3654, 0.7784] | — | — | — | — |
| $1n$ | catboost | split 2999 | 100 | 0.5705 | [0.3371, 0.7224] | — | — | — | — |

| Size | Auditor | Control | $n$ | Median diff. or level | Hierarchical CI | Mean | TOST | W/T/L | Holm $p$ |
|---|---|---|---|---|---|---|---|---|---|
| $4n$ | catboost | split 2999 | 100 | 0.6475 | [0.3684, 0.7774] | — | — | — | — |
| $1n$ | catboost | split 4099 | 100 | 0.5623 | [0.3075, 0.7163] | — | — | — | — |
| $4n$ | catboost | split 4099 | 100 | 0.6399 | [0.3621, 0.7742] | — | — | — | — |
| $1n$ | catboost | split 8191 | 100 | 0.6108 | [0.3366, 0.7559] | — | — | — | — |
| $4n$ | catboost | split 8191 | 100 | 0.6767 | [0.3594, 0.7694] | — | — | — | — |

## G FIDELITY AND EMPIRICAL PRIVACY

Table 32 gives the alpha-precision, beta-recall, DCR-share, and DOMIAS columns of the grouped-validation fidelity ledger of the fit-seed-11 population. On the density cells of the same ledger, DOPE has median logistic two-sample AUC $0.517$ (95% lineage-bootstrap interval $0.504$ to $0.532$, $n = 98$), mean marginal Kolmogorov–Smirnov distance $0.114$ ($0.103$ to $0.135$, $n = 98$), and pairwise correlation fidelity $0.953$ ($0.945$ to $0.961$, $n = 98$); each count is the lineages with a complete three-seed fidelity group, and Table 31 prints the same medians with family-cluster intervals. A logistic detector near one half can still miss nonlinear differences, which is why the main paper reports a CatBoost detector. The DCR column of that ledger is the fraction of synthetic queries whose nearest reference is in the fit view, with ties scored one half. Under an independent draw, the unbalanced share sits near the fit-view fraction of the pooled references. The grouped split keeps about four fifths of those rows in the fit view, so the baseline is near four fifths, not one half. The DOMIAS column is the isolated kernel density-ratio equation from the pinned author expression, evaluated on this numeric projection, which differs from the original DOMIAS training pipeline.

Table 30 is the review privacy panel behind Table 3 of the main paper: for every method on the fit-seed-11 samples, a CatBoost two-sample detector with at most 200 rows per class, the median distance to the closest fit row and to the closest validation row, the nearest-neighbour distance ratio against the fit rows, and a distance-based membership AUC. The fit reference holds about four times as many rows as the validation reference, so the distance to the fit rows is smaller by construction. None of these numbers is a differential-privacy guarantee or a HIPAA de-identification claim. Shadow-model attacks, attribute inference, and an attack on the shipped artifact are defined in predeclare_v2 and are scheduled. Table 29 compares every arm's marginal, pairwise, and logistic-detector fidelity on the five-seed cells, and Table 31 adds the stored fidelity of the fit-seed-11 population ledgers.

Table 29: **Fidelity of every arm at $4n$, five-seed cells.** Median over lineages with the 95% nested bootstrap interval of the mean marginal Kolmogorov–Smirnov distance (lower is closer), the pairwise correlation fidelity (higher is closer), and a logistic two-sample AUC (one half is indistinguishable); $n$ counts lineages with a complete three-seed group. A five-seed arm needs three complete fits per lineage. No Holm family.

| Arm | $n$ | KS distance | Pair correlation | Logistic C2ST |
|---|---|---|---|---|
| DOPE (ours) | 100 | $0.117$ $[0.089, 0.165]$ | $0.953$ $[0.930, 0.964]$ | $0.531$ $[0.516, 0.548]$ |
| DOPE, untuned default | 100 | $0.116$ $[0.090, 0.165]$ | $0.952$ $[0.932, 0.962]$ | $0.532$ $[0.512, 0.547]$ |
| TabSyn (scaled schedule) | 100 | $0.126$ $[0.087, 0.141]$ | $0.958$ $[0.946, 0.962]$ | $0.538$ $[0.518, 0.567]$ |
| ARF (author default) | 100 | $0.103$ $[0.084, 0.140]$ | $0.962$ $[0.948, 0.965]$ | $0.521$ $[0.505, 0.538]$ |
| ARF (native selected) | 100 | $0.108$ $[0.084, 0.138]$ | $0.962$ $[0.946, 0.964]$ | $0.521$ $[0.506, 0.539]$ |
| Gaussian copula | 100 | $0.134$ $[0.114, 0.211]$ | $0.963$ $[0.950, 0.969]$ | $0.520$ $[0.502, 0.563]$ |
| Chow-Liu tree | 100 | $0.127$ $[0.085, 0.229]$ | $0.928$ $[0.891, 0.945]$ | $0.555$ $[0.516, 0.625]$ |
| Independent marginals | 100 | $0.130$ $[0.081, 0.267]$ | $0.899$ $[0.826, 0.934]$ | $0.517$ $[0.494, 0.568]$ |
| TVAE (native selected) | 22 | $0.138$ $[0.117, 0.243]$ | $0.926$ $[0.893, 0.945]$ | $0.582$ $[0.574, 0.629]$ |
| CTGAN (native selected) | 22 | $0.230$ $[0.184, 0.301]$ | $0.912$ $[0.837, 0.943]$ | $0.770$ $[0.667, 0.858]$ |
| Forest-Flow (historical) | 6 | $0.123$ $[0.056, 0.206]$ | $0.969$ $[0.932, 0.981]$ | $0.492$ $[0.462, 0.540]$ |
| Predictor-only control | 100 | $0.105$ $[0.084, 0.141]$ | $0.945$ $[0.882, 0.961]$ | $0.510$ $[0.497, 0.527]$ |
| Real rows resampled | 100 | $0.080$ $[0.069, 0.106]$ | $0.964$ $[0.954, 0.970]$ | $0.503$ $[0.482, 0.530]$ |

Table 24: **Forest-Flow confirmation at $4n$, CatBoost, fit seed 11.** The denominator is 6 lineages. Holm family size is three.

| Lineage | DOPE bytes | DOPE | Forest-Flow bytes | Forest-Flow |
|---|---|---|---|---|
| 1193_BNG_lowbwt | 2,436 | 0.970 | 416,921,273 | 1.004 |
| 503_wind | 3,846 | 0.996 | 637,056,689 | 1.016 |
| 582_fri_c1_500_25 | 5,670 | 0.909 | 1,085,155,062 | 0.883 |
| 596_fri_c2_250_5 | 1,698 | 0.807 | 234,776,554 | 0.968 |
| feynman_I_12_1 | 1,032 | 0.991 | 123,784,681 | 1.001 |
| strogatz_vdp1 | 984 | 0.581 | 115,202,253 | 1.036 |

Table 25: **Published ARF matched receipts at $4n$, CatBoost.** The denominator is the paired complete informative lineages in the generated row. No Holm family in this table.

| Publication | Lineages | DOPE | Receipt | Difference |
|---|---|---|---|---|
| ARF, author_default | 97 | 0.940 | 0.840 | 0.077 |
| ARF, native_selected | 97 | 0.940 | 0.826 | 0.095 |

Table 30: **Holdout privacy and CatBoost detection, fit seed 11.** Each value is the median over lineages of each lineage's median over three sample seeds, with the family-cluster bootstrap interval. Own rows use each method's lineages whose synthetic row count equals fit rows times size; matched-density rows use the lineages shared by DOPE, the Gaussian copula, Chow–Liu, and independent marginals. Fit DCR and NNDR use the fit rows as the reference set and holdout DCR the validation rows, about a quarter as many. Distance MIA AUC is an empirical membership attack, and c2st_catboost_auc is the CatBoost detector. These are empirical attack scores, not formal differential privacy and not HIPAA de-identification. Unavailable cells are no wins: CTGAN 468 (sample_csv_unresolved); DOPE 12 (sample_csv_unresolved); ForestDiffusion/Forest-Flow 36 (sample_csv_absent); TVAE 468 (sample_csv_unresolved). No Holm family.

| Method | Configuration | Size | Metric | $n$ lineages | Median | Hierarchical CI |
|---|---|---|---|---|---|---|
| ARF | author_default | $1n$ | dcr_validation_median | 100 | 0.1095 | [0.0532, 0.1639] |
| ARF | author_default | $1n$ | dcr_fit_median | 100 | 0.0826 | [0.0332, 0.1257] |
| ARF | author_default | $1n$ | nndr_fit_median | 100 | 0.8983 | [0.8617, 0.9278] |
| ARF | author_default | $1n$ | distance_mia_auc | 100 | 0.5452 | [0.5298, 0.5580] |
| ARF | author_default | $1n$ | c2st_catboost_auc | 100 | 0.6159 | [0.5751, 0.6928] |
| ARF | author_default | $4n$ | dcr_validation_median | 100 | 0.1088 | [0.0517, 0.1617] |
| ARF | author_default | $4n$ | dcr_fit_median | 100 | 0.0820 | [0.0331, 0.1253] |
| ARF | author_default | $4n$ | nndr_fit_median | 100 | 0.9069 | [0.8639, 0.9282] |
| ARF | author_default | $4n$ | distance_mia_auc | 100 | 0.5563 | [0.5358, 0.5754] |
| ARF | author_default | $4n$ | c2st_catboost_auc | 100 | 0.6120 | [0.5696, 0.6877] |
| ARF | native_selected | $1n$ | dcr_validation_median | 100 | 0.1108 | [0.0544, 0.1637] |
| ARF | native_selected | $1n$ | dcr_fit_median | 100 | 0.0823 | [0.0361, 0.1316] |
| ARF | native_selected | $1n$ | nndr_fit_median | 100 | 0.9079 | [0.8687, 0.9271] |
| ARF | native_selected | $1n$ | distance_mia_auc | 100 | 0.5422 | [0.5309, 0.5576] |
| ARF | native_selected | $1n$ | c2st_catboost_auc | 100 | 0.6100 | [0.5789, 0.7089] |
| ARF | native_selected | $4n$ | dcr_validation_median | 100 | 0.1109 | [0.0547, 0.1635] |
| ARF | native_selected | $4n$ | dcr_fit_median | 100 | 0.0825 | [0.0389, 0.1295] |
| ARF | native_selected | $4n$ | nndr_fit_median | 100 | 0.9107 | [0.8647, 0.9281] |
| ARF | native_selected | $4n$ | distance_mia_auc | 100 | 0.5408 | [0.5303, 0.5562] |
| ARF | native_selected | $4n$ | c2st_catboost_auc | 100 | 0.6091 | [0.5766, 0.7075] |
| CTGAN | native_selected | $1n$ | dcr_validation_median | 22 | 0.1863 | [0.1163, 0.2214] |
| CTGAN | native_selected | $1n$ | dcr_fit_median | 22 | 0.1538 | [0.0903, 0.1899] |
| CTGAN | native_selected | $1n$ | nndr_fit_median | 22 | 0.9395 | [0.9106, 0.9610] |
| CTGAN | native_selected | $1n$ | distance_mia_auc | 22 | 0.4804 | [0.4514, 0.5101] |
| CTGAN | native_selected | $1n$ | c2st_catboost_auc | 21 | 0.9522 | [0.8778, 0.9841] |
| CTGAN | native_selected | $4n$ | dcr_validation_median | 22 | 0.1869 | [0.1157, 0.2225] |
| CTGAN | native_selected | $4n$ | dcr_fit_median | 22 | 0.1529 | [0.0867, 0.1906] |
| CTGAN | native_selected | $4n$ | nndr_fit_median | 22 | 0.9413 | [0.9117, 0.9581] |
| CTGAN | native_selected | $4n$ | distance_mia_auc | 22 | 0.5024 | [0.4721, 0.5168] |
| CTGAN | native_selected | $4n$ | c2st_catboost_auc | 22 | 0.9535 | [0.8487, 0.9777] |
| Chow-Liu | native_selected | $1n$ | dcr_validation_median | 100 | 0.1334 | [0.0976, 0.2006] |
| Chow-Liu | native_selected | $1n$ | dcr_fit_median | 100 | 0.1078 | [0.0786, 0.1671] |
| Chow-Liu | native_selected | $1n$ | nndr_fit_median | 100 | 0.9228 | [0.8889, 0.9638] |
| Chow-Liu | native_selected | $1n$ | distance_mia_auc | 100 | 0.5077 | [0.4962, 0.5202] |
| Chow-Liu | native_selected | $1n$ | c2st_catboost_auc | 100 | 0.6868 | [0.6025, 0.8986] |
| Chow-Liu | native_selected | $4n$ | dcr_validation_median | 100 | 0.1291 | [0.1053, 0.1901] |
| Chow-Liu | native_selected | $4n$ | dcr_fit_median | 100 | 0.1080 | [0.0858, 0.1661] |
| Chow-Liu | native_selected | $4n$ | nndr_fit_median | 100 | 0.9238 | [0.8883, 0.9644] |
| Chow-Liu | native_selected | $4n$ | distance_mia_auc | 100 | 0.5196 | [0.5037, 0.5350] |
| Chow-Liu | native_selected | $4n$ | c2st_catboost_auc | 99 | 0.7061 | [0.5910, 0.9369] |
| DOPE | features12_steps2048 | $1n$ | dcr_validation_median | 98 | 0.1054 | [0.0529, 0.1560] |
| DOPE | features12_steps2048 | $1n$ | dcr_fit_median | 98 | 0.0800 | [0.0423, 0.1254] |
| DOPE | features12_steps2048 | $1n$ | nndr_fit_median | 98 | 0.9040 | [0.8465, 0.9368] |
| DOPE | features12_steps2048 | $1n$ | distance_mia_auc | 98 | 0.5132 | [0.5012, 0.5249] |
| DOPE | features12_steps2048 | $1n$ | c2st_catboost_auc | 98 | 0.6344 | [0.5705, 0.8368] |

| Method | Configuration | Size | Metric | $n$ lineages | Median | Hierarchical CI |
|---|---|---|---|---|---|---|
| DOPE | features12_steps2048 | $4n$ | dcr_validation_median | 98 | 0.1086 | [0.0528, 0.1521] |
| DOPE | features12_steps2048 | $4n$ | dcr_fit_median | 98 | 0.0797 | [0.0421, 0.1280] |
| DOPE | features12_steps2048 | $4n$ | nndr_fit_median | 98 | 0.9053 | [0.8482, 0.9360] |
| DOPE | features12_steps2048 | $4n$ | distance_mia_auc | 98 | 0.5052 | [0.4921, 0.5160] |
| DOPE | features12_steps2048 | $4n$ | c2st_catboost_auc | 98 | 0.6405 | [0.5830, 0.8312] |
| GaussianCopula | native_selected | $1n$ | dcr_validation_median | 100 | 0.1094 | [0.0780, 0.1543] |
| GaussianCopula | native_selected | $1n$ | dcr_fit_median | 100 | 0.0868 | [0.0621, 0.1268] |
| GaussianCopula | native_selected | $1n$ | nndr_fit_median | 100 | 0.9177 | [0.8814, 0.9603] |
| GaussianCopula | native_selected | $1n$ | distance_mia_auc | 100 | 0.5050 | [0.4960, 0.5293] |
| GaussianCopula | native_selected | $1n$ | c2st_catboost_auc | 100 | 0.7343 | [0.6789, 0.8981] |
| GaussianCopula | native_selected | $4n$ | dcr_validation_median | 100 | 0.1074 | [0.0774, 0.1563] |
| GaussianCopula | native_selected | $4n$ | dcr_fit_median | 100 | 0.0863 | [0.0627, 0.1280] |
| GaussianCopula | native_selected | $4n$ | nndr_fit_median | 100 | 0.9201 | [0.8867, 0.9610] |
| GaussianCopula | native_selected | $4n$ | distance_mia_auc | 100 | 0.5022 | [0.4851, 0.5158] |
| GaussianCopula | native_selected | $4n$ | c2st_catboost_auc | 99 | 0.7346 | [0.6902, 0.8689] |
| TVAE | native_selected | $1n$ | dcr_validation_median | 22 | 0.1122 | [0.0853, 0.1943] |
| TVAE | native_selected | $1n$ | dcr_fit_median | 22 | 0.0815 | [0.0533, 0.1535] |
| TVAE | native_selected | $1n$ | nndr_fit_median | 22 | 0.9056 | [0.8479, 0.9264] |
| TVAE | native_selected | $1n$ | distance_mia_auc | 22 | 0.5312 | [0.4992, 0.5585] |
| TVAE | native_selected | $1n$ | c2st_catboost_auc | 22 | 0.7479 | [0.7221, 0.8859] |
| TVAE | native_selected | $4n$ | dcr_validation_median | 22 | 0.1118 | [0.0855, 0.1938] |
| TVAE | native_selected | $4n$ | dcr_fit_median | 22 | 0.0799 | [0.0535, 0.1516] |
| TVAE | native_selected | $4n$ | nndr_fit_median | 22 | 0.9058 | [0.8489, 0.9312] |
| TVAE | native_selected | $4n$ | distance_mia_auc | 22 | 0.5315 | [0.4915, 0.5424] |
| TVAE | native_selected | $4n$ | c2st_catboost_auc | 22 | 0.7309 | [0.7032, 0.8893] |
| independent_marginals | native_selected | $1n$ | dcr_validation_median | 100 | 0.1360 | [0.1010, 0.1877] |
| independent_marginals | native_selected | $1n$ | dcr_fit_median | 100 | 0.1132 | [0.0890, 0.1587] |
| independent_marginals | native_selected | $1n$ | nndr_fit_median | 100 | 0.9457 | [0.9120, 0.9757] |
| independent_marginals | native_selected | $1n$ | distance_mia_auc | 100 | 0.4982 | [0.4883, 0.5096] |
| independent_marginals | native_selected | $1n$ | c2st_catboost_auc | 99 | 0.8694 | [0.7602, 0.9749] |
| independent_marginals | native_selected | $4n$ | dcr_validation_median | 100 | 0.1319 | [0.0995, 0.1848] |
| independent_marginals | native_selected | $4n$ | dcr_fit_median | 100 | 0.1134 | [0.0886, 0.1580] |
| independent_marginals | native_selected | $4n$ | nndr_fit_median | 100 | 0.9453 | [0.9141, 0.9702] |
| independent_marginals | native_selected | $4n$ | distance_mia_auc | 100 | 0.5081 | [0.4906, 0.5180] |
| independent_marginals | native_selected | $4n$ | c2st_catboost_auc | 100 | 0.8578 | [0.7553, 0.9727] |
| DOPE | matched density | $1n$ | dcr_validation_median | 98 | 0.1054 | [0.0529, 0.1560] |
| GaussianCopula | matched density | $1n$ | dcr_validation_median | 98 | 0.1094 | [0.0770, 0.1543] |
| Chow-Liu | matched density | $1n$ | dcr_validation_median | 98 | 0.1304 | [0.0976, 0.1913] |
| independent_marginals | matched density | $1n$ | dcr_validation_median | 98 | 0.1360 | [0.1000, 0.1872] |
| DOPE | matched density | $1n$ | dcr_fit_median | 98 | 0.0800 | [0.0423, 0.1254] |
| GaussianCopula | matched density | $1n$ | dcr_fit_median | 98 | 0.0859 | [0.0617, 0.1268] |
| Chow-Liu | matched density | $1n$ | dcr_fit_median | 98 | 0.1075 | [0.0786, 0.1613] |
| independent_marginals | matched density | $1n$ | dcr_fit_median | 98 | 0.1128 | [0.0890, 0.1578] |
| DOPE | matched density | $1n$ | nndr_fit_median | 98 | 0.9040 | [0.8495, 0.9368] |
| GaussianCopula | matched density | $1n$ | nndr_fit_median | 98 | 0.9166 | [0.8814, 0.9603] |
| Chow-Liu | matched density | $1n$ | nndr_fit_median | 98 | 0.9219 | [0.8888, 0.9638] |
| independent_marginals | matched density | $1n$ | nndr_fit_median | 98 | 0.9422 | [0.9120, 0.9729] |
| DOPE | matched density | $1n$ | distance_mia_auc | 98 | 0.5132 | [0.5012, 0.5249] |
| GaussianCopula | matched density | $1n$ | distance_mia_auc | 98 | 0.5035 | [0.4949, 0.5235] |
| Chow-Liu | matched density | $1n$ | distance_mia_auc | 98 | 0.5077 | [0.4958, 0.5203] |
| independent_marginals | matched density | $1n$ | distance_mia_auc | 98 | 0.4982 | [0.4883, 0.5070] |
| DOPE | matched density | $1n$ | c2st_catboost_auc | 97 | 0.6234 | [0.5705, 0.8368] |
| GaussianCopula | matched density | $1n$ | c2st_catboost_auc | 97 | 0.7296 | [0.6596, 0.9162] |
| Chow-Liu | matched density | $1n$ | c2st_catboost_auc | 97 | 0.6866 | [0.6055, 0.8986] |
| independent_marginals | matched density | $1n$ | c2st_catboost_auc | 97 | 0.8694 | [0.7656, 0.9749] |
| DOPE | matched density | $4n$ | dcr_validation_median | 98 | 0.1086 | [0.0528, 0.1521] |
| GaussianCopula | matched density | $4n$ | dcr_validation_median | 98 | 0.1074 | [0.0766, 0.1554] |
| Chow-Liu | matched density | $4n$ | dcr_validation_median | 98 | 0.1262 | [0.1020, 0.1874] |
| independent_marginals | matched density | $4n$ | dcr_validation_median | 98 | 0.1308 | [0.0995, 0.1841] |
| DOPE | matched density | $4n$ | dcr_fit_median | 98 | 0.0797 | [0.0404, 0.1280] |
| GaussianCopula | matched density | $4n$ | dcr_fit_median | 98 | 0.0859 | [0.0627, 0.1273] |
| Chow-Liu | matched density | $4n$ | dcr_fit_median | 98 | 0.1042 | [0.0853, 0.1642] |
| independent_marginals | matched density | $4n$ | dcr_fit_median | 98 | 0.1124 | [0.0884, 0.1569] |
| DOPE | matched density | $4n$ | nndr_fit_median | 98 | 0.9053 | [0.8492, 0.9360] |
| GaussianCopula | matched density | $4n$ | nndr_fit_median | 98 | 0.9198 | [0.8858, 0.9603] |
| Chow-Liu | matched density | $4n$ | nndr_fit_median | 98 | 0.9221 | [0.8883, 0.9644] |
| independent_marginals | matched density | $4n$ | nndr_fit_median | 98 | 0.9435 | [0.9119, 0.9702] |
| DOPE | matched density | $4n$ | distance_mia_auc | 98 | 0.5052 | [0.4919, 0.5160] |
| GaussianCopula | matched density | $4n$ | distance_mia_auc | 98 | 0.5004 | [0.4850, 0.5152] |
| Chow-Liu | matched density | $4n$ | distance_mia_auc | 98 | 0.5196 | [0.5023, 0.5380] |
| independent_marginals | matched density | $4n$ | distance_mia_auc | 98 | 0.5077 | [0.4891, 0.5176] |
| DOPE | matched density | $4n$ | c2st_catboost_auc | 96 | 0.6465 | [0.5863, 0.8508] |
| GaussianCopula | matched density | $4n$ | c2st_catboost_auc | 96 | 0.7348 | [0.6902, 0.8804] |
| Chow-Liu | matched density | $4n$ | c2st_catboost_auc | 96 | 0.7062 | [0.6017, 0.9302] |
| independent_marginals | matched density | $4n$ | c2st_catboost_auc | 96 | 0.8578 | [0.7555, 0.9727] |

Table 26: **Raw auditor medians on the informative lineages, fit seed 11.** The denominator is the informative lineages for that auditor. No Holm family.

| Auditor | $n$ | RMSE real | RMSE synth. | training-mean skill, real | training-mean skill, synth. |
|---|---|---|---|---|---|
| CatBoost | 97 | 0.053 | 0.081 | 0.881 | 0.792 |
| Linear | 92 | 0.105 | 0.112 | 0.585 | 0.551 |
| MLP | 75 | 0.098 | 0.083 | 0.666 | 0.785 |

Table 27: **Charged bytes, fit seed 11.** For DOPE artifacts the charge is the encoded model file plus `projection.json`. The DOPE, density, Forest-Flow, and ARF rows count every artifact with a stored charge; the CTGAN and TVAE rows count native-selected fits whose three $4n$ sample cells all succeeded, so main Table 1, which counts every charged lineage, prints different medians for them. No Holm family.

| Method | Median bytes | Within cap |
|---|---|---|
| DOPE reference | 1,895.5 | 98/100 |
| Gaussian copula | 5,464.5 | 62/100 |
| Chow–Liu | 29,865 | 31/100 |
| Independent marginals | 2,549.5 | 94/100 |
| CTGAN | 741,834 | 0/22 |
| TVAE | 317,141.5 | 0/22 |
| Forest-Flow | 325,848,913.5 | 0/6 |
| ARF (native selected) | 1,627,718.5 | 0/100 |

Table 31: **Stored fidelity from the population ledgers, fit seed 11.** Logistic two-sample AUC, mean marginal Kolmogorov–Smirnov distance, and pair-correlation fidelity; $n$ counts lineages with three finite stored values, with the family-cluster bootstrap interval. Cohorts differ across methods, so the rows are no ranking. The stored detector is logistic, not CatBoost, and TabSyn is absent from these ledgers. No Holm family.

| Method | Configuration | Size | Metric | $n$ lineages | Median | Hierarchical CI |
|---|---|---|---|---|---|---|
| DOPE | features12_steps2048 | $1n$ | stored logistic C2ST | 98 | 0.5172 | [0.5000, 0.5376] |
| DOPE | features12_steps2048 | $1n$ | marginal KS | 98 | 0.1216 | [0.0950, 0.1680] |
| DOPE | features12_steps2048 | $1n$ | pair correlation | 98 | 0.9532 | [0.9317, 0.9610] |
| DOPE | features12_steps2048 | $4n$ | stored logistic C2ST | 98 | 0.5172 | [0.5000, 0.5376] |
| DOPE | features12_steps2048 | $4n$ | marginal KS | 98 | 0.1138 | [0.0917, 0.1583] |
| DOPE | features12_steps2048 | $4n$ | pair correlation | 98 | 0.9531 | [0.9346, 0.9645] |
| GaussianCopula | native_selected | $1n$ | stored logistic C2ST | 100 | 0.5246 | [0.5100, 0.5518] |
| GaussianCopula | native_selected | $1n$ | marginal KS | 100 | 0.1393 | [0.1171, 0.2132] |
| GaussianCopula | native_selected | $1n$ | pair correlation | 100 | 0.9614 | [0.9516, 0.9652] |
| GaussianCopula | native_selected | $4n$ | stored logistic C2ST | 100 | 0.5246 | [0.5100, 0.5537] |
| GaussianCopula | native_selected | $4n$ | marginal KS | 100 | 0.1336 | [0.1137, 0.2115] |
| GaussianCopula | native_selected | $4n$ | pair correlation | 100 | 0.9635 | [0.9488, 0.9687] |
| Chow-Liu | native_selected | $1n$ | stored logistic C2ST | 100 | 0.5423 | [0.5040, 0.6292] |
| Chow-Liu | native_selected | $1n$ | marginal KS | 100 | 0.1293 | [0.0892, 0.2361] |
| Chow-Liu | native_selected | $1n$ | pair correlation | 100 | 0.9273 | [0.8834, 0.9418] |
| Chow-Liu | native_selected | $4n$ | stored logistic C2ST | 100 | 0.5404 | [0.5185, 0.6303] |
| Chow-Liu | native_selected | $4n$ | marginal KS | 100 | 0.1266 | [0.0852, 0.2292] |
| Chow-Liu | native_selected | $4n$ | pair correlation | 100 | 0.9283 | [0.8906, 0.9465] |
| independent_marginals | native_selected | $1n$ | stored logistic C2ST | 100 | 0.5213 | [0.5000, 0.5347] |
| independent_marginals | native_selected | $1n$ | marginal KS | 100 | 0.1363 | [0.0866, 0.2847] |
| independent_marginals | native_selected | $1n$ | pair correlation | 100 | 0.8994 | [0.8099, 0.9289] |
| independent_marginals | native_selected | $4n$ | stored logistic C2ST | 100 | 0.5166 | [0.4988, 0.5488] |
| independent_marginals | native_selected | $4n$ | marginal KS | 100 | 0.1297 | [0.0813, 0.2672] |
| independent_marginals | native_selected | $4n$ | pair correlation | 100 | 0.8986 | [0.8258, 0.9329] |
| CTGAN | native_selected | $1n$ | stored logistic C2ST | 22 | 0.7839 | [0.6997, 0.8819] |
| CTGAN | native_selected | $1n$ | marginal KS | 22 | 0.2319 | [0.1887, 0.2943] |
| CTGAN | native_selected | $1n$ | pair correlation | 22 | 0.9091 | [0.8313, 0.9406] |
| CTGAN | native_selected | $4n$ | stored logistic C2ST | 22 | 0.7839 | [0.6997, 0.8821] |
| CTGAN | native_selected | $4n$ | marginal KS | 22 | 0.2300 | [0.1839, 0.2947] |
| CTGAN | native_selected | $4n$ | pair correlation | 22 | 0.9123 | [0.8367, 0.9431] |
| TVAE | native_selected | $1n$ | stored logistic C2ST | 22 | 0.5918 | [0.5616, 0.6392] |
| TVAE | native_selected | $1n$ | marginal KS | 22 | 0.1500 | [0.1199, 0.2598] |
| TVAE | native_selected | $1n$ | pair correlation | 22 | 0.9240 | [0.8905, 0.9418] |
| TVAE | native_selected | $4n$ | stored logistic C2ST | 22 | 0.6033 | [0.5544, 0.6681] |
| TVAE | native_selected | $4n$ | marginal KS | 22 | 0.1382 | [0.1174, 0.2431] |
| TVAE | native_selected | $4n$ | pair correlation | 22 | 0.9258 | [0.8913, 0.9451] |
| ARF | author_default | $1n$ | stored logistic C2ST | 100 | 0.5321 | [0.5156, 0.5442] |
| ARF | author_default | $1n$ | marginal KS | 100 | 0.1128 | [0.0895, 0.1433] |
| ARF | author_default | $1n$ | pair correlation | 100 | 0.9595 | [0.9471, 0.9644] |
| ARF | author_default | $4n$ | stored logistic C2ST | 100 | 0.5268 | [0.5069, 0.5388] |
| ARF | author_default | $4n$ | marginal KS | 100 | 0.1035 | [0.0846, 0.1340] |
| ARF | author_default | $4n$ | pair correlation | 100 | 0.9618 | [0.9487, 0.9653] |

| Method | Configuration | Size | Metric | $n$ lineages | Median | Hierarchical CI |
|---|---|---|---|---|---|---|
| ARF | native_selected | $1n$ | stored logistic C2ST | 100 | 0.5255 | [0.5072, 0.5449] |
| ARF | native_selected | $1n$ | marginal KS | 100 | 0.1122 | [0.0869, 0.1445] |
| ARF | native_selected | $1n$ | pair correlation | 100 | 0.9599 | [0.9418, 0.9626] |
| ARF | native_selected | $4n$ | stored logistic C2ST | 100 | 0.5296 | [0.5094, 0.5375] |
| ARF | native_selected | $4n$ | marginal KS | 100 | 0.1046 | [0.0845, 0.1382] |
| ARF | native_selected | $4n$ | pair correlation | 100 | 0.9620 | [0.9464, 0.9643] |
| ForestDiffusion/Forest-Flow | native_selected | $1n$ | stored logistic C2ST | 6 | 0.5187 | [0.4139, 0.7000] |
| ForestDiffusion/Forest-Flow | native_selected | $1n$ | marginal KS | 6 | 0.1254 | [0.0612, 0.1992] |
| ForestDiffusion/Forest-Flow | native_selected | $1n$ | pair correlation | 6 | 0.9548 | [0.9242, 0.9799] |
| ForestDiffusion/Forest-Flow | native_selected | $4n$ | stored logistic C2ST | 6 | 0.5187 | [0.4139, 0.7000] |
| ForestDiffusion/Forest-Flow | native_selected | $4n$ | marginal KS | 6 | 0.1231 | [0.0557, 0.2062] |
| ForestDiffusion/Forest-Flow | native_selected | $4n$ | pair correlation | 6 | 0.9691 | [0.9321, 0.9811] |

### G.1 Configuration-aware fidelity and detectors

The side-by-side columns use the common numeric projection and the published pilot KS and pair-fidelity formulas, including the projected target. Alpha-precision and beta-recall use the fixed unembedded equations. The stored logistic detector remains distinct from the shared grouped CatBoost detector: five disjoint row-group folds, the fixed iteration and depth budget, and the predeclared equal per-class cap. Every metric gives its own complete-three-seed lineage count and family-cluster interval. These marginal summaries use different availability cohorts and establish no cross-method ranking. ARF author-default and native-selected configurations stay separate. Forest-Flow and originally unmeasured size-$n$ alpha/beta entries retain unavailable cells. The retained mixed-type TabSyn alpha/beta cohort remains separate from the common-numeric cohort. No Holm family is applied to these descriptive tables.

## H Ablations and replay

**Step budget.** `features12_steps8192` is the same residual with four times as many AdamW steps. Where both budgets have a complete three-seed group, the CatBoost median paired difference, longer budget minus shorter, is $0.034$ ($68$ lineages); the linear medians are $0.990$ and $0.987$ ($64$ lineages; difference $-0.000283$) and the MLP medians $1.065$ and $1.057$ ($52$ lineages; difference $0.00451$). Appendix J.2 lists every lineage. **Profile and default.** The main paper's untuned trainer default is a second axis (width and step budget together). The full width, step, readout, screening-size, and byte-budget grid, and the interaction stress suite (XOR, parity, products, many relevant features), are defined in predeclare_v2 and scheduled.

**Replay.** The scored runs did not store per-step losses; a later logged replay of both step budgets, before the readout refit, did. All $100$ lineages finished. Median training loss starts at $0.321$ and ends at $0.00274$ and $0.000763$; median validation loss starts at $0.271$ and ends at $0.00443$ and $0.00181$. The replay files are distinct from the scored artifacts.

## I Baseline coverage and retained evidence

Table 35 records which baseline publications exist and on how many lineages. The ARF follow-on supplies complete five-fit measurements on the same validation lineages. TabSyn's full scaled-schedule panel is in the main paper; the earlier eight-lineage TabSyn subset below is a different cohort. Forest-Flow confirmation is distinct from the Forest-Diffusion pilot modes, and no full-population diffusion result is inferred from the Flow subset. Receipt paths and digests are in `docs/whitepaper/generated/baseline-coverage.json` and `docs/whitepaper/generated/retained-evidence.json`.

**Retained validation.** Tables 36 and 37 add the merged retained-sample evidence. Each lineage averages all three sample seeds before a lineage-bootstrap median, which differs from the median-of-seeds rule of the main tables. Table 36 covers ARF and the copula and Chow–Liu defaults and native configurations; Table 37 includes only the eight complete scaled-schedule TabSyn lineages of an earlier run and the classical methods on identical inputs. Table 38 gives exploratory TabSyn-minus-comparator differences on those lineages, with conservative simultaneous intervals (Bonferroni over four metrics times six configurations), which are

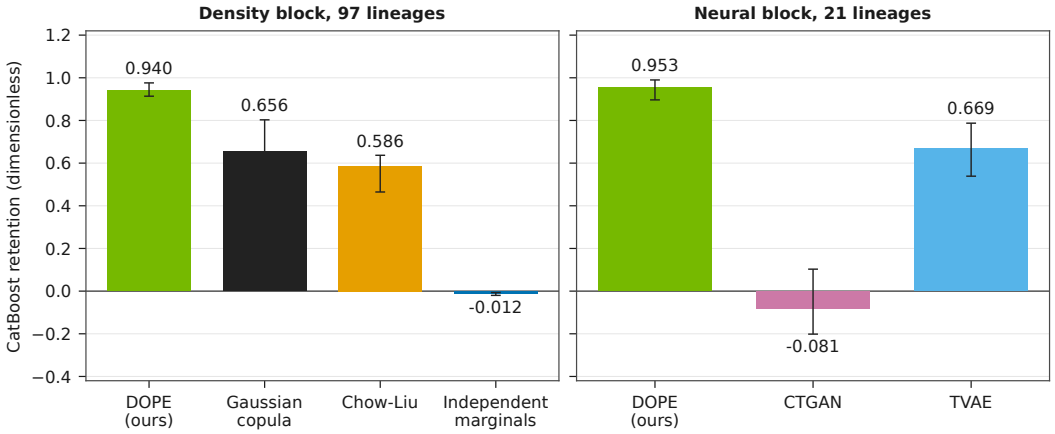



Figure 4: **CatBoost retention at $4n$, fit seed 11.** Each bar is a method's paired-lineage retention median and its lineage-bootstrap interval. Density uses 97 lineages and the neural panel 21; the panels are separate cohorts. No Holm family.

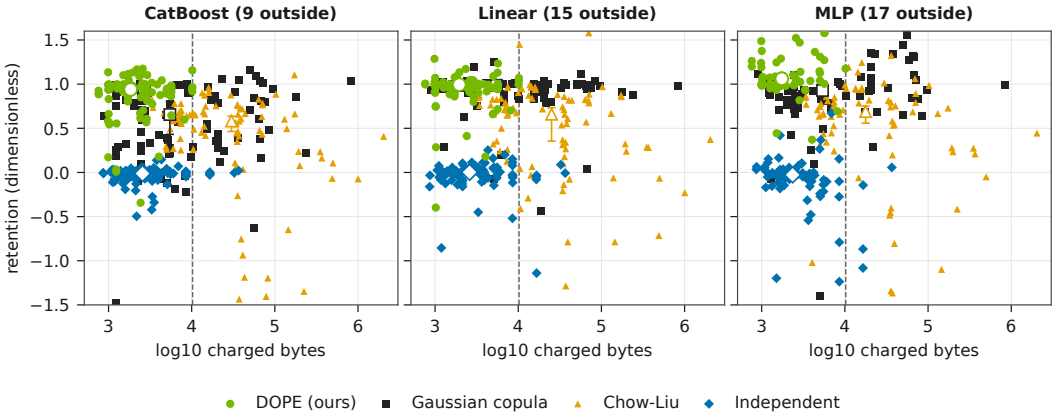



Figure 5: **Density-block retention at $4n$ against charged bytes, per lineage, fit seed 11.** Each point is one lineage's retention level (vertical) against its charged bytes on a log axis. Values outside $[-1.5, 1.6]$ are counted in the panel title and omitted from the frame. The dashed line is the 10,240-byte cap. No Holm family.

unbounded at this count. Table 39 extends the published DOPE–ARF comparison with lineage intervals under its six-test family.

**First-batch diagnostics.** The canonical-job batch below is incomplete at population and sample-group level, so its values are individual physical evaluations without a population interval or ranking. DOPE jobs in this batch use frozen research profiles that differ from the reported profile. Pearson entries are the pinned source mean, alpha and beta use its pinned unembedded variant, and MIA is an empirical distance attack. Per-column marginals, detection folds, DCR and NNDR, costs, and job digests are in the first-batch source listed in `docs/whitepaper/generated/retained-evidence.json`.

**BeyondArena** (`2ecfe882ccfb814fc27c4de10a64ceefd5d7655c`) is a separate fit check outside the 100 PMLB lineages. 5 of the 12 families in that check finished inside the cap, and 7 did not prepare (4 outside the projected range, 3 overlapping); the median elapsed times per family are 5.00 seconds for `steps2048` and 14.0 seconds for `steps8192`. The inventory of all 142 families is in Appendix J.3. No retention auditor ran on BeyondArena.

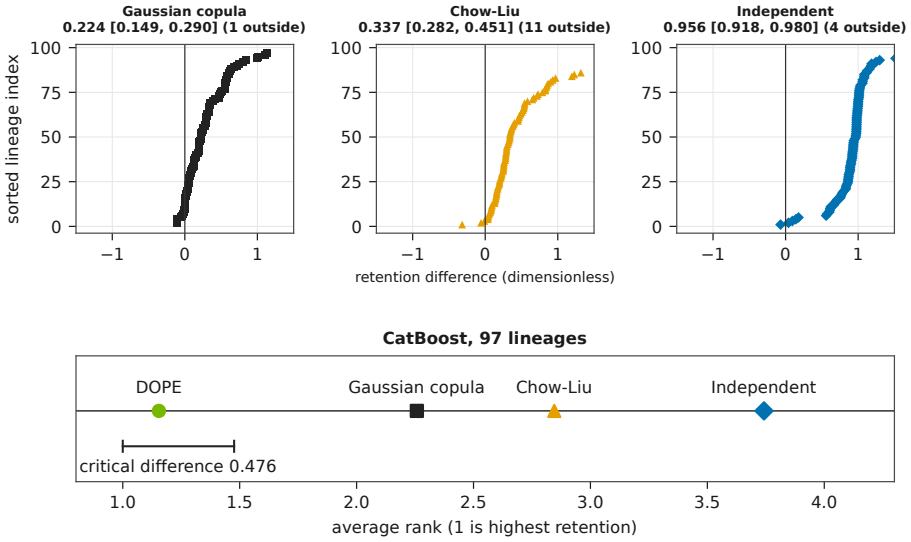



Figure 6: **CatBoost paired differences, sorted, and the Nemenyi scale, fit seed 11.** Rank 1 is the higher retention; the segment length is the critical difference. This display is outside the nine-test Holm family.

Table 32: **Grouped-validation fidelity and empirical privacy, fit seed 11.** Each cell is a median with its 95% lineage-bootstrap interval. The fidelity denominator is the complete three-seed group count, distinct from the 97 retention lineages. Holm families are 12 density tests and 8 neural tests, separate from retention.

| Method | $\alpha$-prec. | $\beta$-recall | DCR share (unbalanced) | DOMIAS |
|---|---|---|---|---|
| *Median [interval] (lineages).* | | | | |
| DOPE | 0.905 [0.889, 0.918] (98) | 0.437 [0.398, 0.473] (98) | 0.811 [0.806, 0.814] (98) | 0.508 [0.498, 0.520] (94) |
| Gaussian | 0.812 [0.786, 0.832] (100) | 0.430 [0.378, 0.453] (100) | 0.806 [0.802, 0.811] (100) | 0.513 [0.494, 0.527] (94) |
| Chow–Liu | 0.858 [0.825, 0.904] (100) | 0.359 [0.312, 0.400] (100) | 0.810 [0.806, 0.816] (100) | 0.512 [0.495, 0.522] (94) |
| Indep. | 0.885 [0.868, 0.897] (100) | 0.284 [0.197, 0.347] (100) | 0.801 [0.797, 0.806] (100) | 0.507 [0.496, 0.521] (94) |
| DOPE | 0.902 [0.882, 0.940] (22) | 0.441 [0.381, 0.489] (22) | 0.811 [0.802, 0.819] (22) | 0.508 [0.480, 0.538] (22) |
| CTGAN | 0.719 [0.520, 0.856] (22) | 0.170 [0.142, 0.267] (22) | 0.820 [0.800, 0.842] (22) | 0.502 [0.463, 0.534] (22) |
| TVAE | 0.850 [0.731, 0.896] (22) | 0.356 [0.283, 0.401] (22) | 0.825 [0.813, 0.843] (22) | 0.527 [0.487, 0.563] (22) |

| Contrast | $\alpha$-prec. | $\beta$-recall | DCR share (unbalanced) | DOMIAS |
|---|---|---|---|---|
| *Line 2: Holm p; rank-biserial r; n. DOPE minus comparator.* | | | | |
| | 0.083 [0.045, 0.113] | 0.024 [0.011, 0.039] | -0.0025 [-0.00625, 0.005] | 0.00157 [-0.0069, 0.0074] |
| DOPE−Gaussian | $1.86 \times 10^{-10}$; 0.781; 98 | 0.394; 0.215; 98 | 1.000; 0.000206; 98 | 1.000; -0.085; 94 |
| | 0.019 [0.00782, 0.056] | 0.048 [0.029, 0.072] | -0.00125 [-0.00562, 0.00125] | 0.000581 [-0.00802, 0.011] |
| DOPE−Chow–Liu | $2.00 \times 10^{-5}$; 0.551; 98 | $1.32 \times 10^{-10}$; 0.789; 98 | 0.806; -0.169; 98 | 1.000; 0.00572; 94 |
| | 0.018 [0.0072, 0.028] | 0.104 [0.068, 0.138] | 0.00625 [0.0025, 0.011] | 0.00244 [-0.00595, 0.015] |
| DOPE−Indep. | $5.47 \times 10^{-5}$; 0.523; 98 | $9.71 \times 10^{-15}$; 0.937; 98 | 0.005; 0.398; 98 | 1.000; 0.080; 94 |
| | 0.168 [0.054, 0.368] | 0.219 [0.131, 0.256] | -0.00812 [-0.029, 0.015] | -0.00028 [-0.014, 0.031] |
| DOPE−CTGAN | 0.004; 0.794; 22 | $3.81 \times 10^{-6}$; 1.000; 22 | 0.532; -0.277; 22 | 0.610; 0.130; 22 |
| | 0.071 [0.00713, 0.141] | 0.062 [0.00546, 0.135] | -0.025 [-0.030, -0.00641] | -0.021 [-0.036, -0.00972] |
| DOPE−TVAE | 0.040; 0.644; 22 | 0.069; 0.573; 22 | 0.059; -0.613; 22 | 0.108; -0.510; 22 |

## J PER-LINEAGE TABLES

The tables below list every prepared lineage of the fit-seed-11 population, the sufficiency budget, and the BeyondArena inventory. They are the rows behind the medians of the historical block tables, and a blank cell is an unavailable fit or an uninformative auditor rather than a zero. The five-seed v2 cells are in the committed scalar ledger `research/benchmark/results/review-fixes-v2/scalar-cells.jsonl`, which holds one row per fit, sample seed, size, and auditor for every arm.

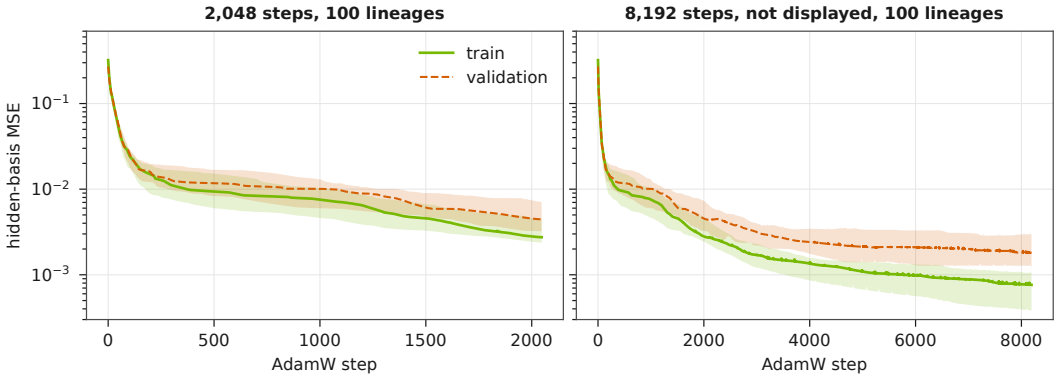



Figure 7: **Logged replay of both step budgets before the readout refit.** The solid line is the training median and the dashed line the validation median; each band is a 95% lineage-bootstrap interval. The longer budget is not the reported model.

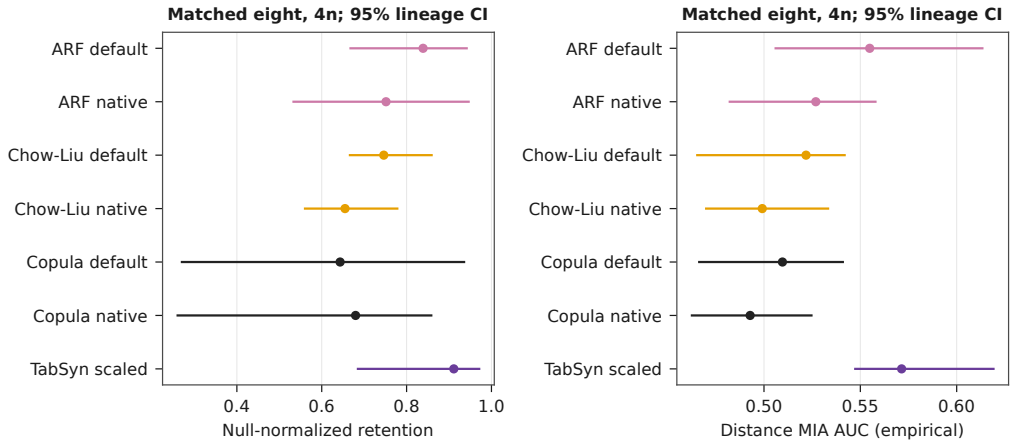



Figure 8: **Earlier eight-lineage TabSyn subset: medians and marginal lineage-bootstrap intervals.** No Holm family. Privacy AUC is an empirical diagnostic.

### J.1 Reported profile, all 100 lineages (fit seed 11)

These rows are the historical fit-seed-11 population of the reported profile. The cap column marks charged bytes at or under 10,240; it is a byte check, not a release certificate. Retention columns are CatBoost at $n$ and $4n$, linear at $4n$, and the MLP at $4n$, each the median of the three sample seeds; the identifier is the lineage's dataset hash prefix and the name is its PMLB name.

Table 42: Displayed profile, all 100 lineages. The cap column is charged bytes at or under 10,240. It is not a release certificate. No Holm family.

| id | name | rows | feat. | bytes | cap | CB $n$ | CB $4n$ | lin $4n$ | MLP $4n$ |
|---|---|---|---|---|---|---|---|---|---|
| 02a45777900441ce | 215_2dplanes | 800 | 10 | 2,583 | pass | 0.9930 | 1.0001 | 0.9990 | 1.0477 |
| 047e62b52a3f4ede | 695_chatfield_4 | 156 | 12 | 3,311 | pass | 1.0041 | 1.0155 | 1.0048 | — |
| 04c80944e971cb3e | first_principles_supernovae_zr | 157 | 1 | 762 | pass | 0.8753 | 0.8846 | 0.9959 | 1.2627 |
| 0561d673cb73c27e | 666_rmftsa_ladata | 339 | 10 | 2,921 | pass | 1.0095 | 1.1026 | 1.0338 | 1.1394 |
| 060b9fff8092db17 | 589_fri_c2_1000_25 | 667 | 25 | 5,717 | pass | 0.8893 | 0.9534 | 0.9911 | 1.5794 |
| 06b0312a585b436e | 1196_BNG_pharynx | 800 | 10 | 2,631 | pass | 0.8601 | 0.8964 | 1.0458 | 1.7049 |
| 0779d3fd5b3133a2 | nikuradse_2 | 241 | 1 | 758 | pass | 0.9096 | 0.9261 | 2.1723 | 2.2082 |
| 07f38946316c039b | 593_fri_c1_1000_10 | 667 | 10 | 2,876 | pass | 0.5650 | 0.6068 | 0.9617 | 1.6005 |
| 0a8428ef25a69175 | 537_houses | 800 | 8 | 2,609 | pass | 0.6257 | 0.7036 | 1.0150 | 1.0256 |
| 0d482c1dd7cc6592 | 210_cloud | 72 | 5 | 1,844 | pass | 1.0374 | 0.9772 | 1.0944 | 1.0239 |
| 0db59fd2579600d0 | 665_sleuth_case2002 | 98 | 6 | 1,962 | pass | 1.0371 | 1.1324 | 1.1345 | — |
| 0f86a714ec5f3ac6 | 1191_BNG_pbc | 800 | 18 | 4,183 | pass | 0.8097 | 0.8761 | 0.9173 | — |
| 167578f5deba4977 | 547_no2 | 333 | 7 | 2,443 | pass | 0.8073 | 0.8513 | 1.0097 | 0.9898 |

| id | name | rows | feat. | bytes | cap | CB $n$ | CB $4n$ | lin $4n$ | MLP $4n$ |
|---|---|---|---|---|---|---|---|---|---|
| 19f2c3c2d67defd7 | feynman_II_11_27 | 800 | 4 | 1,494 | pass | 0.9704 | 0.9890 | 0.9960 | 1.0048 |
| 19f4780b53b3fa41 | 678_visualizing_environmental | 74 | 3 | 1,515 | pass | — | — | — | 0.4431 |
| 1a4456d1cc71a26a | feynman_test_1 | 800 | 7 | 2,191 | pass | 1.0608 | 1.1671 | 0.9816 | — |
| 1a803b27c2400dc4 | feynman_III_10_19 | 800 | 4 | 1,520 | pass | 0.9882 | 0.9952 | 0.9942 | 0.9965 |
| 1cb97b1d7314d43d | 229_pwLinear | 133 | 10 | 2,587 | pass | 0.8417 | 0.9878 | 0.9104 | — |
| 205f4b4982b9ea49 | 608_fri_c3_1000_10 | 667 | 10 | 2,845 | pass | 0.5263 | 0.5671 | 1.0451 | 0.8182 |
| 26eff843be35d831 | 581_fri_c3_500_25 | 333 | 25 | 5,647 | pass | 0.7602 | 0.9020 | 1.1297 | 1.5792 |
| 282f8ddb43b5e677 | 595_fri_c0_1000_10 | 667 | 10 | 2,904 | pass | 0.8974 | 0.9234 | 0.9962 | 1.0250 |
| 297b7688a045c583 | 603_fri_c0_250_50 | 167 | 50 | 10,171 | pass | 0.8848 | 1.1562 | 1.0700 | 2.6883 |
| 2c8d0985f3557d2b | feynman_II_11_28 | 800 | 2 | 1,026 | pass | 0.9831 | 0.9899 | 1.0031 | 0.9976 |
| 30ca2887550f0d8d | 201_pol | 800 | 26 | 8,145 | pass | 0.5742 | 0.5976 | 0.7015 | 0.6923 |
| 3408a6a513c627a3 | strogatz_barmag1 | 267 | 2 | 1,011 | pass | 0.9750 | 0.9775 | 0.9969 | 1.0058 |
| 35a9aa9b88f0e4d1 | 344_mv | 800 | 10 | 2,737 | pass | 0.9985 | 1.0016 | 0.9662 | 1.0347 |
| 3d3321e27d14f9aa | 599_fri_c2_1000_5 | 667 | 5 | 1,707 | pass | 0.8359 | 0.8611 | 0.9668 | 1.6797 |
| 44eaabe4cb923766 | 613_fri_c3_250_5 | 167 | 5 | 1,712 | pass | 0.8671 | 0.9276 | 0.8599 | — |
| 46b25db9396c8209 | 227_cpu_small | 800 | 12 | 3,327 | pass | 0.8569 | 0.9072 | 0.9018 | 0.9364 |
| 46c5a59e5ae42037 | 1193_BNG_lowbwt | 800 | 9 | 2,436 | pass | 0.9635 | 0.9703 | 1.0030 | 0.9871 |
| 471f11cc2d38853a | feynman_test_11 | 800 | 4 | 1,481 | pass | 0.9645 | 0.9760 | 0.9975 | 0.9826 |
| 47c581af5de1136b | 582_fri_c1_500_25 | 333 | 25 | 5,670 | pass | 0.7510 | 0.9088 | 0.7173 | — |
| 48b7a3bf81e5f3e5 | 605_fri_c2_250_25 | 167 | 25 | 5,692 | pass | 0.4046 | 0.6465 | 1.9138 | — |
| 4a354d98c965b01f | 597_fri_c2_500_5 | 333 | 5 | 1,713 | pass | 0.7550 | 0.8055 | 0.9814 | 1.8381 |
| 4aa84fff8ad0cd26 | feynman_III_13_18 | 800 | 4 | 1,501 | pass | 0.8748 | 0.9137 | 0.8850 | 0.9348 |
| 51730b348b1bbd14 | 557_analcatdata_apnea1 | 316 | 3 | 1,237 | pass | -0.0032 | 0.0086 | — | — |
| 54b221668673164c | feynman_III_14_14 | 800 | 5 | 1,726 | pass | 1.0346 | 1.1164 | 1.0011 | 1.0611 |
| 552665868f8f72ed | strogatz_vdp2 | 267 | 2 | 989 | pass | 0.9945 | 0.9963 | 0.9960 | 0.9969 |
| 585fef2a1cb6b33f | strogatz_lv2 | 267 | 2 | 1,018 | pass | 0.8286 | 0.9859 | 0.7502 | — |
| 642738a361f1c63a | feynman_I_12_1 | 800 | 2 | 1,032 | pass | 0.9892 | 0.9910 | 0.9892 | 0.9933 |
| 64c64f0f9983c48f | strogatz_bacres2 | 267 | 2 | 998 | pass | 0.9793 | 0.9832 | 0.9783 | 0.9862 |
| 654566a1f91408aa | 4544_GeographicalOriginalofMusic | 706 | 117 | 22,233 | fail | — | — | — | — |
| 693890aae9dc860b | first_principles_supernovae_zg | 162 | 1 | 761 | pass | 0.8492 | 0.8908 | 1.0066 | 1.1997 |
| 6cacb11611345a24 | feynman_I_11_19 | 800 | 6 | 1,947 | pass | 0.9524 | 0.9812 | 0.9780 | 0.9800 |
| 7291b347fe0238fb | feynman_II_11_20 | 800 | 5 | 1,757 | pass | 0.9928 | 1.0432 | 1.0016 | 1.0674 |
| 7423783f7e38bb9b | strogatz_shearflow1 | 267 | 2 | 1,021 | pass | 0.4900 | 0.5614 | -0.3978 | — |
| 7a3d123dd555110d | 598_fri_c0_1000_25 | 667 | 25 | 5,668 | pass | 0.8613 | 0.9166 | 1.0046 | 1.0034 |
| 7c8943b160464add | 601_fri_c1_250_5 | 167 | 5 | 1,719 | pass | 0.7859 | 0.8552 | 1.0773 | — |
| 7e51711baa1b1fb1 | feynman_I_12_11 | 800 | 5 | 1,718 | pass | 0.9267 | 0.9320 | 1.0015 | 0.9860 |
| 80589368fa78f21a | 218_house_8L | 800 | 8 | 2,392 | pass | 0.5835 | 0.6708 | 0.9616 | 1.1040 |
| 8164950b155f7f56 | feynman_I_12_2 | 800 | 4 | 1,511 | pass | 0.9702 | 0.9986 | 0.9485 | 0.9753 |
| 85a31ae409944a1b | feynman_II_10_9 | 800 | 3 | 1,269 | pass | 0.9330 | 0.9508 | 0.9589 | 0.9515 |
| 8d859cfa7b51a351 | feynman_III_15_12 | 800 | 3 | 1,248 | pass | 0.5172 | 0.5542 | 1.0228 | 1.0749 |
| 923483635406b213 | 1203_BNG_pwLinear | 800 | 10 | 2,581 | pass | 1.2079 | 1.1750 | 0.9941 | 1.5230 |
| 9312d813229df035 | 690_visualizing_galaxy | 216 | 4 | 1,458 | pass | 0.9725 | 0.9755 | 0.9853 | 1.0870 |
| 94a00ea67f49bf84 | 596_fri_c2_250_5 | 167 | 5 | 1,698 | pass | 0.7546 | 0.8072 | — | — |
| 94bd4520b8ceb3f0 | feynman_test_12 | 800 | 5 | 1,747 | pass | 0.9688 | 0.9804 | 0.9913 | 1.0067 |
| 956d34f13e7af830 | 586_fri_c3_1000_25 | 667 | 25 | 5,701 | pass | 0.9325 | 0.9849 | 1.2339 | 2.1457 |
| 9a10c15885022034 | _deprecated_195_auto_price | 106 | 15 | 5,025 | pass | 0.8717 | 0.9485 | 0.9598 | 1.3107 |
| a04f964bc53281e0 | strogatz_lv1 | 267 | 2 | 985 | pass | -3.8784 | -3.4212 | — | — |
| a23f4bd626b76590 | feynman_test_13 | 800 | 5 | 1,740 | pass | 0.9178 | 0.9605 | 0.9900 | 1.0991 |
| a32a69e7c701eb14 | 588_fri_c4_1000_100 | 667 | 100 | 19,019 | fail | — | — | — | — |
| a5ba682572f53add | 602_fri_c3_250_10 | 167 | 10 | 2,900 | pass | 0.5355 | 0.8109 | 0.7988 | — |
| a863cd42be4d05d4 | 663_rabe_266 | 80 | 2 | 1,014 | pass | 0.9781 | 0.9933 | 0.9967 | 1.0521 |
| a97065df07a47eb0 | 592_fri_c4_1000_25 | 667 | 25 | 5,655 | pass | 0.8791 | 0.9340 | 0.9578 | 1.2714 |
| ab279eded3b3c52b | 574_house_16H | 800 | 16 | 4,035 | pass | 0.1181 | 0.1796 | 0.1795 | 0.3705 |
| aebf5d861a8365b4 | 522_pm10 | 333 | 7 | 2,409 | pass | 0.5082 | 0.6149 | 0.4125 | 2.3076 |
| b6adf79118a58cea | 562_cpu_small | 800 | 12 | 3,353 | pass | 0.6968 | 0.7747 | 0.9834 | 0.9307 |
| b78d323566930bd7 | 583_fri_c1_1000_50 | 667 | 50 | 10,098 | pass | 0.8607 | 0.9444 | 0.9960 | 1.6211 |
| b7f9774e2142a758 | strogatz_shearflow2 | 267 | 2 | 1,019 | pass | 0.4707 | 0.5439 | 0.7450 | 1.8955 |
| bc85856c012bcc1b | 607_fri_c4_1000_50 | 667 | 50 | 10,209 | pass | 0.8856 | 0.9640 | 3.9394 | — |
| c107ee39eb333a9d | 584_fri_c4_500_25 | 333 | 25 | 5,638 | pass | 0.8062 | 0.8619 | 0.9417 | — |
| c30ff9405c916718 | 579_fri_c0_250_5 | 167 | 5 | 1,692 | pass | 0.9475 | 1.0923 | 0.9961 | 0.9783 |
| c4f8e81b70d33a13 | 590_fri_c0_1000_50 | 667 | 50 | 10,180 | pass | 0.8839 | 0.9172 | 1.0359 | 1.1762 |
| c649a4cee87c4321 | 529_pollen | 800 | 4 | 1,506 | pass | 1.0356 | 1.1012 | 0.9717 | 1.2459 |
| cb5bf66ef2ee09ed | nikuradse_1 | 241 | 2 | 992 | pass | 0.9346 | 0.9373 | 0.9997 | 1.1909 |
| cba8393879afd1dc | 230_machine_cpu | 140 | 6 | 2,338 | pass | 1.0026 | 1.0891 | 1.1718 | — |
| ce1a79e2d658eb8a | strogatz_glider2 | 267 | 2 | 1,003 | pass | 0.6382 | 0.6625 | 1.0720 | 1.3756 |
| ce837ccd24a3a18c | strogatz_bacres1 | 267 | 2 | 1,004 | pass | 0.7352 | 0.8309 | 0.9014 | 0.9826 |
| cee6f811d0bf199d | feynman_II_11_3 | 800 | 5 | 1,728 | pass | 0.9385 | 0.9666 | 0.8950 | 1.0149 |
| d255d3a2e1c8dbf4 | feynman_III_12_43 | 800 | 2 | 1,028 | pass | 0.9920 | 0.9935 | 0.9966 | 0.9940 |
| d52bfd2b6ab1c850 | feynman_I_10_7 | 800 | 3 | 1,250 | pass | 0.9920 | 0.9932 | 0.9940 | 0.9959 |
| d9a3408a840c9f48 | 560_bodyfat | 168 | 14 | 4,364 | pass | 1.0247 | 1.0436 | 0.9930 | 1.3651 |
| da290fae0d6c072f | feynman_test_10 | 800 | 3 | 1,267 | pass | 0.9955 | 0.9990 | 0.9990 | 1.0015 |
| da78606950f100cd | 1199_BNG_echoMonths | 800 | 9 | 2,544 | pass | 0.9962 | 1.0086 | 0.9655 | 1.0694 |
| dd16ac872ffb8673 | strogatz_glider1 | 267 | 2 | 991 | pass | 0.1145 | 0.1723 | 0.9123 | — |
| de2868ac837699fd | 564_fried | 800 | 10 | 2,861 | pass | 0.9477 | 0.9829 | 0.9543 | 1.4707 |
| e08f0b9931f93686 | 225_puma8NH | 800 | 8 | 2,439 | pass | 1.0639 | 1.1043 | 0.9951 | 2.8498 |
| e18350fb51a9d1b0 | strogatz_barmag2 | 267 | 2 | 1,027 | pass | 0.8897 | 0.9399 | 0.2855 | — |
| e5a3a7b3668a1cc2 | 556_analcatdata_apnea2 | 316 | 3 | 1,238 | pass | 0.0339 | 0.0341 | — | — |
| e68fdd96cc63271b | 573_cpu_act | 800 | 21 | 4,954 | pass | 0.8080 | 0.9045 | 0.6593 | 1.1465 |
| ed2efb82fce4b098 | strogatz_vdp1 | 267 | 2 | 984 | pass | 0.5228 | 0.5812 | 1.2867 | 1.2401 |

| id | name | rows | feat. | bytes | cap | CB $n$ | CB $4n$ | lin $4n$ | MLP $4n$ |
|---|---|---|---|---|---|---|---|---|---|
| edc9d41a298fdf60 | 612_fri_c1_1000_5 | 667 | 5 | 1,705 | pass | 0.8224 | 0.8596 | 0.9670 | 1.2891 |
| efd53e9c0f36d608 | _deprecated_207_autoPrice | 106 | 15 | 5,017 | pass | 0.9020 | 0.9357 | 1.0899 | 1.0398 |
| f0a3d6881458f990 | 604_fri_c4_500_10 | 333 | 10 | 2,913 | pass | 0.6868 | 0.8360 | 0.8552 | — |
| f24e9383331c43e2 | 503_wind | 800 | 14 | 3,846 | pass | 0.9690 | 0.9962 | 1.0017 | 1.0367 |
| f652ea96eff62654 | strogatz_predprey1 | 267 | 2 | 989 | pass | 0.9964 | 1.1345 | 1.0218 | 1.4876 |
| f8cf362e371e8960 | strogatz_predprey2 | 267 | 2 | 1,030 | pass | 0.9876 | 0.9847 | 0.9718 | 1.0815 |
| fc3812567b5fdd64 | 197_cpu_act | 800 | 21 | 5,011 | pass | 0.7680 | 0.8922 | 0.6669 | 1.1253 |
| fe6d7ac3fcd3129e | 1201_BNG_breastTumor | 800 | 9 | 2,427 | pass | -0.6141 | -0.3437 | — | — |

## J.2 SUFFICIENCY BUDGET (FIT SEED 11)

These rows are the `features12_steps8192` budget check on the lineages where it produced a charged byte count or a retention; the remaining lineages have neither and are omitted. The columns match the reported-profile table, so the two tables can be read side by side for the step-budget axis of Appendix H. The longer budget is a check on the reported profile and is not the reported model.

Table 43: Sufficiency budget. 69 lineages have a charged byte count or a retention. 31 lineages have neither and are omitted. The cap column is charged bytes at or under 10,240. It is not a release certificate. No Holm family.

| id | name | rows | feat. | bytes | cap | CB $n$ | CB $4n$ | lin $4n$ | MLP $4n$ |
|---|---|---|---|---|---|---|---|---|---|
| 047e62b52a3f4ede | 695_chatfield_4 | 156 | 12 | 3,353 | pass | 1.0106 | 1.0287 | 1.0110 | — |
| 04c80944e971cb3e | first_principles_supernovae_zr | 157 | 1 | 750 | pass | 0.9372 | 0.9454 | 1.0346 | 1.3271 |
| 060b9fff8092db17 | 589_fri_c2_1000_25 | 667 | 25 | 5,734 | pass | 0.9237 | 0.9886 | 1.0258 | 1.5430 |
| 06b0312a585b436e | 1196_BNG_pharynx | 800 | 10 | 2,681 | pass | 0.8188 | 0.8768 | 1.0632 | 0.7968 |
| 07f38946316c039b | 593_fri_c1_1000_10 | 667 | 10 | 2,869 | pass | 0.9780 | 1.0222 | 0.9929 | 1.8097 |
| 0a8428ef25a69175 | 537_houses | 800 | 8 | 2,636 | pass | 0.7243 | 0.7841 | 1.0073 | 1.0334 |
| 0d482c1dd7cc6592 | 210_cloud | 72 | 5 | 1,855 | pass | 0.9561 | 0.8815 | 1.0056 | 0.9224 |
| 0f86a714ec5f3ac6 | 1191_BNG_pbc | 800 | 18 | 4,218 | pass | 0.8069 | 0.8576 | 0.8783 | — |
| 167578f5deba4977 | 547_no2 | 333 | 7 | 2,474 | pass | 0.8821 | 0.9276 | 0.9850 | 1.0276 |
| 1a4456d1cc71a26a | feynman_test_1 | 800 | 7 | 2,185 | pass | 1.1162 | 1.3026 | 0.6721 | — |
| 1a803b27c2400dc4 | feynman_III_10_19 | 800 | 4 | 1,522 | pass | 0.9698 | 0.9777 | 0.9732 | 0.9807 |
| 1cb97b1d7314d43d | 229_pwLinear | 133 | 10 | 2,598 | pass | 0.8275 | 1.0471 | 0.9727 | — |
| 26eff843be35d831 | 581_fri_c3_500_25 | 333 | 25 | 5,655 | pass | 0.8660 | 0.9960 | 1.0900 | 1.6067 |
| 282f8ddb43b5e677 | 595_fri_c0_1000_10 | 667 | 10 | 2,913 | pass | 0.9645 | 1.0058 | 1.0029 | 1.1269 |
| 297b7688a045c583 | 603_fri_c0_250_50 | 167 | 50 | 10,183 | pass | 0.8257 | 1.1292 | 1.0240 | 2.8999 |
| 30ca2887550f0d8d | 201_pol | 800 | 26 | 8,215 | pass | 0.6214 | 0.6436 | 0.7350 | 0.6603 |
| 3d3321e27d14f9aa | 599_fri_c2_1000_5 | 667 | 5 | 1,713 | pass | 0.9820 | 1.0044 | 1.0441 | 1.6148 |
| 44eaabe4cb923766 | 613_fri_c3_250_5 | 167 | 5 | 1,708 | pass | 0.8605 | 0.9611 | 0.9427 | — |
| 46b25db9396c8209 | 227_cpu_small | 800 | 12 | 3,338 | pass | 0.7033 | 0.8730 | 0.9142 | 0.9908 |
| 46c5a59e5ae42037 | 1193_BNG_lowbwt | 800 | 9 | 2,468 | pass | 0.8393 | 0.8390 | 0.8798 | 0.8103 |
| 471f11cc2d38853a | feynman_test_11 | 800 | 4 | 1,472 | pass | 0.9340 | 0.9481 | 0.9730 | 0.9578 |
| 47c581af5de1136b | 582_fri_c1_500_25 | 333 | 25 | 5,676 | pass | 0.8232 | 0.9820 | 0.8086 | — |
| 48b7a3bf81e5f3e5 | 605_fri_c2_250_25 | 167 | 25 | 5,696 | pass | 0.2608 | 0.5247 | 1.4268 | — |
| 4a354d98c965b01f | 597_fri_c2_500_5 | 333 | 5 | 1,715 | pass | 0.9578 | 1.0001 | 1.0224 | 1.3200 |
| 51730b348b1bbd14 | 557_analcatdata_apnea1 | 316 | 3 | 1,246 | pass | 0.6556 | 0.6654 | — | — |
| 54b2216686731 64c | feynman_III_14_14 | 800 | 5 | 1,724 | pass | 1.0175 | 1.1226 | 1.0021 | 1.0727 |
| 585fef2a1cb6b33f | strogatz_lv2 | 267 | 2 | 1,021 | pass | 0.9239 | 1.0451 | 0.8041 | — |
| 642738a361f1c63a | feynman_I_12_1 | 800 | 2 | 1,034 | pass | 0.9936 | 0.9961 | 0.9993 | 0.9989 |
| 64c64f0f9983c48f | strogatz_bacres2 | 267 | 2 | 996 | pass | 0.9743 | 0.9826 | 0.9816 | 0.9923 |
| 693890aae9dc860b | first_principles_supernovae_zg | 162 | 1 | 766 | pass | 0.9415 | 0.9642 | 1.0277 | 1.1876 |
| 6cacb11611345a24 | feynman_I_11_19 | 800 | 6 | 1,952 | pass | 0.9911 | 1.0176 | 1.0047 | 1.0089 |
| 7423783f7e38bb9b | strogatz_shearflow1 | 267 | 2 | 1,023 | pass | 0.4860 | 0.5973 | -4.3352 | — |
| 7a3d123dd555110d | 598_fri_c0_1000_25 | 667 | 25 | 5,672 | pass | 0.8570 | 0.9206 | 1.0010 | 1.0168 |
| 7e51711baa1b1fb1 | feynman_I_12_11 | 800 | 5 | 1,720 | pass | 0.9780 | 0.9927 | 0.9910 | 1.0559 |
| 80589368fa78f21a | 218_house_8L | 800 | 8 | 2,410 | pass | 0.6661 | 0.7528 | 0.6830 | 1.0728 |
| 85a31ae409944a1b | feynman_II_10_9 | 800 | 3 | 1,264 | pass | 0.9803 | 0.9961 | 0.9881 | 1.0061 |
| 8d859cfa7b51a351 | feynman_III_15_12 | 800 | 3 | 1,280 | pass | 0.7428 | 0.8572 | 0.8771 | 1.2306 |
| 9312d813229df035 | 690_visualizing_galaxy | 216 | 4 | 1,470 | pass | 0.9770 | 0.9875 | 0.9843 | 1.0661 |
| 94a00ea67f49bf84 | 596_fri_c2_250_5 | 167 | 5 | 1,718 | pass | 0.8262 | 0.8934 | — | — |
| 94bd4520b8ceb3f0 | feynman_test_12 | 800 | 5 | 1,745 | pass | 0.9677 | 0.9845 | 0.9937 | 1.0157 |
| 9a10c15885022034 | _deprecated_195_auto_price | 106 | 15 | 5,038 | pass | 0.8965 | 0.9544 | 0.9901 | 1.3165 |
| a04f964bc53281e0 | strogatz_lv1 | 267 | 2 | 986 | pass | -3.0531 | -2.4502 | — | — |
| a32a69e7c701eb14 | 588_fri_c4_1000_100 | 667 | 100 | 19,041 | fail | — | — | — | — |
| a5ba682572f53add | 602_fri_c3_250_10 | 167 | 10 | 2,902 | pass | 0.7774 | 0.9773 | 0.9471 | — |
| a863cd42be4d05d4 | 663_rabe_266 | 80 | 2 | 1,014 | pass | 0.9849 | 0.9996 | 1.0018 | 1.0573 |
| ab279eded3b3c52b | 574_house_16H | 800 | 16 | 4,076 | pass | 0.0805 | 0.1211 | -0.7472 | -0.3199 |
| aebf5d861a8365b4 | 522_pm10 | 333 | 7 | 2,450 | pass | 0.6372 | 0.7959 | 0.6203 | 3.4124 |
| b78d323566930bd7 | 583_fri_c1_1000_50 | 667 | 50 | 10,098 | pass | 0.8817 | 0.9737 | 0.8733 | 0.9465 |
| b7f9774e2142a758 | strogatz_shearflow2 | 267 | 2 | 1,022 | pass | 0.7388 | 0.8339 | -0.4873 | 1.5681 |
| c107ee39eb333a9d | 584_fri_c4_500_25 | 333 | 25 | 5,641 | pass | 0.8307 | 0.9024 | 0.9623 | — |
| c30ff9405c916718 | 579_fri_c0_250_5 | 167 | 5 | 1,700 | pass | 0.9081 | 1.0750 | 0.9949 | 0.9830 |

| id | name | rows | feat. | bytes | cap | CB $n$ | CB $4n$ | lin $4n$ | MLP $4n$ |
|---|---|---|---|---|---|---|---|---|---|
| c649a4cee87c4321 | 529_pollen | 800 | 4 | 1,506 | pass | 1.0619 | 1.1198 | 0.9983 | 1.2801 |
| cb5bf66ef2ee09ed | nikuradse_1 | 241 | 2 | 990 | pass | 0.9662 | 0.9677 | 0.9970 | 1.1953 |
| ce1a79e2d658eb8a | strogatz_glider2 | 267 | 2 | 1,004 | pass | 0.5396 | 0.6099 | 1.1474 | 1.2017 |
| ce837ccd24a3a18c | strogatz_bacres1 | 267 | 2 | 1,004 | pass | 0.7807 | 0.8568 | 0.8858 | 0.9570 |
| d255d3a2e1c8dbf4 | feynman_III_12_43 | 800 | 2 | 1,023 | pass | 0.9947 | 0.9964 | 0.9984 | 0.9967 |
| d52bfd2b6ab1c850 | feynman_I_10_7 | 800 | 3 | 1,248 | pass | 0.9978 | 0.9988 | 0.9994 | 1.0005 |
| da290fae0d6c072f | feynman_test_10 | 800 | 3 | 1,257 | pass | 0.9681 | 0.9718 | 0.9737 | 0.9744 |
| da78606950f100cd | 1199_BNG_echoMonths | 800 | 9 | 2,578 | pass | 0.9473 | 0.9673 | 0.9447 | 1.0028 |
| de2868ac837699fd | 564_fried | 800 | 10 | 2,864 | pass | 0.9826 | 1.0201 | 0.9548 | 1.5094 |
| e08f0b9931f93686 | 225_puma8NH | 800 | 8 | 2,440 | pass | 1.0168 | 1.0664 | 0.9433 | 2.7529 |
| e5a3a7b3668a1cc2 | 556_analcatdata_apnea2 | 316 | 3 | 1,241 | pass | 0.6282 | 0.6707 | — | — |
| e68fdd96cc63271b | 573_cpu_act | 800 | 21 | 4,958 | pass | 0.8507 | 0.9689 | 0.5661 | 1.1177 |
| ed2efb82fce4b098 | strogatz_vdp1 | 267 | 2 | 991 | pass | 0.5397 | 0.6472 | -0.1705 | 1.1154 |
| efd53e9c0f36d608 | _deprecated_207_autoPrice | 106 | 15 | 5,024 | pass | 0.9035 | 0.9215 | 1.0781 | 1.0481 |
| f0a3d6881458f990 | 604_fri_c4_500_10 | 333 | 10 | 2,918 | pass | 0.8895 | 1.0064 | 1.0421 | — |
| f24e9383331c43e2 | 503_wind | 800 | 14 | 3,877 | pass | 0.9992 | 0.9953 | 0.9918 | 1.0034 |
| f8cf362e371e8960 | strogatz_predprey2 | 267 | 2 | 1,030 | pass | 0.9865 | 0.9827 | 0.9658 | 1.0750 |
| fc3812567b5fdd64 | 197_cpu_act | 800 | 21 | 5,006 | pass | 0.8164 | 0.9529 | 0.6632 | 1.1726 |

### J.3 BeyondArena families in the dataset bucket

The inventory below lists every BeyondArena family name in the dataset bucket with its task, row count, feature count, recorded license, and split type. Only the families with a cleared license entered the fit check of Appendix I, and none of them was scored for retention. The inventory is a list of candidate tables for later panels and makes no claim about any of them.

Table 44: All 142 BeyondArena families. No Holm family.

| name | task | rows | cols | license | split |
|---|---|---|---|---|---|
| 5g_energy_consumption | regression | 92,629 | 22 | MIT | grouped |
| acquire_valued_shoppers_challenge | binary | 160,057 | 112 | not cleared | temporal |
| airfoil_self_noise | regression | 1,503 | 6 | CC-BY-4.0 | iid |
| allstate_claims_severity | regression | 188,317 | 131 | not cleared | iid |
| amazon_employee_access | binary | 32,769 | 10 | public-domain | iid |
| amex_non_iid | binary | 1,249,605 | 191 | not cleared | grouped |
| anes_voting_2026 | binary | 48,587 | 319 | not cleared | temporal |
| aps_failure | binary | 76,000 | 171 | CC-BY-4.0 | iid |
| asp_potassco_classification | multiclass | 1,212 | 138 | not cleared | grouped |
| audiology_diagnosis | multiclass | 199 | 69 | CC-BY-4.0 | iid |
| bad_customer_detection | binary | 1,723 | 14 | public-domain | iid |
| bank_customer_churn | binary | 10,000 | 11 | public-domain | iid |
| bank_marketing | binary | 45,211 | 14 | CC-BY-4.0 | iid |
| biogeographical_ancestry_prediction | multiclass | 635 | 105 | not cleared | iid |
| biomechanical_orthopaedic_prediction | multiclass | 310 | 7 | CC-BY-4.0 | iid |
| bioresponse | binary | 3,751 | 1777 | public-domain | iid |
| blood_tests_drink_prediction | regression | 345 | 6 | CC-BY-4.0 | iid |
| blood_transfusion | binary | 748 | 5 | CC-BY-4.0 | iid |
| body_density_prediction | regression | 252 | 14 | not cleared | iid |
| california_house_prices_2020 | regression | 41,528 | 42 | not cleared | temporal |
| cardiotocography | multiclass | 2,126 | 24 | CC-BY-4.0 | grouped |
| churn | binary | 5,000 | 20 | public-domain | iid |
| cirrhosis_patient_survival_prediction | regression | 161 | 18 | not cleared | iid |
| climate_model_weather_forecasting | regression | 1,250,000 | 101 | not cleared | temporal |
| clock_protein_toxicity | binary | 171 | 1118 | CC-BY-4.0 | iid |
| coffee_rating_prediction | regression | 2,369 | 13 | not cleared | temporal |
| coil_2000 | binary | 9,822 | 86 | CC-BY-4.0 | iid |
| concrete_compressive_strength | regression | 1,030 | 9 | CC-BY-4.0 | iid |
| consumer_complaints | multiclass | 1,226,140 | 13 | us-gov | temporal |
| cooking_time | regression | 1,250,000 | 197 | not cleared | temporal |
| covertype | multiclass | 512,625 | 15 | CC-BY-4.0 | grouped |
| credit_approval | binary | 690 | 16 | CC-BY-4.0 | iid |
| credit_card_clients_default | binary | 30,000 | 24 | CC-BY-4.0 | iid |
| credit_g | binary | 1,000 | 21 | CC-BY-4.0 | iid |
| customer_satisfaction_in_airline | binary | 129,880 | 22 | public-domain | iid |
| delivery_eta | regression | 1,250,000 | 226 | not cleared | temporal |
| dementia_prediction | multiclass | 370 | 10 | not cleared | grouped |
| diabetes_130_us | binary | 71,518 | 45 | CC-BY-4.0 | iid |
| diamonds | regression | 53,940 | 10 | MIT | iid |
| drug_induced_autoimmunity_prediction | binary | 597 | 178 | CC-BY-4.0 | iid |
| early_learning_predictors | regression | 18,874 | 745 | CC-BY-4.0 | grouped |
| early_stage_diabetes_risk_prediction | binary | 251 | 17 | CC-BY-4.0 | iid |
| ecoli_proteins | multiclass | 327 | 7 | CC-BY-4.0 | iid |
| ecommerce_shipping | binary | 10,999 | 11 | public-domain | iid |
| electric_motor_temperature_prediction | regression | 1,296,316 | 111 | not cleared | grouped |

Table 44, continued

| name | task | rows | cols | license | split |
|---|---|---|---|---|---|
| emscad | binary | 17,460 | 18 | CC0-1.0 | iid |
| eryhemato_squamous_disease | multiclass | 366 | 35 | CC-BY-4.0 | iid |
| fiat_500 | regression | 1,538 | 8 | CC0-1.0 | iid |
| fitness_club | binary | 1,500 | 7 | public-domain | iid |
| food_delivery_time | regression | 45,451 | 10 | not cleared | iid |
| forensic_glass_identification | multiclass | 214 | 10 | CC-BY-4.0 | iid |
| forest_fires | regression | 517 | 13 | CC-BY-4.0 | iid |
| gallstone_disease | binary | 319 | 39 | CC-BY-4.0 | iid |
| garments_worker_productivity | regression | 1,197 | 16 | CC-BY-4.0 | temporal |
| ghanas_indigenous_intel | multiclass | 10,928 | 11 | not cleared | temporal |
| give_me_some_credit | binary | 150,000 | 11 | not cleared | iid |
| hazelnut_spread_contaminant_detection | binary | 2,400 | 31 | not cleared | iid |
| healthcare_insurance_expenses | regression | 1,338 | 7 | not cleared | iid |
| heart_disease_cleveland | binary | 303 | 14 | CC-BY-4.0 | iid |
| heart_disease_hungary | binary | 294 | 14 | CC-BY-4.0 | iid |
| heart_disease_va_long_beach | binary | 200 | 14 | CC-BY-4.0 | iid |
| heart_failure_followup_survival | binary | 299 | 13 | CC-BY-4.0 | iid |
| heloc | binary | 10,459 | 24 | not cleared | iid |
| hepatitis_c_prediction | multiclass | 608 | 13 | CC-BY-4.0 | iid |
| hepatitis_survival_prediction | binary | 155 | 20 | CC-BY-4.0 | iid |
| hiva_agnostic | binary | 3,845 | 1519 | public-domain | iid |
| home_credit_default_risk | binary | 307,507 | 505 | not cleared | iid |
| home_credit_default_stability | binary | 1,224,927 | 712 | not cleared | temporal |
| homeq_default_prediction | binary | 5,708 | 13 | not cleared | iid |
| homesite_quote_conversion | binary | 260,753 | 296 | not cleared | iid |
| horse_colic_survival | multiclass | 344 | 21 | CC-BY-4.0 | iid |
| hotel_booking_demand | binary | 81,418 | 32 | CC-BY-4.0 | temporal |
| houses | regression | 19,675 | 9 | not cleared | iid |
| hr_analytics | binary | 19,158 | 13 | CC0-1.0 | iid |
| ieee_fraud_detection | binary | 590,540 | 436 | not cleared | temporal |
| immoscout_german_house_prices | regression | 10,317 | 24 | not cleared | iid |
| in_vehicle_coupon_recommendation | binary | 12,684 | 25 | CC-BY-4.0 | iid |
| indian_liver_patient_dataset | binary | 583 | 11 | CC-BY-4.0 | iid |
| iranian_churn | binary | 2,850 | 14 | CC-BY-4.0 | iid |
| jm1 | binary | 10,885 | 22 | public-domain | iid |
| kdd_cup_09_appetency | binary | 50,000 | 213 | public-domain | iid |
| kick | binary | 72,983 | 33 | public-domain | temporal |
| kickstarter | binary | 187,118 | 16 | not cleared | temporal |
| labour_inspection_compliance | binary | 63,634 | 377 | CC0-1.0 | iid |
| lending_club | binary | 1,064,751 | 97 | CC0-1.0 | temporal |
| ljubljana_breast_cancer | binary | 286 | 10 | CC-BY-4.0 | iid |
| ljubljana_primary_tumor | multiclass | 302 | 18 | CC-BY-4.0 | iid |
| lung_cancer | multiclass | 197 | 12601 | not cleared | iid |
| lung_cancer_epithelial_genexp | binary | 187 | 22216 | public-domain | iid |
| maps_router_eta | regression | 1,250,000 | 989 | not cleared | temporal |
| marketing_campaign | binary | 2,240 | 26 | CC-BY-4.0 | iid |
| maternal_health_risk | multiclass | 1,014 | 7 | CC-BY-4.0 | iid |
| mercari_price_suggestion | regression | 1,250,000 | 7 | not cleared | iid |
| mercedes_benz_greener_manufacturing | regression | 4,204 | 372 | not cleared | temporal |
| miami_housing | regression | 13,776 | 16 | not cleared | iid |
| mic | multiclass | 1,699 | 112 | CC-BY-4.0 | iid |
| mice_protein_trisomy_discriminant | multiclass | 1,080 | 78 | CC-BY-4.0 | grouped |
| micro_mass | multiclass | 571 | 1084 | CC-BY-4.0 | grouped |
| musk | binary | 6,598 | 168 | CC-BY-4.0 | grouped |
| mutual_funds_india | regression | 793 | 13 | CC0-1.0 | iid |
| naticusdroid_android_permissions_dataset | binary | 7,491 | 86 | CC-BY-4.0 | iid |
| obesity_estimation | regression | 498 | 15 | CC-BY-4.0 | iid |
| online_shoppers_purchasing_intention_dataset | binary | 12,330 | 18 | CC-BY-4.0 | iid |
| otto_group_product_classification_challenge | multiclass | 61,878 | 94 | not cleared | iid |
| pancreatic_cancer_mouse_detection | binary | 181 | 6773 | not cleared | grouped |
| parkinsons_biomedical_voice_measurements | binary | 195 | 25 | CC-BY-4.0 | grouped |
| physiochemical_protein | regression | 45,730 | 10 | CC-BY-4.0 | iid |
| polish_companies_bankruptcy | binary | 5,790 | 65 | CC-BY-4.0 | iid |
| porto_seguro | binary | 595,206 | 38 | not cleared | iid |
| predict_students_dropout_and_academic_success | multiclass | 4,424 | 37 | CC-BY-4.0 | iid |
| prostate_cancer_detection | binary | 322 | 15155 | not cleared | iid |
| pva_revenue_prediction_kddcup98 | binary | 144,095 | 478 | not cleared | iid |
| qsar_aquatic_toxicity | regression | 546 | 9 | CC-BY-4.0 | iid |
| qsar_biodeg | binary | 1,054 | 42 | CC-BY-4.0 | iid |
| qsar_fish_toxicity | regression | 908 | 7 | CC-BY-4.0 | iid |
| qsar_tid_11 | regression | 5,741 | 1025 | public-domain | iid |
| regensburg_pediatric_appendicitis | binary | 763 | 52 | not cleared | iid |
| rossmann_store_sales | regression | 844,392 | 16 | not cleared | temporal |
| santander_customer_satisfaction | binary | 71,080 | 308 | not cleared | iid |
| santander_customer_transaction_prediction | binary | 200,000 | 601 | not cleared | iid |
| santander_transaction_value | regression | 4,447 | 541 | not cleared | iid |
| sat11_hand_algo_runtime | regression | 2,960 | 171 | not cleared | grouped |
| sberbank_housing_market_forecasting | regression | 27,195 | 387 | not cleared | temporal |

Table 44, continued

| name | task | rows | cols | license | split |
|---|---|---|---|---|---|
| sdss_17 | multiclass | 78,053 | 12 | public-domain | iid |
| seismic_bumps | binary | 2,584 | 16 | CC-BY-4.0 | iid |
| sepsis_prediction | binary | 1,228,686 | 44 | not cleared | grouped |
| sepsis_survival_minimal_clinical_records | binary | 110,204 | 4 | CC-BY-4.0 | iid |
| sf_permit_time | regression | 116,954 | 38 | not cleared | temporal |
| south_africa_coronary_heart_disease | binary | 462 | 10 | not cleared | iid |
| splice | multiclass | 3,190 | 61 | CC-BY-4.0 | iid |
| student_portuguese_performance | regression | 649 | 31 | CC-BY-4.0 | iid |
| superconductivity | regression | 21,263 | 82 | CC-BY-4.0 | iid |
| taiwanese_bankruptcy_prediction | binary | 6,819 | 93 | CC-BY-4.0 | iid |
| telemonitoring_parkinsons_biomedical_voice_measurements | regression | 502 | 21 | CC-BY-4.0 | grouped |
| thyroid_discordant | binary | 3,711 | 27 | CC-BY-4.0 | iid |
| tour_travels_churn | binary | 954 | 7 | CC0-1.0 | iid |
| video_game_fps_prediction | regression | 12,288 | 40 | CC-BY-4.0 | grouped |
| video_transcoding_time_prediction | regression | 68,784 | 20 | CC-BY-4.0 | grouped |
| website_phishing | multiclass | 1,353 | 10 | CC-BY-4.0 | iid |
| wids_diabetes_mellitus | binary | 127,358 | 182 | not cleared | iid |
| wine_quality | regression | 6,497 | 13 | CC-BY-4.0 | iid |
| wine_world_cost | regression | 1,279 | 15 | CC0-1.0 | iid |

# K EXTENDED RELATED WORK

**Generators by family.** Parametric copulas (Patki et al., 2016) and Chow–Liu trees (Chow & Liu, 1968) are the classical density models, and pair-copula constructions (Aas et al., 2009) generalize them. Conditional GANs and variational autoencoders form a second family: CTGAN, TVAE, TableGAN, CTAB-GAN, and CTAB-GAN+ (Xu et al., 2019; Park et al., 2018; Zhao et al., 2021; 2024). Diffusion and score-based models form a third: TabDDPM, TabSyn, STASy, CoDi, TabDiff, CDTD, Forest-Diffusion, and GATD (Kotelnikov et al., 2023; Zhang et al., 2024; Kim et al., 2023; Lee et al., 2023; Shi et al., 2025; Mueller et al., 2025; Jolicoeur-Martineau et al., 2024; Zagardo, 2026). Autoregressive and language-model generators, and prior-data fitted generators, form a fourth (Borisov et al., 2023; Solatorio & Dupriez, 2023; Ma et al., 2023; Hollmann et al., 2025). Graphical neural generators are separate (Liu et al., 2023); ARF estimates density with a forest (Watson et al., 2023); CART synthesis is the synthpop line (Nowok et al., 2016); SMOTE interpolates between neighbouring rows (Chawla et al., 2002); and a survey collects these design choices (Borisov et al., 2024). Probabilistic-program synthesis (Saad et al., 2019) and CrossCat (Mansinghka et al., 2016) produce small, inspectable joint models.

**How DOPE differs from each line.** A parametric copula or a Chow–Liu tree optimizes a joint density of a fixed form; DOPE adds an explicit model of the target's conditional mean and charges the result against a byte cap. A conditional GAN optimizes adversarial agreement and a VAE reconstruction plus a latent regularizer; both ship weights orders of magnitude larger than DOPE's cap. A diffusion model optimizes a denoising or score objective over the joint table; DOPE optimizes the target's conditional mean directly. A language-model generator optimizes serialized-row likelihood with a pretrained backbone; DOPE stores a screened residual of fixed width. A forest density model such as ARF optimizes axis-aligned splits of the joint table; CART synthesis optimizes sequential conditional draws, which also model the target given the features but store the trees. A prior-data fitted network optimizes a pretraining objective on other tables, whereas DOPE is fit on the lineage alone. A probabilistic-program synthesizer targets posterior inference over the joint distribution. A differentially private synthesizer optimizes under a privacy budget; DOPE's privacy evidence is empirical, and its release gate is the MFS contract.

**What this paper measured.** The measured generators are the Gaussian copula, a Chow–Liu baseline and independent marginals written for this comparison, CTGAN, TVAE, ARF at five fit seeds, TabSyn on the full panel with a scaled schedule at five fit seeds, and a historical Forest-Flow block; TabDDPM, TabDiff, a language-model generator, synthpop CART, and SMOTE are scheduled under a common budget in predeclare_v2. This panel scores regression retention.

**Utility ratios, fidelity, and privacy.** Retention is a null-normalized TSTR/TRTR ratio; Esteban, Hyland, and Rätsch introduced train-on-synthetic, test-on-real and train-on-real, test-on-real (Esteban et al., 2017). Utility reviews separate marginal fidelity from downstream prediction (Snoke et al., 2018; Raab et al., 2018); alpha-precision, beta-recall, SDMetrics, and

synthcity are the usual fidelity reports (Alaa et al., 2022; Zhang et al., 2020; Qian et al., 2023). Differentially private Bayesian networks and marginal synthesizers optimize a privacy budget (Zhang et al., 2014; McKenna et al., 2021; 2022; Cai et al., 2021). Membership inference, shadow-model attacks, and holdout-referenced distances form the empirical privacy literature (Shokri et al., 2017; Hayes et al., 2018; van Breugel et al., 2023; Stadler et al., 2022; Platzer & Reutterer, 2021); formal differential privacy is a different claim from an empirical resemblance score (Dwork & Roth, 2014; Abadi et al., 2016).

**Auditors and rank tests.** Tree models remain strong tabular predictors, which is why CatBoost is one of the three auditors (Grinsztajn et al., 2022; Prokhorenkova et al., 2018; Shwartz-Ziv & Armon, 2022). TabZilla and TabArena show how brittle a single split can be (McElfresh et al., 2023; Erickson et al., 2026). The regression lineages come from PMLB (Olson et al., 2017; Romano et al., 2022). Comparing several methods on the same lineages needs a rank test and a critical difference, or a Bayesian signed-rank alternative (Demšar, 2006; Benavoli et al., 2017).

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Table 33: **Fidelity and detection by configuration and size.** Median, hierarchical interval, and lineage count in each measured cell. Common-numeric cohorts, with separate availability by metric. Stored logistic AUC and grouped CatBoost AUC have distinct columns. AUC alone establishes neither privacy nor release safety. No Holm family.

| Method | Configuration | Size | Marginal KS | Pair fidelity | Logistic AUC | CatBoost AUC |
|---|---|---|---|---|---|---|
| | | | 0.1128 | 0.9595 | 0.5321 | 0.6159 |
| | | | [0.0895, 0.1433] | [0.9471, 0.9644] | [0.5156, 0.5442] | [0.5751, 0.6928] |
| ARF | author default | $1n$ | $n = 100$ | $n = 100$ | $n = 100$ | $n = 100$ |
| | | | 0.1035 | 0.9618 | 0.5268 | 0.6120 |
| | | | [0.0846, 0.1340] | [0.9487, 0.9653] | [0.5069, 0.5388] | [0.5696, 0.6877] |
| ARF | author default | $4n$ | $n = 100$ | $n = 100$ | $n = 100$ | $n = 100$ |
| | | | 0.1122 | 0.9599 | 0.5255 | 0.6100 |
| | | | [0.0869, 0.1445] | [0.9418, 0.9626] | [0.5072, 0.5449] | [0.5789, 0.7089] |
| ARF | native selected | $1n$ | $n = 100$ | $n = 100$ | $n = 100$ | $n = 100$ |
| | | | 0.1046 | 0.9620 | 0.5296 | 0.6091 |
| | | | [0.0845, 0.1382] | [0.9464, 0.9643] | [0.5094, 0.5375] | [0.5766, 0.7075] |
| ARF | native selected | $4n$ | $n = 100$ | $n = 100$ | $n = 100$ | $n = 100$ |
| | | | 0.2319 | 0.9091 | 0.7839 | 0.9522 |
| | | | [0.1887, 0.2943] | [0.8313, 0.9406] | [0.6997, 0.8819] | [0.8778, 0.9841] |
| CTGAN | native selected | $1n$ | $n = 22$ | $n = 22$ | $n = 22$ | $n = 21$ |
| | | | 0.2300 | 0.9123 | 0.7839 | 0.9535 |
| | | | [0.1839, 0.2947] | [0.8367, 0.9431] | [0.6997, 0.8821] | [0.8487, 0.9777] |
| CTGAN | native selected | $4n$ | $n = 22$ | $n = 22$ | $n = 22$ | $n = 22$ |
| | | | 0.1293 | 0.9273 | 0.5423 | 0.6868 |
| | | | [0.0892, 0.2361] | [0.8834, 0.9418] | [0.5040, 0.6292] | [0.6025, 0.8986] |
| Chow-Liu | native selected | $1n$ | $n = 100$ | $n = 100$ | $n = 100$ | $n = 100$ |
| | | | 0.1266 | 0.9283 | 0.5404 | 0.7061 |
| | | | [0.0852, 0.2292] | [0.8906, 0.9465] | [0.5185, 0.6303] | [0.5910, 0.9369] |
| Chow-Liu | native selected | $4n$ | $n = 100$ | $n = 100$ | $n = 100$ | $n = 99$ |
| | | | 0.1216 | 0.9532 | 0.5172 | 0.6344 |
| | | | [0.0950, 0.1680] | [0.9317, 0.9610] | [0.5000, 0.5376] | [0.5705, 0.8368] |
| DOPE | DOPE profile | $1n$ | $n = 98$ | $n = 98$ | $n = 98$ | $n = 98$ |
| | | | 0.1138 | 0.9531 | 0.5172 | 0.6405 |
| | | | [0.0917, 0.1583] | [0.9346, 0.9645] | [0.5000, 0.5376] | [0.5830, 0.8312] |
| DOPE | DOPE profile | $4n$ | $n = 98$ | $n = 98$ | $n = 98$ | $n = 98$ |
| | | | 0.1254 | 0.9548 | 0.5187 | |
| | | | [0.0612, 0.1992] | [0.9242, 0.9799] | [0.4139, 0.7000] | |
| Forest-Flow | native selected | $1n$ | $n = 6$ | $n = 6$ | $n = 6$ | — |
| | | | 0.1231 | 0.9691 | 0.5187 | |
| | | | [0.0557, 0.2062] | [0.9321, 0.9811] | [0.4139, 0.7000] | |
| Forest-Flow | native selected | $4n$ | $n = 6$ | $n = 6$ | $n = 6$ | — |
| | | | 0.1393 | 0.9614 | 0.5246 | 0.7343 |
| | | | [0.1171, 0.2132] | [0.9516, 0.9652] | [0.5100, 0.5518] | [0.6789, 0.8981] |
| GaussianCopula | native selected | $1n$ | $n = 100$ | $n = 100$ | $n = 100$ | $n = 100$ |
| | | | 0.1336 | 0.9635 | 0.5246 | 0.7346 |
| | | | [0.1137, 0.2115] | [0.9488, 0.9687] | [0.5100, 0.5537] | [0.6902, 0.8689] |
| GaussianCopula | native selected | $4n$ | $n = 100$ | $n = 100$ | $n = 100$ | $n = 99$ |
| | | | 0.1500 | 0.9240 | 0.5918 | 0.7479 |
| | | | [0.1199, 0.2598] | [0.8905, 0.9418] | [0.5616, 0.6392] | [0.7221, 0.8859] |
| TVAE | native selected | $1n$ | $n = 22$ | $n = 22$ | $n = 22$ | $n = 22$ |
| | | | 0.1382 | 0.9258 | 0.6033 | 0.7309 |
| | | | [0.1174, 0.2431] | [0.8913, 0.9451] | [0.5544, 0.6681] | [0.7032, 0.8893] |
| TVAE | native selected | $4n$ | $n = 22$ | $n = 22$ | $n = 22$ | $n = 22$ |
| | | | 0.1244 | 0.9543 | | 0.6242 |
| | | | [0.0828, 0.1416] | [0.9412, 0.9598] | | [0.5734, 0.7373] |
| TabSyn | scaled schedule | $1n$ | $n = 99$ | $n = 99$ | — | $n = 99$ |
| | | | 0.1196 | 0.9581 | | 0.6168 |
| | | | [0.0829, 0.1351] | [0.9458, 0.9631] | | [0.5866, 0.7772] |
| TabSyn | scaled schedule | $4n$ | $n = 99$ | $n = 99$ | — | $n = 98$ |
| | | | 0.1363 | 0.8994 | 0.5213 | 0.8694 |
| | | | [0.0866, 0.2847] | [0.8099, 0.9289] | [0.5000, 0.5347] | [0.7602, 0.9749] |
| Independent marginals | native selected | $1n$ | $n = 100$ | $n = 100$ | $n = 100$ | $n = 99$ |
| | | | 0.1297 | 0.8986 | 0.5166 | 0.8578 |
| | | | [0.0813, 0.2672] | [0.8258, 0.9329] | [0.4988, 0.5488] | [0.7553, 0.9727] |
| Independent marginals | native selected | $4n$ | $n = 100$ | $n = 100$ | $n = 100$ | $n = 100$ |

Table 34: **Alpha-precision and beta-recall.** Fixed unembedded equations, median over complete sample triples, then hierarchical interval and lineage count. Configuration and cohort are explicit; unavailable cells carry no scalar. The retained TabSyn subset remains separate. No Holm family.

| Method | Configuration | Size | Cohort | $\alpha$ precision | $\beta$ recall |
|---|---|---|---|---|---|
| ARF | author default | $1n$ | common numeric | 0.8838 [0.8644, 0.9144] $n = 100$ | 0.4269 [0.3237, 0.4727] $n = 100$ |
| ARF | author default | $4n$ | common numeric | 0.8786 [0.8447, 0.9103] $n = 100$ | 0.4227 [0.3150, 0.4663] $n = 100$ |
| ARF | native selected | $1n$ | common numeric | 0.8713 [0.8502, 0.9013] $n = 100$ | 0.4360 [0.3010, 0.4646] $n = 100$ |
| ARF | native selected | $4n$ | common numeric | 0.8767 [0.8465, 0.9069] $n = 100$ | 0.4208 [0.3180, 0.4607] $n = 100$ |
| CTGAN | native selected | $1n$ | common numeric | unavailable $n = 0$ | unavailable $n = 0$ |
| CTGAN | native selected | $4n$ | common numeric | 0.7193 [0.4848, 0.8751] $n = 22$ | 0.1695 [0.1140, 0.2667] $n = 22$ |
| Chow-Liu | native selected | $1n$ | common numeric | unavailable $n = 0$ | unavailable $n = 0$ |
| Chow-Liu | native selected | $4n$ | common numeric | 0.8585 [0.7403, 0.9238] $n = 100$ | 0.3595 [0.1793, 0.4230] $n = 100$ |
| DOPE | DOPE profile | $1n$ | common numeric | unavailable $n = 0$ | unavailable $n = 0$ |
| DOPE | DOPE profile | $4n$ | common numeric | 0.9051 [0.8751, 0.9254] $n = 98$ | 0.4366 [0.2566, 0.4863] $n = 98$ |
| Forest-Flow | native selected | $1n$ | common numeric | unavailable $n = 0$ | unavailable $n = 0$ |
| Forest-Flow | native selected | $4n$ | common numeric | unavailable $n = 0$ | unavailable $n = 0$ |
| GaussianCopula | native selected | $1n$ | common numeric | 0.8125 [0.7516, 0.8499] $n = 100$ | 0.4301 [0.2173, 0.4593] $n = 100$ |
| GaussianCopula | native selected | $4n$ | common numeric | unavailable $n = 0$ | unavailable $n = 0$ |
| TVAE | native selected | $1n$ | common numeric | 0.8496 [0.7411, 0.8966] $n = 22$ | 0.3562 [0.2778, 0.3991] $n = 22$ |
| TVAE | native selected | $4n$ | common numeric | 0.8714 [0.8033, 0.9182] $n = 99$ | 0.4807 [0.3965, 0.5177] $n = 99$ |
| TabSyn | scaled schedule | $1n$ | common numeric | 0.8418 [0.4700, 0.9326] $n = 8$ | 0.4591 [0.3520, 0.5267] $n = 8$ |
| TabSyn | scaled schedule | $1n$ | retained subset | 0.8920 [0.7919, 0.9280] $n = 99$ | 0.4633 [0.4100, 0.5242] $n = 99$ |
| TabSyn | scaled schedule | $4n$ | common numeric | 0.8262 [0.4467, 0.9352] $n = 8$ | 0.4466 [0.3420, 0.5686] $n = 8$ |
| TabSyn | scaled schedule | $4n$ | retained subset | unavailable $n = 0$ | unavailable $n = 0$ |
| Independent marginals | native selected | $1n$ | common numeric | 0.8849 [0.8141, 0.9181] $n = 100$ | 0.2837 [0.0949, 0.3963] $n = 100$ |

Table 35: **Baseline coverage from committed publications.** Each row summarizes the retained publication for its method, whose path and digest are in `baseline-coverage.json`; that publication is the row's denominator. The TabSyn row is the earlier retained run; main Tables 1 and 2 report the later full scaled-schedule panel. No Holm family.

| Method | Published measured coverage | Remaining campaign |
|---|---|---|
| TabSyn | 49 scaled-default lineages, 143 measured cells at fit11: eight complete n/4n lineages (48 cells), plus 95 n cells on 41 lineages (19 complete/22 partial groups). Native tuning/full population pending. | Pending |
| TabDDPM | 12 retained lineages; 126 physical/132 logical cells; 21 models, fit seed 11. Default/native cohorts differ; native search incomplete. | Pending |
| Forest-Flow | 13 default lineages at five fit seeds; 65 models/390 cells. Separate legacy six-lineage default/native panel; full population pending. | Pending |
| Forest-Diffusion | Pilot only; separate 100-lineage matched diffusion panel pending. | Pending |
| ARF | 100 default/native lineages at five fit seeds, n/4n, three sample seeds: 6,000 logical cells / 5,892 physical receipts (108 aliases). Between-fit SD available; official-test certification pending. | Pending |
| GaussianCopula | 100 default/native lineages at fit seed 11; 1200 logical cells. Five-fit common coverage pending. | Pending |
| Chow-Liu | 100 default/native lineages at fit seed 11; 1200 logical cells. Five-fit common coverage pending. | Pending |

Table 36: **Retained classical validation at $4n$.** Median [95% lineage-bootstrap interval] (informative lineages). No Holm family in this table.

| Method | Config. | CB retention | KS/TV | C2ST CB AUC | Distance MIA AUC |
|---|---|---|---|---|---|
| ARF | Default | 0.820 [0.787, 0.872] (99) | 0.103 [0.097, 0.114] (100) | 0.604 [0.582, 0.626] (100) | 0.553 [0.538, 0.564] (100) |
| ARF | Native | 0.806 [0.736, 0.855] (99) | 0.105 [0.098, 0.114] (100) | 0.610 [0.591, 0.627] (100) | 0.542 [0.536, 0.559] (100) |
| Chow–Liu | Default | 0.643 [0.552, 0.672] (99) | 0.128 [0.106, 0.154] (100) | 0.721 [0.652, 0.773] (100) | 0.513 [0.503, 0.525] (100) |
| Chow–Liu | Native | 0.566 [0.460, 0.631] (99) | 0.126 [0.105, 0.157] (100) | 0.702 [0.641, 0.761] (99) | 0.513 [0.504, 0.523] (100) |
| Copula | Default | 0.594 [0.427, 0.762] (99) | 0.126 [0.094, 0.138] (100) | 0.615 [0.591, 0.671] (100) | 0.514 [0.503, 0.524] (100) |
| Copula | Native | 0.641 [0.431, 0.803] (99) | 0.134 [0.125, 0.158] (100) | 0.730 [0.696, 0.779] (99) | 0.507 [0.494, 0.521] (100) |

Table 37: **Earlier eight-lineage TabSyn subset at $4n$.** Same inputs and real-data controls for every method; median [95% interval] (lineages). A separate cohort from the full TabSyn panel. No Holm family in this table.

| Method | Config. | CB retention | KS/TV | C2ST CB AUC | Distance MIA AUC |
|---|---|---|---|---|---|
| ARF | Default | 0.839 [0.668, 0.942] (8) | 0.129 [0.084, 0.165] (8) | 0.623 [0.547, 0.969] (8) | 0.555 [0.506, 0.613] (8) |
| ARF | Native | 0.752 [0.534, 0.946] (8) | 0.132 [0.084, 0.163] (8) | 0.606 [0.583, 0.812] (8) | 0.527 [0.482, 0.558] (8) |
| Chow–Liu | Default | 0.746 [0.666, 0.859] (8) | 0.186 [0.068, 0.255] (8) | 0.756 [0.525, 0.999] (8) | 0.522 [0.465, 0.542] (8) |
| Chow–Liu | Native | 0.655 [0.561, 0.778] (8) | 0.188 [0.087, 0.248] (8) | 0.707 [0.525, 0.999] (8) | 0.499 [0.470, 0.533] (8) |
| Copula | Default | 0.643 [0.270, 0.935] (8) | 0.146 [0.067, 0.242] (8) | 0.687 [0.554, 0.987] (8) | 0.510 [0.466, 0.541] (8) |
| Copula | Native | 0.680 [0.260, 0.858] (8) | 0.157 [0.097, 0.242] (8) | 0.698 [0.642, 0.987] (8) | 0.493 [0.463, 0.525] (8) |
| TabSyn | Scaled default | 0.911 [0.685, 0.971] (8) | 0.136 [0.120, 0.157] (8) | 0.670 [0.597, 0.869] (8) | 0.571 [0.547, 0.619] (8) |

Table 38: **Exploratory eight-lineage subset differences, CatBoost.** The denominator is the shared lineages in the cell. Holm family size is four metrics times six configurations. The corrected simultaneous intervals are unbounded.

| TabSyn minus | $n$ | Median CB difference | Simultaneous median CI | Holm $p$ |
|---|---|---|---|---|
| ARF/default | 8 | 0.010 | Unbounded | 1.000 |
| ARF/native | 8 | 0.159 | Unbounded | 0.469 |
| Chow–Liu/default | 8 | 0.157 | Unbounded | 1.000 |
| Chow–Liu/native | 8 | 0.149 | Unbounded | 0.188 |
| Copula/default | 8 | 0.242 | Unbounded | 1.000 |
| Copula/native | 8 | 0.245 | Unbounded | 1.000 |

Table 39: **DOPE–ARF legacy matched receipt at $4n$: exploratory paired differences with conservative simultaneous median intervals.** The denominator is the shared-lineage intersection for that auditor. Holm family size is six. Median-of-seed aggregation.

| ARF config. | Auditor | $n$ | DOPE minus ARF | Simultaneous median CI | Holm $p$ |
|---|---|---|---|---|---|
| Default | catboost | 97 | 0.077 | [0.019, 0.154] | $1.31 \times 10^{-5}$ |
| Native | catboost | 97 | 0.095 | [0.028, 0.175] | $5.47 \times 10^{-7}$ |
| Default | linear | 92 | 0.027 | [0.0056, 0.090] | $8.53 \times 10^{-5}$ |
| Native | linear | 92 | 0.037 | [0.00582, 0.113] | $1.62 \times 10^{-5}$ |
| Default | mlp | 75 | 0.115 | [0.043, 0.236] | $4.56 \times 10^{-8}$ |
| Native | mlp | 75 | 0.143 | [0.040, 0.253] | $8.01 \times 10^{-8}$ |

Table 40: **First-batch diagnostics, one row per physical job.** Individual evaluations without a population interval. No Holm family.

| Job prefix | Method | Pearson diff. | Alpha | Beta | Distance MIA AUC |
|---|---|---|---|---|---|
| 0002af084a92 | independent_marginals | 0.805 | 0.602 | 0 | 0.474 |
| 000377f02a18 | DOPE | 0.071 | 0.836 | 0.558 | 0.504 |
| 000f66aa8642 | DOPE | 0.266 | 0.876 | 0.187 | 0.595 |
| 001897658278 | Chow-Liu | 0.194 | 0.804 | 0.450 | 0.632 |
| 0021656cfc57 | Forest-Flow | 0.046 | 0.780 | 0.555 | 0.614 |
| 00514b7a27e2 | DOPE | 0.143 | 0.875 | 0.239 | 0.517 |
| 0056ef8c748b | independent_marginals | 0.149 | 0.696 | 0.361 | 0.637 |
| 006a72169115 | GaussianCopula | 0.064 | 0.878 | 0.383 | 0.452 |
| 006cafa72e1a | DOPE | 0.069 | 0.838 | 0.725 | 0.710 |
| 006dfe86527f | independent_marginals | 0.072 | 0.805 | 0.491 | 0.478 |
| 007f4ac594ee | TVAE | 0.134 | 0.944 | 0.523 | 0.615 |
| 0090be69a85e | GaussianCopula | 0.098 | 0.867 | 0.480 | 0.393 |

Table 41: **BeyondArena families that prepared.** The denominator is the 12-family fit check, distinct from the 142-family inventory. No retention auditor and no Holm family.

| Family | Fit rows | 2,048 B | 8,192 B | Val 2,048 | Val 8,192 |
|---|---|---|---|---|---|
| airfoil | 802 | 804 | 818 | 0.010 | 0.00448 |
| garments | 939 | 2,317 | 2,372 | 0.028 | 0.025 |
| musk | 3,738 | 9,273 | 9,350 | 0.146 | 0.090 |
| telemonitoring | 257 | 1,852 | 1,890 | 0.033 | 0.019 |
| video | 35,918 | 5,054 | 5,071 | 0.00247 | 0.00195 |